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Hawkes Process Models of Limit Order Books for Optimal Market Making:

A Stochastic Control Approach

Limit Order Book modelling with Hawkes processes and optimal market making

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Contents

1 Introduction **1**
- 1.1 Motivation: liquidity provision in clustered electronic markets . . . 1
- 1.2 From empirical realism to a well-posed control problem . . . 1
- 1.3 Why the theory is nonstandard . . . 2
- 1.4 Contributions and what is new in this thesis . . . 2
- 1.5 Thesis outline . . . 2

2 State of the Art **3**
- 2.1 LOB modelling families: a brief map . . . 3
- 2.2 Point-process and Hawkes models for order flow . . . 3
- 2.3 Optimal market making and stochastic control . . . 3
- 2.4 Gap and thesis positioning . . . 4

3 Mathematical preliminaries: point processes and Hawkes dynamics **4**
- 3.1 Counting processes, conditional intensities, and compensation . . . 4
- 3.2 Linear Hawkes processes . . . 4
- 3.3 Stability, integrability, and non-explosion: terminology . . . 5
- 3.4 Hawkes stability, integrability, and non-explosion . . . 5
- 3.5 Markov embedding for exponential Hawkes kernels . . . 7
- 3.6 Predictable Bernoulli marking and thinning . . . 7

4 Market model and control problem **8**
- 4.1 State, information, and controls . . . 8
- 4.2 Market-wide aggressive order flow: a bivariate Hawkes model . . . 9
- 4.3 Executions against the market maker: predictable thinnings . . . 9
- 4.4 Midprice dynamics via Bernoulli marking . . . 10
- 4.5 Inventory, cash, and marked-to-market wealth . . . 11
- 4.6 Controlled state dynamics and Markov property . . . 11
- 4.7 Objective functional . . . 12
- 4.8 Value function . . . 13
- 4.9 Well-definedness of the CARA objective . . . 13
- 4.10 Modelling scope: midprice realism vs tractability . . . 14

5 Semimartingale calculus and controlled generators **14**
- 5.1 A generic controlled jump–diffusion driven by counting processes . . . 14
- 5.2 Poisson embedding and the merged Poisson random measure formulation . . . 15
- 5.3 Well-posedness and integrability via a triangular construction . . . 16
- 5.4 Itô's formula and the drift/generator decomposition . . . 23
- 5.5 Controlled generator: definition . . . 24
- 5.6 Dynkin's formula . . . 25
- 5.7 Specialisation: generator for the Hawkes market-making state . . . 26

6 Dynamic Programming Principle **27**
- 6.1 Viscosity solutions . . . 27
- 6.2 Stopping-time DPP for a terminal criterion . . . 27
- 6.3 Proof of the stopping-time DPP . . . 28

7 Derivation of the Hamilton–Jacobi–Bellman equation **29**
- 7.1 Classical derivation via a two-inequalities argument . . . 29

8 Verification theorem (classical solution implies optimality) **31**

9 CARA factorization (dimension reduction) 32
9.1 CARA factorization and terminal condition . . . 32
9.2 Recap: controlled generator in our Hawkes market-making state . . . 32
9.3 Derivation of the reduced HJB . . . 33
9.4 Algebraic lifting of classical solutions . . . 34
9.5 Integrability for Dynkin and verification under CARA . . . 35

10 Empirical pipeline and implementation conventions 36
10.1 Episode definition and time origin . . . 37
10.2 Event extraction: a LOBSTER-consistent event dictionary . . . 37
10.3 Reduction to a 2D aggressive execution flow . . . 38
10.4 Empirical evidence supporting the modelling reductions . . . 38
10.5 Hawkes calibration on intraday windows . . . 41
10.6 Reduced microstructure calibration . . . 44
10.7 Numerical HJB solution: discretization choices that match the data . . . 46
10.8 Backtesting modes and sign-consistent bookkeeping . . . 46
10.9 End-to-end pipeline: what is fixed, what is estimated, what is assumed . . . 50
10.10 Code availability and reproducibility . . . 51

11 Results and empirical findings 51

12 Conclusion 51

13 Future work and development directions 53
13.1 A more realistic midprice model (beyond one-tick Bernoulli marking) . . . 53
13.2 From 2D aggressive flow to richer event dictionaries . . . 53
13.3 Control and execution realism: queue position, latency, and partial fills . . . 53
13.4 Stronger mathematical generality: viscosity solutions and weak DPP . . . 54
13.5 Numerics: higher dimensions, approximation, and verification . . . 54
13.6 Empirical robustness and cross-asset generalization . . . 54

1 Introduction

1.1 Motivation: liquidity provision in clustered electronic markets

Modern equity markets are almost entirely electronically traded and organized around *limit order books* (LOBs), where liquidity is supplied through posted limit orders and consumed by marketable orders. At sub-second horizons, the book evolves through a dense stream of discrete events—submissions, cancellations, and executions—generated by a heterogeneous population of algorithmic traders. For a liquidity provider, profitability is tied to a simple mechanism (earning the spread when quotes are hit), but the associated risks are fundamentally *microstructural*: inventory accumulates, short-term price moves can make recently-filled inventory unprofitable, whilst execution opportunities are not uniformly distributed in time.

A key empirical feature is that order flow is *bursty* rather than memoryless. Executions (and more generally book events) tend to cluster and to exhibit persistence and cross-dependence across sides. This is precisely the regime in which the standard independent-Poisson simplification becomes most fragile: when activity is endogenous and self-reinforcing, both execution probabilities and adverse selection are time-varying in a structured way. As a consequence, quoting is not merely a static choice of spread; it is a *dynamic* decision under a time-dependent, feedback-driven flow of arrivals.

This thesis studies market making through that lens. The guiding practical question is:

> *How should a market maker choose bid and ask quote offsets over time, balancing spread capture against inventory risk, when the order-arrival mechanism is strongly time-dependent and exhibits self- and cross-excitation?*

The ambition is not to build an all-encompassing simulator of every LOB mechanism, but to construct a microstructure-consistent model that is sufficiently realistic to reproduce key event-time statistics while remaining sufficiently structured to support a rigorous stochastic-control treatment.

1.2 From empirical realism to a well-posed control problem

A recurring tension in the literature lies in the gap between (i) models that are empirically rich enough to reproduce high-frequency LOB features, and (ii) models that are analytically transparent enough to yield tractable optimal quoting policies via dynamic programming. This thesis aims to bridge that gap by committing to an explicit probabilistic structure: *event-time modelling with predictable intensities*.

The modelling backbone is a multivariate point-process representation of order flow, with Hawkes-type dynamics used to capture endogenous clustering and cross-dependence. Hawkes processes provide a disciplined way to turn "bursty activity" into mathematics: intensities respond to past events through excitation kernels, so that periods of high activity tend to be followed by further activity. Crucially, with exponential kernels the intensity dynamics admit a *finite-dimensional Markov embedding*, which allows one to enlarge the state and recover Markovianity. This single modelling choice is what makes it possible to carry out dynamic programming with a Hawkes-driven flow: without a Markov representation, the value function would depend on the full event history.

On top of the arrival model, the market maker's interaction with the flow must be described in a way that is both microstructure-consistent and mathematically controllable. In this thesis, two predictable mechanisms play a central role: (i) *execution thinning*, which turns market-order flow into "hits my quotes" versus "misses me" as a function of quote offsets, and (ii) *price-move marking*, which associates a short-horizon directional price-move event to executions, capturing adverse selection in reduced form. Both are implemented as predictable transformations of point processes, preserving intensity-based structure while encoding the economic effect of quoting.

The resulting controlled state is a Hawkes-driven jump–diffusion of the form

$$Z_t = (S_t, Q_t, X_t, \lambda_t^+, \lambda_t^-),$$

where S is the midprice proxy, Q the inventory, X the cash (or wealth component), and (λ^+, λ^-) the Markovian Hawkes intensity coordinates governing the aggressive flow. Controls are the bid/ask quote offsets $\delta_t = (\delta_t^a, \delta_t^b)$, restricted to a compact admissible set to exclude pathologies and match tick-level quoting. The objective is posed through a terminal liquidation-style criterion with inventory penalty and CARA utility, so that the optimisation problem is well-defined under explicit integrability conditions and bounded inventory.

1.3 Why the theory is nonstandard

From a control-theoretic perspective, the hard part is not writing down a formal HJB equation but ensuring that the objects needed by dynamic programming actually exist and behave properly when the jump mechanism has *stochastic predictable intensities*.

Many textbook derivations start from controlled diffusions or from Lévy jump models with independent increments. A Hawkes-driven setting is different: increments are not independent, and the correct calculus is compensator-based (Doob–Meyer) point-process theory rather than Lévy–Itô machinery. In particular, a rigorous DPP/HJB pipeline requires three ingredients:

1. a *well-posed controlled state equation* (existence and uniqueness of a strong solution, and integrability);
2. a *Dynkin/Itô identity* for the controlled semimartingale with jump terms driven by compensated counting-process martingales;
3. a *stability* of the admissible control class under stopping-time concatenation, which is essential for a stopping-time DPP.

This thesis makes these steps explicit in a Hawkes-driven environment by (a) leveraging the Markov embedding induced by exponential kernels, and (b) building the semimartingale and generator calculus in a way that is compatible with predictable thinning/marking. With these tools in place, the thesis states and proves a stopping-time DPP for the finite-horizon problem and derives the associated HJB equation.

Because the HJB includes a supremum over controls and *nonlocal* jump terms (state shifts due to executions and intensity updates), global smoothness of the value function cannot be taken for granted. For full mathematical generality, the natural notion is that of viscosity solutions; in this thesis the viscosity perspective is discussed to clarify the proper PDE framework, while the main derivation and verification argument is presented in the classical smooth-candidate setting (under sufficient regularity of the proposed solution). This keeps the economic structure transparent while making clear where additional analytical machinery would be needed to remove smoothness assumptions.

1.4 Contributions and what is new in this thesis

The thesis contributions are best read as an *end-to-end chain* that connects data, modelling, and control, with particular emphasis on the mathematical legwork that makes the control layer legitimate.

Theoretical contributions. First, the thesis formulates a Hawkes-driven market-making model in which the *intensity state is part of the controlled Markov state*, enabling dynamic programming in an otherwise history-dependent environment. Second, it develops the compensator-based semimartingale calculus required in this setting (controlled generators, Dynkin identities, and the martingale structure of compensated point-process integrals), and uses it to prove a stopping-time DPP and justify the HJB characterisation in a feedback-control class. Third, it connects the classical verification logic to the jump–diffusion/Hawkes structure and clarifies the role of viscosity solutions as the correct general framework when smoothness is not available.

Empirical and modelling contributions. On the empirical side, the thesis constructs a microstructure-consistent event representation from LOBSTER-style messages and calibrates a Hawkes model for aggressive flow, targeting the clustering and cross-excitation patterns that are absent under independent-Poisson specifications. Reduced-form microstructure components map quote offsets to fill propensity and to a short-horizon adverse-move probability; these are designed to preserve the intensity-based control structure without introducing latent queue position as an additional state. Finally, the thesis implements a numerical solution of the reduced HJB representation with discretisation choices that are consistent with tick size and with empirically calibrated intensity scales, and evaluates the resulting feedback quoting policies in replay/simulation experiments using execution/inventory and P&L diagnostics.

1.5 Thesis outline

The remainder of the thesis is organised to move from motivation and modelling choices, to a well-posed control formulation, and finally to numerical and empirical validation.

Section 2 reviews the literature on limit order book modelling, Hawkes processes in finance, and optimal market making, and frames the central realism–tractability tension that motivates our approach. Section 3 then fixes the probabilistic setting and the point-process notation used throughout.

With this foundation in place, Section 4 introduces the full modelling-and-control layer in a single, coherent step: it specifies the market variables and the market maker's control (quote distances), constructs the Hawkes-driven

order-flow dynamics, links executions and adverse selection to the control through predictable thinning/marking, and defines the resulting state process and terminal objective.
The analytical part starts from the semimartingale viewpoint. Section 5 develops the compensator and generator machinery for the controlled jump structure, which is then used to formalise dynamic programming. Sections 6, 8, and 9 establish the stopping-time DPP, derive and reduce the HJB equation (including the CARA-based factorisation), and prove a verification result linking a classical HJB solution to optimality.
The final chapters report the empirical pipeline (event extraction, calibration, numerical solution, and backtesting), present the results, discuss modelling limitations, and conclude with the development directions collected in Section 13.

2 State of the Art

A large body of work documents empirical regularities of order-driven markets and links them to execution costs, volatility, and liquidity provision (see, e.g., (Abergel et al., 2016; Cartea et al., 2015)).
This section provides a structured overview of the modelling approaches most relevant to this thesis and highlights the gap between realistic LOB simulators and tractable stochastic-control formulations for market making.

2.1 LOB modelling families: a brief map

A useful way to organize the LOB modelling literature is by increasing statistical realism and model complexity.

Zero-intelligence and queueing models. Early "zero-intelligence" approaches model order arrivals as independent Poisson processes and study the induced queueing dynamics (Smith et al., 2003; Cont et al., 2010). They are analytically tractable and yield clean stationary implications, but the conditional-independence assumption conflicts with the strong dependence and clustering observed in real order flow.

State-dependent (queue-reactive) intensity models. A second strand makes intensities depend on the current book state (queue sizes, spread, imbalance). Queue-reactive models can be calibrated from ultra-high-frequency data and often provide effective simulators; a prominent example is the queue-reactive framework of (Huang et al., 2015).

Agent-based and learning-based simulators. Agent-based models simulate heterogeneous trader populations and are useful for "policy experiments", but they can be high-dimensional and difficult to calibrate. More recently, deep generative approaches (GANs, VAEs, neural point-process models) have been proposed to reproduce complex empirical distributions; see the survey (Jain et al., 2024). For the purposes of an optimisation layer, their main drawback is typically reduced transparency and interpretability relative to explicit intensity-based models.

2.2 Point-process and Hawkes models for order flow

Point-process models provide a natural language for event-time dynamics. Within this class, multivariate Hawkes processes are attractive because they preserve an explicit probabilistic structure (predictable intensities, compensators, branching interpretation) while capturing endogenous clustering via self- and cross-excitation (Hawkes and Oakes, 1974; Bacry et al., 2015).
In the LOB context, Hawkes specifications model the arrival times of multiple event types (submissions, cancellations, executions; bid/ask; possibly price-changing vs. non-price-changing). High-dimensional Hawkes models can reproduce key stylised facts such as conditional inter-event-time behaviour and signature plots, provided calibration is handled carefully (Lu and Abergel, 2018). Recent extensions incorporate richer marks (e.g. order sizes), intraday seasonality, and inhibitory effects, improving empirical realism and simulation quality (Jain et al., 2024).

2.3 Optimal market making and stochastic control

Optimal market making has a mature stochastic-control literature, synthesised for example in (Pham, 2009; Cartea et al., 2015). A central tool is dynamic programming and the associated Hamilton–Jacobi–Bellman (HJB) equation, which characterises the value function and yields optimal feedback controls under suitable regularity.

A canonical benchmark is Avellaneda–Stoikov (Avellaneda and Stoikov, 2008), where the midprice is diffusive and executions at the dealer's quotes arrive as Poisson processes with quote-dependent intensities; the dealer chooses bid/ask offsets to balance spread capture against inventory risk. Many extensions preserve the same control-theoretic structure while refining modelling ingredients e.g. (Guéant et al., 2013).

2.4 Gap and thesis positioning

Taken together, the literature reveals a recurring tension: realistic microstructure models (queue-reactive, Hawkes, and data-driven simulators) capture rich dependence in order flow, while classical HJB-based market-making formulations often rely on simplified arrival dynamics for tractability.
This thesis aims to bridge the two by combining a calibrated Hawkes-type order-flow model with an HJB-based market-making problem on an augmented state. After this review, we formalise the mathematical setting and notation, introduce the market model and calibration pipeline, and then formulate and solve the corresponding control problem.

3 Mathematical preliminaries: point processes and Hawkes dynamics

This section introduces the event-time objects used throughout the thesis. We focus on the minimal toolkit needed later for the market model and the control problem: conditional intensities, compensated martingales, Hawkes processes, stability/subcriticality, Markov embeddings, and predictable (Bernoulli) thinning.

3.1 Counting processes, conditional intensities, and compensation

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ satisfy the usual conditions. A (one-dimensional) counting process $N = (N_t)_{t\in[0,T]}$ is an increasing, integer-valued, càdlàg process with $N_0 = 0$.

Predictable intensity. A nonnegative $\mathbb{F}$-predictable process $\lambda = (\lambda_t)_{t\in[0,T]}$ is called an *intensity* of N if

$$\mathbb{E}\left[\int_{(0,T]} H_t \, dN_t\right] = \mathbb{E}\left[\int_0^T H_t \, \lambda_t \, dt\right] \quad \text{for all bounded predictable } H. \tag{1}$$

Informally, this corresponds to the heuristic identity $\mathbb{E}[dN_t \mid \mathcal{F}_{t^-}] = \lambda_{t^-}\, dt$.

Doob–Meyer compensation. Equivalently (Doob–Meyer), N admits the decomposition

$$N_t = M_t + \int_0^t \lambda_s \, ds, \qquad M_t := N_t - \int_0^t \lambda_s \, ds, \tag{2}$$

where M is an $\mathbb{F}$-(local) martingale; see Brémaud (1981) and Daley and Vere-Jones (2003, Ch. 7). For a multivariate counting process $N = (N^1, \dots, N^d)$, the definition applies componentwise.

3.2 Linear Hawkes processes

Hawkes processes are self- and cross-exciting point processes: past events increase the current intensity. They provide a structured way to encode endogenous clustering in order flow (Hawkes, 1971; Bacry et al., 2015).

Definition 3.1 (Univariate linear Hawkes process)**.** A counting process N is a (univariate) linear Hawkes process if it admits an intensity process λ (càdlàg), whose predictable intensity is the left limit λ_{t^-}, of the form

$$\lambda_t = \mu + \int_0^t \phi(t-s) \, dN_s, \qquad t \in [0,T], \tag{3}$$

where $\mu > 0$ and $\phi : \mathbb{R}_+ \to \mathbb{R}_+$ is a kernel. Equivalently, if $(\tau_k)_k$ are the jump times of N, then $\lambda_t = \mu + \sum_{\tau_k < t} \phi(t - \tau_k)$.

Definition 3.2 (Multivariate Hawkes process)**.** A multivariate Hawkes process is a counting process $N = (N^1, \dots, N^d)$ with predictable intensity $\lambda = (\lambda^1, \dots, \lambda^d)$ satisfying

$$\lambda_t = \mu + \int_0^t \Phi(t-s) \, dN_s, \tag{4}$$

where $\mu \in \mathbb{R}_+^d$ and $\Phi : \mathbb{R}_+ \to \mathbb{R}_+^{d\times d}$ is a kernel matrix.

3.3 Stability, integrability, and non-explosion: terminology

For the remainder of the thesis we work with a d-dimensional *linear Hawkes process* $N = (N^1, \dots, N^d)$ with a càdlàg intensity state $\lambda = (\lambda^1, \dots, \lambda^d)$ whose predictable intensity is λ_{t^-}, of the form

$$\lambda_t = \mu + \int_0^t \Phi(t-s)\, dN_s, \qquad \mu \in \mathbb{R}_+^d, \quad \Phi : \mathbb{R}_+ \to \mathbb{R}_+^{d\times d}. \tag{5}$$

We adopt the càdlàg convention because λ is part of the Markov state later used for dynamic programming; whenever a predictable intensity is required (compensators, thinnings), we use the left limit λ_{t^-}.
We assume Φ is componentwise nonnegative and integrable on $\mathbb{R}_+$, so that Tonelli/Fubini applies without ambiguity.

Branching matrix and subcriticality. Define the *branching matrix*

$$K := \int_0^\infty \Phi(u)\, du \in \mathbb{R}_+^{d\times d}, \tag{6}$$

whose entry K_{ij} can be interpreted as the expected number of type-i offspring events directly triggered by a type-j event in the cluster (branching) representation (Hawkes and Oakes, 1974; Daley and Vere-Jones, 2003).

Definition 3.3 (Spectral radius)**.** The spectral radius of a square matrix A is:
$\rho(A) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } A\}$.

Assumption 3.4 (Hawkes stability / subcriticality)**.** We assume $\rho(K) < 1$.

In the univariate exponential-kernel case $\phi(u) = \alpha e^{-\beta u}$, one has $K = \int_0^\infty \phi(u)\, du = \alpha/\beta$, so Assumption 3.4 reduces to $\alpha/\beta < 1$.

Non-explosion and first-moment integrability. On any finite horizon $[0, T]$, we say that N is *non-explosive* if $N_T^i < \infty$ a.s. for every component i. We say that the model is *integrable at first order* if $\mathbb{E}[N_T^i] < \infty$ for all i, equivalently if $\mathbb{E}[\int_0^T \lambda_t^i\, dt] < \infty$ by (2). Assumption 3.4 is a standard sufficient condition guaranteeing both properties (Brémaud and Massoulié, 1996; Bacry et al., 2015).

Stationary mean intensity (heuristic). If a stationary (time-homogeneous) version of the Hawkes process exists with finite first moment, then its mean intensity $\bar\lambda := \mathbb{E}[\lambda_t]$ is constant in t and satisfies the fixed-point relation

$$\bar\lambda \;=\; \mu + K\,\bar\lambda, \qquad \text{so that} \qquad \bar\lambda \;=\; (I-K)^{-1}\mu.$$

This identity should be read as *intuition* rather than a proof of stationarity: it shows that the long-run average activity is the baseline rate μ amplified by endogenous feedback through the branching matrix K. The condition $\rho(K) < 1$ is exactly what makes this amplification finite (equivalently, it ensures that the Neumann series $(I-K)^{-1} = \sum_{n\ge 0} K^n$ converges), hence it is the natural "subcritical" regime.

3.4 Hawkes stability, integrability, and non-explosion

Existence and non-explosion of linear Hawkes processes under Assumption 3.4 are classical; see, e.g., Brémaud and Massoulié (1996) and Daley and Vere-Jones (2003, Ch. 7). We record here a short first-moment bound that will be used later.

Lemma 3.5 (Finite intensity mass and non-explosion on $[0, T]$)**.** *Under Assumption 3.4, for every finite horizon $T < \infty$,*

$$\mathbb{E}\left[\int_0^T \|\lambda_t\|_1\, dt\right] < \infty, \qquad \mathbb{E}[N_T^i] < \infty \quad \forall i, \qquad N_T^i < \infty \quad \text{a.s.} \quad \forall i.$$

For a d-dimensional intensity vector $\lambda_t = (\lambda_t^1, \dots, \lambda_t^d) \in \mathbb{R}_+^d$, we write $\|\lambda_t\|_1 := \sum_{i=1}^d |\lambda_t^i| = \sum_{i=1}^d \lambda_t^i$.
In particular, each component has finitely many jumps on $[0, T]$ (no explosion).

Proof. By the cited results, the Hawkes model is well-defined and non-explosive under Assumption 3.4, and the mean intensity $m(t) := \mathbb{E}[\lambda_t] \in \mathbb{R}_+^d$ is finite and satisfies the Volterra equation

$$m(t) = \mu + \int_0^t \Phi(t-s)\, m(s)\, ds. \tag{7}$$

Integrating (7) over $t \in [0,T]$ and swapping integrals (Tonelli, nonnegativity) gives

$$\int_0^T m(t)\, dt = T\mu + \int_0^T \left(\int_s^T \Phi(t-s)\, dt \right) m(s)\, ds \le T\mu + \int_0^T K\, m(s)\, ds,$$

with K as in (6). Rearranging yields

$$(I-K) \int_0^T m(t)\, dt \le T\mu.$$

Since $\rho(K) < 1$, $(I-K)$ is invertible and $(I-K)^{-1} = \sum_{n=0}^{\infty} K^n$ (Neumann series), hence it is componentwise nonnegative. Multiplying by $(I-K)^{-1}$ gives the bound

$$\int_0^T m(t)\, dt \le (I-K)^{-1} T\mu < \infty.$$

Finally, by (2) and Fubini,

$$\mathbb{E}[N_T^i] = \mathbb{E}\left[\int_0^T \lambda_t^i\, dt \right] = \int_0^T m_i(t)\, dt < \infty,$$

and $\mathbb{E}[N_T^i] < \infty$ implies $N_T^i < \infty$ a.s. (otherwise $\mathbb{E}[N_T^i] = \infty$). □

Lemma 3.6 (Finite fourth moments on finite horizons)**.** *Assume the d-dimensional linear Hawkes process $N = (N^1, \dots, N^d)$ is subcritical, i.e. $\rho(K) < 1$, and has exponential kernel $\Phi(u) = Ae^{-Bu}\mathbf{1}_{\{u \ge 0\}}$ as in Assumption 3.7. Then for every finite horizon $T < \infty$,*

$$\mathbb{E}\left[\sum_{i=1}^d (N_T^i)^4 \right] < \infty, \qquad \text{and} \qquad \mathbb{E}\left[\int_0^T \sum_{i=1}^d (\lambda_t^i)^4\, dt \right] < \infty.$$

Consequently, for any predictable thinning factor $g_t \in [0,1]$, the thinned intensity $\Lambda_t := g_t\, \lambda_t^i$ satisfies $\mathbb{E} \int_0^T \Lambda_t^4\, dt < \infty$.

Proof. Under $\rho(K) < 1$, a linear Hawkes process admits the Poisson cluster (branching) representation (Hawkes and Oakes, 1974). In particular, the number of points in any finite horizon has finite moments of all orders; see Cui et al. (2020) for an explicit moment derivation (and references). Hence $\mathbb{E}[(N_T^{\pm})^4] < \infty$.
For exponential kernels, the intensity can be written componentwise as

$$\lambda_t^i = \mu^i + \sum_{j=1}^d \sum_{\tau_k^j < t} A_{ij} e^{-\beta_i (t - \tau_k^j)} \le \mu^i + \sum_{j=1}^d A_{ij} N_t^j \le \mu^i + \sum_{j=1}^d A_{ij} N_T^j, \qquad t \le T,$$

since $e^{-\beta_i(\cdot)} \le 1$ and $N_t^j \le N_T^j$ on $[0,T]$. Therefore,

$$\sup_{t \le T} (\lambda_t^i)^4 \le C\Big(1 + \sum_{j=1}^d (N_T^j)^4\Big)$$

for a constant C depending only on (μ, A). Taking expectations gives $\mathbb{E}[\sup_{t \le T} (\lambda_t^i)^4] < \infty$, hence

$$\mathbb{E} \int_0^T (\lambda_t^i)^4\, dt \le T\, \mathbb{E}\big[\sup_{t \le T} (\lambda_t^i)^4 \big] < \infty.$$

Finally, if $\Lambda_t = g_t \lambda_t^i$ with $0 \le g_t \le 1$, then $\Lambda_t^4 \le (\lambda_t^i)^4$, so $\mathbb{E} \int_0^T \Lambda_t^4\, dt < \infty$. □

3.5 Markov embedding for exponential Hawkes kernels

Assume now Φ is exponential. This yields a finite-dimensional Markov representation, which is the bridge to dynamic programming.

Assumption 3.7 (Exponential kernel Markovianity)**.** The Hawkes kernel is exponential: $\Phi(u) = Ae^{-Bu}\mathbf{1}_{\{u\geq 0\}}$ with $A \in \mathbb{R}_+^{d\times d}$ and $B = \mathrm{diag}(\beta_1, \ldots, \beta_d)$, $\beta_i > 0$.

Kernel choice and finite-dimensional state representation For a general Hawkes kernel ϕ, the intensity has the convolution form $\lambda_t = \mu + \int_0^t \phi(t-s)\, dN_s$, so λ_t depends on the *entire* past event history. In control terms, this makes the problem path-dependent: the value function would naturally depend on the whole past trajectory (an infinite-dimensional state), and the associated dynamic programming equation would become a functional/infinite-dimensional HJB. Choosing an exponential (or a sum of exponentials) kernel is therefore a tractability commitment: the convolution admits a finite-dimensional Markov embedding, so the augmented state $(S_t, Q_t, X_t, \lambda_t)$, that will be properly defined in section 4, is Markov and admits a controlled generator. This is the key property that allows a stopping-time DPP, a finite-dimensional HJB, and a numerically implementable feedback policy on a grid.

Proposition 3.1 (Finite-dimensional Markov dynamics for λ)**.** *Under Assumption 3.7, the intensity process satisfies the càdlàg SDE*

$$d\lambda_t = B(\mu - \lambda_t)\, dt + A\, dN_t, \tag{8}$$

and $(N_t, \lambda_t)_{t\geq 0}$ is a time-homogeneous Markov process.
If the full (controlled) state is $Z_t = (\text{price}, \text{inventory}, \text{cash}, \lambda_t)$ with coefficients depending only on (t, Z_{t^-}, δ_t), then Z is Markov under any Markov feedback *control of the form $\delta_t = \hat{\delta}(t, Z_{t^-})$ (Borel $\hat{\delta}$). In general, for arbitrary $\mathbb{F}$-predictable open-loop controls, the controlled process need not be Markov.*

Proof. Substituting $\Phi(u) = Ae^{-Bu}$ into (5), write $\lambda_t = \mu + Y_t$ with

$$Y_t := \int_0^t Ae^{-B(t-s)}\, dN_s.$$

Since $t \mapsto e^{-Bt}$ is continuous of finite variation, integration-by-parts yields $dY_t = -BY_t\, dt + A\, dN_t$. As $\lambda_t = \mu + Y_t$, this is equivalent to (8). The Markov property follows because λ evolves autonomously given its current value and the jump increment dN_t, and λ determines the jump law of N via the intensity. □

Sum of exponentials (multi-scale embedding). If the calibrated kernel uses multiple time scales,

$$\Phi(u) = \sum_{m=1}^{M} A^{(m)} e^{-B^{(m)}u}\mathbf{1}_{\{u\geq 0\}},$$

then one introduces auxiliary states $Y_t^{(m)} := \int_0^t A^{(m)} e^{-B^{(m)}(t-s)}\, dN_s$ satisfying $dY_t^{(m)} = -B^{(m)}Y_t^{(m)}\, dt + A^{(m)}\, dN_t$, and $\lambda_t = \mu + \sum_{m=1}^M Y_t^{(m)}$. The augmented state $(N, (Y^{(m)})_{m=1}^M)$ is finite-dimensional Markov, with dimension increased by a factor M.

3.6 Predictable Bernoulli marking and thinning

We will repeatedly model "subflows" of a base order-flow process via *predictable thinning*. This covers, for example, (i) distinguishing price-moving vs. non-price-moving events, and (ii) executions at the market maker's quotes.
Let N be a counting process with predictable intensity λ. Let $J_t \in \{0, 1\}$ be an $\mathbb{F}$-predictable process (a Bernoulli *marking rule*), interpreted as "accept/reject" at the instant of a jump.
Define the thinned counting process

$$N_t^{(1)} := \int_{(0,t]} J_s\, dN_s \qquad (t \in [0, T]). \tag{9}$$

Proposition 3.2 (Predictable thinning)**.** *The process $N^{(1)}$ admits predictable intensity $\lambda_t^{(1)} := J_t\, \lambda_t$. Equivalently, $M_t^{(1)} := N_t^{(1)} - \int_0^t J_s\lambda_s ds$ is an $\mathbb{F}$-(local) martingale.*

Proof. For any bounded predictable process H, by definition of the stochastic integral w.r.t. N,

$$\mathbb{E}\left[\int_{(0,T]} H_s \, dN_s^{(1)}\right] = \mathbb{E}\left[\int_{(0,T]} H_s J_s \, dN_s\right].$$

Using the compensator of N (2), the right-hand side equals $\mathbb{E}\left[\int_0^T H_s J_s \lambda_s \, ds\right]$, so the compensator of $N^{(1)}$ is $\int_0^t J_s \lambda_s ds$, hence the intensity is $J_t \lambda_t$ (Brémaud, 1981; Daley and Vere-Jones, 2003). □

State-dependent Bernoulli marking. A common and empirically useful special case is to choose J so that, at a jump time τ,

$$\mathbb{P}(J_\tau = 1 \mid \mathcal{F}_{\tau^-}) = p(Z_{\tau^-}), \qquad p(\cdot) \in [0,1],$$

for a Markov state variable Z. Then $N^{(1)}$ has intensity $p(Z_{t^-})\lambda_t$, and the complementary subflow $N_t^{(0)} := N_t - N_t^{(1)}$ has intensity $(1 - p(Z_{t^-}))\lambda_t$. This device will be used to create price-moving subflows and execution subflows in the market model.

4 Market model and control problem

We now set up a stylised limit order book environment together with the market maker's optimisation problem. The guiding question is:

How should a small market maker choose bid/ask quote distances so as to trade off spread capture against inventory risk and short-horizon adverse selection, when aggressive order flow is clustered and self-exciting?

From market mechanics to an optimisation problem. The purpose of the market model is not forecasting per se, but to evaluate and optimise quoting decisions. A quoting rule δ_t determines which market orders execute against the market maker; executions generate cashflows, inventory exposure, and hence realised P&L. The modelling objective in this section is therefore to construct a controlled state process that links (i) quote distances, (ii) executions, (iii) midprice changes, and (iv) portfolio outcomes. Once this mapping is explicit, the optimisation problem is to choose δ so as to maximise a risk-adjusted terminal performance criterion.
In words, we choose predictable quote offsets $\delta_t = (\delta_t^a, \delta_t^b)$ to maximise the expected utility of terminal marked-to-market wealth, while penalising non-zero terminal inventory. To make this precise, we proceed by: (1) defining the information structure and the control variable; (2) specifying the market dynamics for order flow, executions, and midprice; (3) deriving the induced portfolio dynamics and collecting the controlled state process; and (4) stating the terminal objective, admissible controls, and value function.

4.1 State, information, and controls

Probability space and horizon. We work on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ satisfying the usual conditions, supporting: (i) a Brownian motion W, and (ii) the counting processes introduced below. All controls are required to be $\mathbb{F}$-predictable (so decisions use information available just before time t).

Reference price (midprice). Let $S = (S_t)_{t\in[0,T]}$ denote the midprice process, i.e. the reference price around which the market maker posts quotes. We treat S as an exogenous market variable (price-taking assumption); its dynamics will be specified in Section 4.4 after we introduce the price-moving order-flow marks.

Control variable (quote distances). At each time t, the market maker chooses the bid/ask *distances* from the midprice, $\delta_t = (\delta_t^a, \delta_t^b) \in \mathbb{R}_+^2$, and posts the corresponding quotes

$$p_t^a = S_t + \delta_t^a, \qquad p_t^b = S_t - \delta_t^b.$$

Here δ_t^a (resp. δ_t^b) measures how far the ask (resp. bid) is placed from the midprice. Larger δ corresponds to less aggressive quoting: fills become less likely, but adverse selection is mitigated.
For numerical tractability (and to exclude pathological behaviour), we restrict the control to a closed admissible set

$$\mathcal{D} \subset \mathbb{R}_+^2, \tag{10}$$

typically $\mathcal{D} = [0, \delta_{\max}]^2$. The precise admissibility requirements (predictability, well-posedness, integrability) will be formalised together with the objective in Section 4.7.

4.2 Market-wide aggressive order flow: a bivariate Hawkes model

Let $N^+ = (N_t^+)_{t\in[0,T]}$ and $N^- = (N_t^-)_{t\in[0,T]}$ be the counting processes of buy and sell *market orders* (i.e. aggressive liquidity-taking orders). They admit predictable intensities $(\lambda_{t^-}^+, \lambda_{t^-}^-)$ (equivalently, we work with a càdlàg intensity state $(\lambda_t^+, \lambda_t^-)$ whose left limit is predictable) in the sense of Section 3 (Definition (1) and Doob–Meyer (2)).

To capture endogenous clustering and cross-excitation between the buy/sell streams, we assume (N^+, N^-) is a bivariate Hawkes process (Definition 3.2) satisfying the stability/subcriticality condition in Assumption 3.4.

We work with exponential kernels so that the intensities admit a finite-dimensional Markov representation as proved in (Proposition 3.1). In the bivariate exponential case:

$$\begin{aligned} d\lambda_t^+ &= -\beta_+(\lambda_t^+ - \mu^+)\,dt + a_{++}\,dN_t^+ + a_{+-}\,dN_t^-, \\ d\lambda_t^- &= -\beta_-(\lambda_t^- - \mu^-)\,dt + a_{-+}\,dN_t^+ + a_{--}\,dN_t^-. \end{aligned} \tag{11}$$

4.3 Executions against the market maker: predictable thinnings

A key modelling stance is *price-taking*: the market maker does not affect whether market orders exist, only whether they execute against her quotes. We represent this via predictable thinning.

Disjoint event types. Each market order either executes against the market maker or not. For clean bookkeeping (and later for the generator), we refine the market-wide flows into four disjoint counting processes:

- H^a: buy market order that executes against the market maker's *ask* quote (the maker sells),
- H^b: sell market order that executes against the market maker's *bid* quote (the maker buys),
- $N^{+,0}$: buy market order that does *not* hit the market maker,
- $N^{-,0}$: sell market order that does *not* hit the market maker.

They satisfy the flow conservation identities

$$N_t^+ = H_t^a + N_t^{+,0}, \qquad N_t^- = H_t^b + N_t^{-,0}. \tag{12}$$

Fill probabilities as functions of quote distance. Let $f_a, f_b : \mathbb{R}_+ \to [0,1]$ be decreasing fill-probability functions: $f_a(\delta)$ is the probability that a buy market order reaches the maker's ask at distance δ from mid, and similarly $f_b(\delta)$ for sells hitting the bid. This is a reduced-form execution proxy: deeper quotes fill less often.

We model executions through predictable Bernoulli thinning (Proposition 3.2): conditionally on $\mathcal{F}_{t^-}$, a buy market order hits the ask with probability $f_a(\delta_t^a)$, and a sell market order hits the bid with probability $f_b(\delta_t^b)$. Hence the controlled intensities are

$$\begin{aligned} \Lambda_t^a &= f_a(\delta_t^a)\lambda_t^+, \quad \Lambda_t^{+,0} = (1 - f_a(\delta_t^a))\lambda_t^+, \\ \Lambda_t^b &= f_b(\delta_t^b)\lambda_t^-, \quad \Lambda_t^{-,0} = (1 - f_b(\delta_t^b))\lambda_t^-. \end{aligned} \tag{13}$$

Remark 4.1 (Reduced-form thinning). The thinning functions f_a, f_b are a reduced-form proxy for execution priority: they compress queue position, competition, and hidden liquidity into an empirically calibrated map from quote distance to fill propensity. This choice is deliberate: it keeps executions as a predictable transformation of the market-wide flow, so the controlled state remains Markov and the HJB stays tractable. The cost is realism: f_a, f_b do not model partial fills, latency, or queue depletion explicitly (see Section 4.10 for scope).

In a price–time priority LOB, a newly posted limit order typically joins the back of the queue. Accordingly, f_a, f_b should be read as *effective, unconditional* execution propensities for a freshly submitted quote, aggregating queue position, cancellations, and competing liquidity at the same price level. To avoid overstating execution efficiency, f_a, f_b are calibrated conservatively to match realised fill frequencies.

4.4 Midprice dynamics via Bernoulli marking

Not every aggressive trade moves the midprice. We model price moves as an additional predictable Bernoulli marking of the market-wide order flow.

State-dependent price-move probabilities. Define the instantaneous intensity imbalance

$$\rho_t = \frac{\lambda_t^+ - \lambda_t^-}{\lambda_t^+ + \lambda_t^-} \in [-1, 1], \tag{14}$$

and set the conditional probabilities of an up/down mid move given a buy/sell market order as

$$p_t^+ := p^+(\lambda_{t^-}^+, \lambda_{t^-}^-) = \bar{p}\,\frac{1}{1 + e^{-k(\rho_{t^-} - \theta)}}, \qquad p_t^- := p^-(\lambda_{t^-}^+, \lambda_{t^-}^-) = \bar{p}\,\frac{1}{1 + e^{+k(\rho_{t^-} + \theta)}}, \tag{15}$$

with $\bar{p} \in (0, 1]$, $k > 0$, $\theta \in \mathbb{R}$.

Why a logistic link? Here "logistic" simply means that we map a real-valued signal (the imbalance ρ_{t^-}) into a probability through a sigmoid function. Concretely, we take

$$\text{sigmoid}(x) := \frac{1}{1 + e^{-x}}, \qquad \text{so that} \qquad p_t^{\pm} = \bar{p}\,\text{sigmoid}(\,\cdot\,) \in [0, \bar{p}].$$

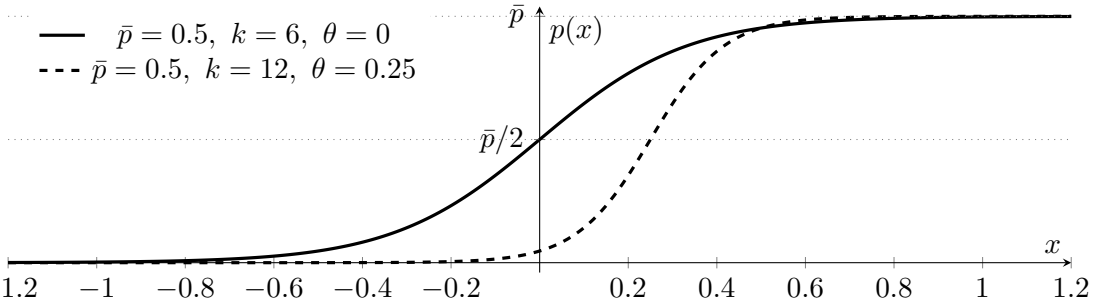


Figure 1: Scaled logistic map $p(x) = \bar{p}/(1 + e^{-k(x-\theta)})$. The parameter $\bar{p}$ sets the upper plateau, k controls steepness (sensitivity), and θ shifts the curve (threshold).

This choice is pragmatic: (i) it guarantees valid probabilities by construction, (ii) it is monotone in imbalance (more buy pressure $\Rightarrow$ larger p_t^+ and smaller p_t^-), with sensitivity controlled by k and a shift/asymmetry controlled by θ, and (iii) it is easy to estimate from event-labeled data. It should be read as a reduced-form adverse-selection channel, not as a structural model of price formation.

Remark 4.2. The use of left limits $\lambda_{t^-}^{\pm}$ ensures predictability in the presence of jumps. We assume $\lambda_t^+ + \lambda_t^- > 0$ (e.g. guaranteed by strictly positive baselines) so that ρ_t is well-defined.

Price-moving subflows. Attach Bernoulli marks $J_t^+, J_t^- \in \{0, 1\}$ at market-order arrivals such that $\mathbb{P}(J_t^+ = 1 \mid \mathcal{F}_{t^-}) = p_t^+$ and $\mathbb{P}(J_t^- = 1 \mid \mathcal{F}_{t^-}) = p_t^-$, and define the price-moving subflows

$$dM_t^+ := J_t^+\, dN_t^+, \qquad dM_t^- := J_t^-\, dN_t^-. \tag{16}$$

By predictable thinning, M^+ and M^- have predictable intensities $p_t^+ \lambda_t^+$ and $p_t^- \lambda_t^-$.

Midprice dynamics. We model the midprice S_t as the sum of a continuous background fluctuation and discrete one-tick moves triggered by *price-moving* market orders. Concretely, recall that N_t^+ (resp. N_t^-) counts buy (resp. sell) market orders with predictable intensities λ_t^+ and λ_t^-. Each such arrival is independently marked by a Bernoulli variable $J_t^{\pm} \in \{0, 1\}$ indicating whether the order moves the midprice, with

$$\mathbb{P}(J_t^+ = 1 \mid \mathcal{F}_{t^-}) = p_t^+, \qquad \mathbb{P}(J_t^- = 1 \mid \mathcal{F}_{t^-}) = p_t^-,$$

where the probabilities $p_t^{\pm}$ are predictable functions of the intensity state (cf. (15)). The resulting price-moving subflows are

$$dM_t^+ := J_t^+ \, dN_t^+, \qquad dM_t^- := J_t^- \, dN_t^-,$$

so M_t^+ counts buy market orders associated with upward midprice moves, and M_t^- counts sell market orders associated with downward midprice moves.
The midprice then evolves as the jump-diffusion

$$dS_t = \sigma \, dW_t + \kappa\big(dM_t^+ - dM_t^-\big), \tag{17}$$

where $\sigma > 0$ controls the diffusive volatility and $\kappa > 0$ is a fixed tick size. Thus, whenever $dM_t^+ = 1$ the midprice jumps up by one tick κ; whenever $dM_t^- = 1$ it jumps down by κ; otherwise it evolves continuously through σdW_t. By construction, the jump component is driven by *market-wide* order flow through (M^+, M^-), while the market maker's control affects only her executions (via H^a, H^b) and does not enter (17).

Price-taker consistency. The control δ affects only executions against the market maker (via f_a, f_b), not the market-wide arrival of aggressive orders (N^+, N^-) nor the market-wide price-moving mechanism (J^+, J^-). Accordingly, f_a, f_b do not enter (17). This is the formal expression of "small market maker / price taker".

Remark 4.3 (Price enters only through relative quantities.). Execution and adverse selection depend on price only through relative objects: quote-to-mid distances δ, and probabilities of midprice *increments* (via $p_t^{\pm}$), not through the absolute price level S_t.

4.5 Inventory, cash, and marked-to-market wealth

A quoting policy $\delta_t = (\delta_t^a, \delta_t^b)$ becomes economically meaningful only once we specify how it translates into positions and profit-and-loss. Since executions against the market maker occur through the maker-specific fill processes H^a and H^b, we track two bookkeeping variables: the inventory Q_t (shares held) and the cash account X_t (cumulative cash gained/spent). These processes will enter the terminal performance criterion and, together with $(S_t, \lambda_t^+, \lambda_t^-)$, form the state variables of the optimisation problem.
Fix a trade size $q_0 > 0$. The inventory and cash evolve only when the market maker is executed: a fill at the bid corresponds to buying q_0 shares, while a fill at the ask corresponds to selling q_0 shares. Accordingly,

$$\begin{aligned} dQ_t &= q_0\big(dH_t^b - dH_t^a\big), \\ dX_t &= (S_{t^-} + \delta_t^a)\, q_0 \, dH_t^a \;-\; (S_{t^-} - \delta_t^b)\, q_0 \, dH_t^b. \end{aligned} \tag{18}$$

The first line states that inventory increases when a sell market order hits the maker's bid ($dH_t^b = 1$) and decreases when a buy market order hits the maker's ask ($dH_t^a = 1$). The second line records the corresponding cashflows: selling at the ask generates a receipt at price $S_{t^-} + \delta_t^a$, while buying at the bid generates a payment at price $S_{t^-} - \delta_t^b$.
To evaluate performance we use marked-to-market wealth, which combines realised cash with the current value of inventory at the midprice:

$$W_t := X_t + Q_t S_t.$$

This decomposition makes explicit the market-making trade-off: spread capture is accumulated through X_t, whereas exposure to midprice moves is carried by $Q_t S_t$.

4.6 Controlled state dynamics and Markov property

Collect the relevant state variables in

$$Z_t = (S_t, Q_t, X_t, \lambda_t^+, \lambda_t^-).$$

Combining (17), (18), and (11), we obtain:

$$\begin{cases} dS_t = \sigma \, dW_t + \kappa\big(dM_t^+ - dM_t^-\big), \\ dQ_t = q_0\big(dH_t^b - dH_t^a\big), \\ dX_t = (S_{t^-} + \delta_t^a) q_0 \, dH_t^a \;-\; (S_{t^-} - \delta_t^b) q_0 \, dH_t^b, \\ d\lambda_t^+ = -\beta_+(\lambda_t^+ - \mu^+)\, dt + a_{++}\, dN_t^+ + a_{+-}\, dN_t^-, \\ d\lambda_t^- = -\beta_-(\lambda_t^- - \mu^-)\, dt + a_{-+}\, dN_t^+ + a_{--}\, dN_t^-, \end{cases} \tag{19}$$

where $dN_t^{\pm}$ are the aggregates in (12) and $dM_t^{\pm}$ are defined in (16). By Proposition 3.1, including (λ^+, λ^-) yields a finite-dimensional Markov representation, so Z is a controlled Markov process under any admissible control.

Remark 4.4 (Event bookkeeping: "who drives what"). The model involves three layers of point processes. Keeping their roles distinct avoids confusion later when we build the generator.

- **Market-wide flow.** The aggressive order flow (N^+, N^-) drives the Hawkes intensity state (λ^+, λ^-) via (11). The market maker is price-taking and does not affect (N^+, N^-) nor (λ^+, λ^-).
- **Price-moving subflow.** Bernoulli marks $J^{\pm}$ convert market-wide arrivals into the price-moving flows $M^{\pm}$ through $dM_t^{\pm} = J_t^{\pm}\, dN_t^{\pm}$ in (16). The midprice dynamics (17) depend only on (M^+, M^-) (and the diffusion term), hence only on market-wide flow and marks.
- **Maker executions.** The control $\delta_t = (\delta_t^a, \delta_t^b)$ acts only through the thinning functions f_a, f_b, which split market-wide flow into maker fills (H^a, H^b) and missed flow $(N^{+,0}, N^{-,0})$ via (12)–(13). Inventory and cash update only on maker fills, as in (18).

$$\delta \xrightarrow[f_a, f_b]{} (H^a, H^b), \qquad (N^+, N^-) \xrightarrow{\text{Hawkes}} (\lambda^+, \lambda^-), \qquad (N^+, N^-) \xrightarrow[J^{\pm}]{} (M^+, M^-) \to S, \qquad (H^a, H^b) \to (Q, X).$$

4.7 Objective functional

Having specified how quotes δ affect executions and how executions propagate to the state $(S, Q, X, \lambda^+, \lambda^-)$, we now define the market maker's performance criterion. Fix an initial time $t \in [0, T]$ and an initial state $Z_t = z$. The control goal is to choose quote offsets δ so as to maximise a risk-adjusted terminal performance that rewards P&L while discouraging residual inventory.

Terminal criterion (liquidation-style). We measure terminal performance through a marked-to-market wealth with an explicit inventory penalty:

$$\widetilde{W}_T := X_T + Q_T S_T - \frac{\eta}{2} Q_T^2, \qquad \eta > 0, \tag{20}$$

where the quadratic term is a reduced-form proxy for inventory risk and/or liquidation costs at the horizon.

Utility and objective functional. We use exponential (CARA) utility with risk aversion $\gamma > 0$,

$$U(w) := -e^{-\gamma w}.$$

For a given control δ, define the corresponding objective functional

$$J(t, z; \delta) := \mathbb{E}\Big[U(\widetilde{W}_T) \,\Big|\, Z_t = z\Big]. \tag{21}$$

CARA utility is standard in market making because it yields an exponential-affine structure that enables the dimension reduction we will discuss later in 9; in particular, the optimal offsets depend on $(t, q, \lambda^+, \lambda^-)$ but not on the absolute cash level. The quadratic penalty is a convex proxy for liquidation risk: it discourages large terminal inventory without adding an explicit end-of-horizon liquidation control, keeping the terminal condition simple.

Definition 4.5 (Admissible controls and optimisation problem)**.** Fix $t \in [0, T]$. A control δ is *admissible* on $[t, T]$ if:

(i) δ is $\mathbb{F}$-predictable and satisfies $\delta_u \in \mathcal{D}$ for all $u \in [t, T]$;
(ii) the controlled state dynamics (19) admit a (strong) solution on $[t, T]$;
(iii) $J(t, z; \delta)$ in (21) is finite for the given initial condition (t, z).

We denote by $\mathcal{A}_t$ the set of admissible controls on $[t, T]$.
The market maker's optimisation problem is:

$$\text{maximise } J(t, z; \delta) \text{ over } \delta \in \mathcal{A}_t.$$

4.8 Value function

Definition 4.6 (Value function)**.** The value function is

$$V(t,z) := \sup_{\delta \in \mathcal{A}_t} J(t,z;\delta), \qquad (t,z) \in [0,T] \times \mathcal{Z}, \tag{22}$$

where $\mathcal{Z}$ denotes the state space of Z.

In the remainder we restrict the optimisation to Markov feedback controls as discussed in Remark 4.7. For notational simplicity we keep the symbol V for this feedback-restricted value function.

Remark 4.7 (Admissible controls and Markov feedback restriction). Although one may define the value process as an essential supremum over all $\mathbb{F}$-predictable controls, in this thesis we restrict attention to *Markov feedback* policies of the form $\delta_t = \hat{\delta}(t, Z_{t^-})$ with $\hat{\delta}$ Borel and $\delta_t \in \mathcal{D}$. Under the Markov property of the state (triangular construction), this restriction yields a deterministic value function $v(t,z) = \sup_{\hat{\delta}} J(t,z;\hat{\delta})$ and justifies conditioning on $Z_t = z$ in the DPP/HJB derivation.

4.9 Well-definedness of the CARA objective

The CARA objective involves expectations of the form $\mathbb{E}[e^{-\gamma \widetilde{W}_T}]$. Since $\widetilde{W}_T$ contains the marked-to-market term $Q_T S_T$ and cash X_T, we must ensure sufficient integrability so that $J(t,z;\delta) = \mathbb{E}[U(\widetilde{W}_T) \mid Z_t = z]$ is finite for admissible controls.

Assumption 4.8 (Integrability for CARA utility)**.** Fix $T < \infty$ and $\gamma > 0$. There exists an inventory bound $Q_{\max} < \infty$ such that $|Q_t| \le Q_{\max}$ a.s. for all $t \in [0,T]$ (e.g. enforced by admissible controls or hard risk limits). Moreover, there exists $r > 1$ such that

$$\mathbb{E}\Big[e^{r\gamma|X_T|}\Big] < \infty, \qquad \mathbb{E}\Big[e^{\frac{r}{r-1}\gamma Q_{\max}|S_T|}\Big] < \infty.$$

Lemma 4.9 (Finiteness of the CARA criterion)**.** *Under Assumption 4.8, for any admissible control δ and any initial condition (t,z),*

$$J(t,z;\delta) > -\infty,$$

hence the value function $V(t,z) = \sup_{\delta \in \mathcal{A}_t} J(t,z;\delta)$ is well-defined and satisfies $V(t,z) \in (-\infty, 0]$.

Proof. Recall $\widetilde{W}_T = X_T + Q_T S_T - \frac{\eta}{2} Q_T^2$ and $U(w) = -e^{-\gamma w}$. Since $|Q_T| \le Q_{\max}$,

$$e^{-\gamma \widetilde{W}_T} = \exp\Big(-\gamma X_T - \gamma Q_T S_T + \frac{\gamma\eta}{2} Q_T^2\Big) \le \exp\Big(\frac{\gamma\eta}{2} Q_{\max}^2\Big)\, e^{\gamma|X_T|}\, e^{\gamma Q_{\max}|S_T|}.$$

Let $r > 1$ and $r' = \frac{r}{r-1}$. By Hölder's inequality,

$$\mathbb{E}\Big[e^{\gamma|X_T|} e^{\gamma Q_{\max}|S_T|}\Big] \le \Big(\mathbb{E}[e^{r\gamma|X_T|}]\Big)^{1/r} \Big(\mathbb{E}[e^{r'\gamma Q_{\max}|S_T|}]\Big)^{1/r'} < \infty,$$

so $\mathbb{E}[e^{-\gamma \widetilde{W}_T}] < \infty$ and therefore $J(t,z;\delta) > -\infty$. Since $U \le 0$, also $V(t,z) \le 0$. □

Remark 4.10 (Why Assumption 4.8 is mild here). The bounded-inventory condition is standard in practical market making and is already aligned with the admissibility constraints used in the numerics. Moreover, in our price specification S_T is a Brownian term plus bounded tick jumps, so S_T admits exponential moments on finite horizons.
The non-trivial requirement is the exponential integrability of X_T, since cash increments scale linearly with S_{t^-} at execution times: $\Delta X = \pm(S_{t^-} \pm \delta)\, q_0$. In our framework, X_T is an integral with respect to the controlled execution processes H^a, H^b; thus its tail behaviour is driven jointly by the price process and by the (thinned) arrival rates. For finite horizons and stable Hawkes intensities (Assumption 3.4), the moment conditions required by CARA can be met under moderate parameter choices (e.g. by choosing γ consistently with the horizon and risk limits) and are automatically satisfied in the bounded/truncated implementations used in the empirical section.

4.10 Modelling scope: midprice realism vs tractability

Real midprice paths are shaped by more than trade arrivals (queue depletion, cancellations, limit submissions). The midprice model (17) is intentionally parsimonious: it captures a qualitative coupling between clustered aggressive flow, imbalance, and price moves, while keeping a tractable Markov generator for the HJB.

Implications and limitations.

- Only a subset of aggressive trades moves the price (via $M^{\pm}$); cancellations/limit submissions are not modelled explicitly.
- Jump sizes are fixed to one tick κ, whereas empirical moves may span multiple ticks.
- The model targets mechanisms rather than high-fidelity midprice forecasting.

Evaluation strategy. Synthetic simulations are interpreted as stress-test scenarios driven by realistic order-flow clustering. For high-fidelity P&L attribution on real days, we use midprice replay and feed the backtester with empirical midprice (and/or empirical event counts), focusing evaluation on the controlled execution mechanism and Hawkes-driven state dependence.

Possible remedies. Improving price fidelity typically requires (i) explicit best-quote move event types, (ii) richer event dictionaries (8D/10D), or (iii) variable jump sizes / queue dynamics, each of which raises modelling and control complexity.

5 Semimartingale calculus and controlled generators

This section collects the mathematical tools used later for the DPP/HJB derivation. We do *not* derive the HJB here. Instead, we assemble the machinery that makes that step legitimate in a setting with Hawkes-driven jump channels: compensators, semimartingale representations, Itô/Dynkin identities, and the controlled infinitesimal generator.

Why the dynamic-programming toolbox is nonstandard

Many textbook dynamic-programming derivations start from controlled diffusions. Here the state is a jump–diffusion whose jump channels are driven by counting processes with *stochastic predictable intensities* (Hawkes processes and predictable thinnings). Since Hawkes processes are not Lévy processes (no independent increments), the right toolbox is Doob–Meyer/compensator calculus for point processes rather than a Lévy–Itô decomposition.
The key input from the Hawkes preliminaries (Section 3) is that exponential kernels yield a finite-dimensional Markov embedding (Proposition 3.1). This turns the augmented Hawkes state into a tractable Markov process and allows us to write down a controlled generator explicitly.

5.1 A generic controlled jump–diffusion driven by counting processes

Let $Z = (Z_t)_{t\in[0,T]}$ be an $\mathbb{F}$-adapted càdlàg process taking values in a state space $\mathcal{Z} \subset \mathbb{R}^m$. We consider dynamics driven by:
(i) a Brownian motion W, and
(ii) a finite family of counting processes $(N^i)_{i=1,\dots,D}$. A convenient schematic form is

$$dZ_t = b(t, Z_{t^-}, \delta_t)\, dt + \Sigma(t, Z_{t^-}, \delta_t)\, dW_t + \sum_{i=1}^{D} \Gamma_i(t, Z_{t^-}, \delta_t)\, dN_t^i, \tag{23}$$

where δ is a predictable control taking values in a closed set $\mathcal{D}$, and each N^i admits a predictable intensity $\lambda^i(t, Z_{t^-}, \delta_t)$. The dependence $\Sigma(t, Z, \delta)$ allows for state-dependent volatility (stochastic volatility/regime dependence). In our market-making application the diffusion coefficient is in fact constant and acts only on the midprice.

Using Doob–Meyer, $dN_t^i = \lambda_t^i\,dt + dM_t^i$, we obtain the compensated form

$$dZ_t = \Big(b(t, Z_{t^-}, \delta_t) + \sum_{i=1}^{D} \Gamma_i(t, Z_{t^-}, \delta_t)\lambda_t^i\Big)\,dt + \Sigma(t, Z_{t^-}, \delta_t)\,dW_t + \sum_{i=1}^{D} \Gamma_i(t, Z_{t^-}, \delta_t)\,dM_t^i, \tag{24}$$

where $M_t^i := N_t^i - \int_0^t \lambda_s^i\,ds$ are (local) martingales.

5.2 Poisson embedding and the merged Poisson random measure formulation

This subsection makes explicit the bridge between the compensated representation (24) and a Poisson-random-measure (PRM) formulation suitable for strong existence arguments.

Embedding principle (Ogata thinning viewpoint). On a possibly enlarged probability space, one may represent each counting process N^i with predictable intensity λ^i as a thinning of a unit-rate Poisson random measure; see Ogata (1981) and Brémaud (1981). Let $\Pi^i(ds, dz)$ be independent PRMs on $\mathbb{R}_+ \times \mathbb{R}_+$ with intensity measure $ds\,dz$, and define

$$N_t^i := \int_0^t \int_0^\infty \mathbf{1}_{\{z \le \lambda^i(s, Z_{s^-}, \delta_s)\}}\,\Pi^i(ds, dz), \qquad i = 1, \dots, D. \tag{25}$$

Then N^i has $\mathbb{F}$-predictable intensity $\lambda^i(\cdot)$, and its compensated martingale admits the PRM form

$$M_t^i = \int_0^t \int_0^\infty \mathbf{1}_{\{z \le \lambda^i(s, Z_{s^-}, \delta_s)\}}\,\tilde{\Pi}^i(ds, dz), \qquad \tilde{\Pi}^i(ds, dz) := \Pi^i(ds, dz) - ds\,dz. \tag{26}$$

From counting processes to PRMs: the key integral identity. By the thinning definition (25), for any predictable scalar (or vector-valued) integrand P_s for which the integrals are well-defined (after localisation if needed), we have pathwise

$$\int_0^t P_s\,dN_s^i = \int_0^t \int_0^\infty P_s\,\mathbf{1}_{\{z \le \lambda^i(s, Z_{s^-}, \delta_s)\}}\,\Pi^i(ds, dz), \qquad i = 1, \dots, D. \tag{27}$$

Indeed, both sides are sums over the same atoms (s, z) of Π^i that satisfy $z \le \lambda^i(s, Z_{s^-}, \delta_s)$.

Merged PRM formulation. Start from the integrated form of (23):

$$Z_t = Z_0 + \int_0^t b(s, Z_{s^-}, \delta_s)\,ds + \int_0^t \Sigma(s, Z_{s^-}, \delta_s)\,dW_s + \sum_{i=1}^{D} \int_0^t \Gamma_i(s, Z_{s^-}, \delta_s)\,dN_s^i.$$

Apply (27) with $P_s = \Gamma_i(s, Z_{s^-}, \delta_s)$ to each jump channel. This yields

$$\begin{aligned} Z_t = Z_0 &+ \int_0^t b(s, Z_{s^-}, \delta_s)\,ds + \int_0^t \Sigma(s, Z_{s^-}, \delta_s)\,dW_s \\ &+ \sum_{i=1}^{D} \int_0^t \int_0^\infty \Gamma_i(s, Z_{s^-}, \delta_s)\,\mathbf{1}_{\{z \le \lambda^i(s, Z_{s^-}, \delta_s)\}}\,\Pi^i(ds, dz), \end{aligned} \tag{28}$$

which is the PRM-driven (merged) equation used later for strong existence arguments.

Conversely: recovering the compensated (Doob–Meyer) form. Assume Z satisfies the merged equation (28) and define N^i by (25). Decompose each PRM as $\Pi^i = \tilde{\Pi}^i + ds\,dz$, with $\tilde{\Pi}^i := \Pi^i - ds\,dz$. Then the jump term splits into

$$\begin{aligned} \int_0^t \int_0^\infty \Gamma_i(s, Z_{s^-}, \delta_s)\mathbf{1}_{\{z \le \lambda_s^i\}}\Pi^i(ds, dz) &= \int_0^t \int_0^\infty \Gamma_i(s, Z_{s^-}, \delta_s)\mathbf{1}_{\{z \le \lambda_s^i\}}\tilde{\Pi}^i(ds, dz) \\ &\quad + \int_0^t \int_0^\infty \Gamma_i(s, Z_{s^-}, \delta_s)\mathbf{1}_{\{z \le \lambda_s^i\}}\,dz\,ds \\ &= \int_0^t \int_0^\infty \Gamma_i(s, Z_{s^-}, \delta_s)\mathbf{1}_{\{z \le \lambda_s^i\}}\tilde{\Pi}^i(ds, dz) + \int_0^t \Gamma_i(s, Z_{s^-}, \delta_s)\lambda_s^i\,ds, \end{aligned}$$

where we wrote $\lambda_s^i := \lambda^i(s, Z_{s^-}, \delta_s)$ and used $\int_0^\infty \mathbf{1}_{\{z \le \lambda_s^i\}} dz = \lambda_s^i$. Define the compensated counting-process martingale

$$M_t^i := \int_0^t \int_0^\infty \mathbf{1}_{\{z \le \lambda^i(s, Z_{s^-}, \delta_s)\}} \tilde{\Pi}^i(ds, dz) = N_t^i - \int_0^t \lambda_s^i \, ds,$$

which matches (26). Plugging this decomposition into (28) gives exactly the compensated form (24).

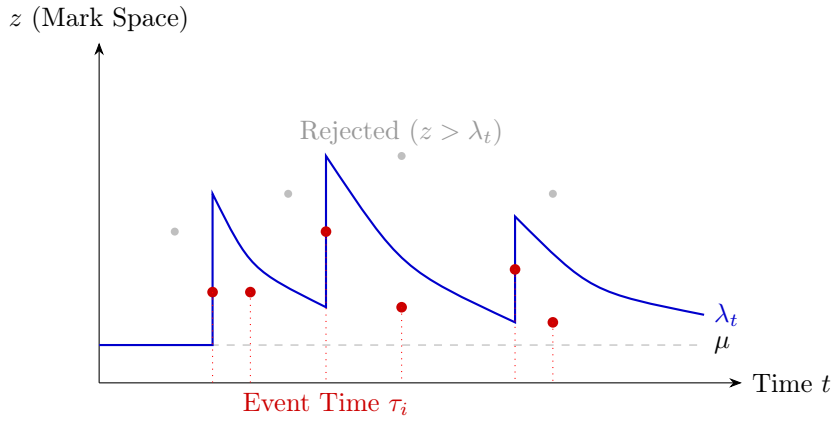


Figure 2: Visual representation of the Ogata Thinning Algorithm. Points in the space $[0, T] \times [0, \bar{\lambda}]$ are generated by a homogeneous Poisson process. Points below the intensity curve λ_t (red) are accepted as real events; points above (gray) are rejected.

5.3 Well-posedness and integrability via a triangular construction

Dynkin's formula and the verification argument require two technical inputs for the controlled state process:

(i) *Strong well-posedness:* for every admissible control, the state admits a pathwise unique càdlàg strong solution on $[0, T]$.

(ii) *Martingale integrability:* in Itô's formula, the Brownian and compensated jump terms are true martingales (in particular in L^2), so their conditional expectations vanish.

This subsection proves both inputs without assuming bounded jump sizes. The key is to exploit the *triangular* structure of the model. Later, when verifying CARA optimality, we will impose bounded-inventory/offset conditions to obtain exponential integrability; this is *not* needed for well-posedness of the state dynamics proved here.

Triangular structure (what is fixed first, what is solved second). The market-making model is built in layers:

- **Layer 1 (market-wide flow):** construct the Hawkes process (N^+, N^-) and its Markovian intensity state (λ^+, λ^-).
- **Layer 2 (derived event channels):** conditional on Layer 1, define maker fills and price-move events as predictable thinnings/markings of (N^+, N^-) using independent auxiliary randomness.
- **Layer 3 (continuous state variables):** define (S, Q, X) as stochastic integrals driven by W and by the counting processes from Layer 2.

The control δ affects only the thinning/marking probabilities (fill propensity and adverse-selection marks). It does not change the existence of market orders (N^+, N^-) nor the Hawkes dynamics of (λ^+, λ^-).

The technical consequence is crucial: once Layer 1 is constructed, the intensities of the derived channels in Layer 2 become *given predictable inputs*. We can therefore treat them as exogenous when proving well-posedness and integrability for the lower layers. The next subsubsection formalizes a generic tool in exactly that setting.

A generic local tool: PRM-driven jump diffusions with predictable intensity inputs

The abstract model. Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space supporting a Brownian motion W and independent Poisson random measures (PRMs) Π^i on $\mathbb{R}_+ \times \mathbb{R}_+$ with intensity $ds\,dz$, $i = 1, \dots, D$. Let δ be an admissible control (predictable, valued in a bounded set $\mathcal{D}$), and let $(\Lambda^i)_{i=1}^D$ be nonnegative $\mathbb{F}$-predictable

processes. We consider the SDE in $\mathbb{R}^m$

$$Z_t = Z_0 + \int_0^t b(s, Z_{s^-}, \delta_s)\, ds + \int_0^t \Sigma(s, Z_{s^-}, \delta_s)\, dW_s + \sum_{i=1}^{D} \int_0^t \int_0^\infty \Gamma_i(s, Z_{s^-}, \delta_s)\, \mathbf{1}_{\{z \le \Lambda_s^i\}}\, \Pi^i(ds, dz). \tag{29}$$

The indicator $\mathbf{1}_{\{z \le \Lambda_s^i\}}$ is the standard thinning device: conditional on $\mathbb{F}$, the resulting jump channel has instantaneous intensity Λ_s^i.

Notation. The state variable is $z \in \mathbb{R}^m$. We use the Euclidean norm on vectors:

$$\|z\| := \Big(\sum_{k=1}^{m} z_k^2\Big)^{1/2}.$$

The symbol $\vee$ means maximum, so

$$\|z\| \vee \|z'\| \le R \quad \Longleftrightarrow \quad \|z\| \le R \text{ and } \|z'\| \le R.$$

The diffusion coefficient is a matrix $\Sigma(t, z, \delta) \in \mathbb{R}^{m \times d}$ and we use the Frobenius norm:

$$\|A\|_F := \Big(\sum_{i=1}^{m}\sum_{j=1}^{d} A_{ij}^2\Big)^{1/2} = \sqrt{Tr(AA^\top)}.$$

The drift is $b(t, z, \delta) \in \mathbb{R}^m$ and each jump size is $\Gamma_i(t, z, \delta) \in \mathbb{R}^m$, for $i = 1, \dots, D$. The control takes values in a set $\mathcal{D}$.

Assumption 5.1 (Local regularity of the coefficients)**. (Local Lipschitz on bounded sets).** For each radius $R > 0$ there exists a constant $L_R < \infty$ such that for all $t \in [0, T]$, all $\delta \in \mathcal{D}$, and all $z, z' \in \mathbb{R}^m$ with $\|z\| \vee \|z'\| \le R$,

$$\|b(t, z, \delta) - b(t, z', \delta)\| + \|\Sigma(t, z, \delta) - \Sigma(t, z', \delta)\|_F + \sum_{i=1}^{D} \|\Gamma_i(t, z, \delta) - \Gamma_i(t, z', \delta)\| \le L_R\, \|z - z'\|.$$

(Local boundedness on bounded sets). For each radius $R > 0$ there exists a constant $C_R < \infty$ such that for all $t \in [0, T]$, all $\delta \in \mathcal{D}$, and all $z \in \mathbb{R}^m$ with $\|z\| \le R$,

$$\|b(t, z, \delta)\|^2 + \|\Sigma(t, z, \delta)\|_F^2 + \sum_{i=1}^{D} \|\Gamma_i(t, z, \delta)\|^2 \le C_R.$$

Notation. We write $\mathbb{S}^2([0, T])$ for the set of càdlàg adapted processes Y such that $\mathbb{E}[\sup_{t \le T} \|Y_t\|^2] < \infty$.
Two different mechanisms force two different integrability requirements.

- The **jump quadratic variation** of the compensated PRM integrals produces terms of the form $\int_0^t \|\Gamma_i(\cdot)\|^2 \Lambda_s^i\, ds$ (Poisson isometry / BDG). This forces control of $\int \Lambda$.
- The **compensated drift** contains $\sum_i \Gamma_i(\cdot)\Lambda_s^i$. In $\mathbb{S}^2$ estimates one squares this drift term and generates $(\Lambda_s^i)^2$. This forces control of $\int \Lambda^2$.

This is why we localize by the *intensity mass* $\int (\Lambda + \Lambda^2)$.

Intensity-mass localization. Define for $m \in \mathbb{N}$ the stopping time

$$\tau_m := \inf\Big\{t \in [0, T] : \int_0^t \sum_{i=1}^{D} \big(\Lambda_s^i + (\Lambda_s^i)^2\big)\, ds \ge m\Big\} \wedge T. \tag{30}$$

Note that for each i, $\int_0^{\tau_m} \Lambda_s^i\, ds \le m$ and $\int_0^{\tau_m} (\Lambda_s^i)^2\, ds \le m$ since each integrand is dominated by the sum inside (30).

Lemma 5.2 (Why $\tau_m \uparrow T$). *If*

$$A_T := \int_0^T \sum_{i=1}^{D} \left(\Lambda_s^i + (\Lambda_s^i)^2\right) ds < \infty \quad a.s.,$$

then $\tau_m \uparrow T$ a.s. as $m \to \infty$. Indeed, for $m > A_T(\omega)$ one has $\tau_m(\omega) = T$.

An exit time such as $\inf\{t : \|Z_t\| \geq R\}$ is only meaningful after Z is constructed. To keep the argument non-circular, we first solve a truncated equation with globally Lipschitz coefficients, then define the exit time on that constructed solution.

Define the radial projection $\pi_R : \mathbb{R}^m \to \mathbb{R}^m$ by

$$\pi_R(z) := \begin{cases} z, & \text{if } \|z\| \leq R, \\ \dfrac{R}{\|z\|}\, z, & \text{if } \|z\| > R, \end{cases}$$

and truncated coefficients

$$b^R(t,z,\delta) := b(t,\pi_R(z),\delta), \quad \Sigma^R(t,z,\delta) := \Sigma(t,\pi_R(z),\delta), \quad \Gamma_i^R(t,z,\delta) := \Gamma_i(t,\pi_R(z),\delta).$$

Because π_R is 1-Lipschitz, Assumption 5.1 implies that $(b^R, \Sigma^R, \Gamma_i^R)$ are *globally* Lipschitz in z (constant L_R) and globally bounded (constant depending on R).

Proposition 5.1 (Local strong well-posedness and $\mathbb{S}^2$ under intensity-mass localization). *Fix $T < \infty$ and a predictable control δ. Assume Assumption 5.1 and $\mathbb{E}\|Z_0\|^2 < \infty$.*

Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space supporting a d-dimensional Brownian motion W and independent Poisson random measures $(\Pi^i)_{i=1,\dots,D}$ on $\mathbb{R}_+ \times \mathbb{R}_+$ with intensity measure $ds\,dz$. Let $(\Lambda^i)_{i=1,\dots,D}$ be nonnegative $\mathbb{F}$-predictable processes.

Consider (29) *and define τ_m by* (30).

Then, for every $m \in \mathbb{N}$ and $R > 0$:

(i) (Existence and uniqueness for the truncated stopped equation). *There exists a unique càdlàg strong solution $Z^{m,R}$ on $[0,T]$ to the truncated stopped dynamics*

$$\begin{aligned} Z_t^{m,R} = Z_0 &+ \int_0^{t\wedge\tau_m} b^R(s, Z_{s^-}^{m,R}, \delta_s)\, ds + \int_0^{t\wedge\tau_m} \Sigma^R(s, Z_{s^-}^{m,R}, \delta_s)\, dW_s \\ &+ \sum_{i=1}^{D} \int_0^{t\wedge\tau_m} \int_0^{\infty} \Gamma_i^R(s, Z_{s^-}^{m,R}, \delta_s)\, \mathbf{1}_{\{z \leq \Lambda_s^i\}}\, \Pi^i(ds, dz). \end{aligned} \tag{31}$$

Define the exit time (well-defined since $Z^{m,R}$ exists)

$$\sigma_R := \inf\{t \in [0,T] : \|Z_t^{m,R}\| \geq R\} \wedge T, \qquad \theta_{m,R} := \tau_m \wedge \sigma_R.$$

(ii) (Reduction to the original equation). *The stopped process $Z_t := Z_{t\wedge\theta_{m,R}}^{m,R}$ solves the original equation* (29) *on $[0, \theta_{m,R}]$.*

(iii) (Pathwise uniqueness up to $\theta_{m,R}$). *If $\bar{Z}$ is another strong solution of* (29) *on $[0,\theta_{m,R}]$ driven by the same noises and with the same Z_0, then $Z_t = \bar{Z}_t$ for all $t \leq \theta_{m,R}$ a.s.*

(iv) ($\mathbb{S}^2$ bound up to $\theta_{m,R}$). *One has $Z_{\cdot\wedge\theta_{m,R}} \in \mathbb{S}^2([0,T])$, i.e.*

$$\mathbb{E}\Big[\sup_{t\leq T} \|Z_{t\wedge\theta_{m,R}}\|^2\Big] < \infty.$$

Proof. Fix $m \in \mathbb{N}$ and $R > 0$ and set $\theta := \theta_{m,R}$.

Step 1: why the truncated stopped SDE admits a unique strong solution. Fix $m \in \mathbb{N}$ and $R > 0$. Consider the stopped SDE with truncated coefficients

$$Z_t^{m,R} = Z_0 + \int_0^{t\wedge\tau_m} b^R(s, Z_{s^-}^{m,R}, \delta_s)\, ds + \int_0^{t\wedge\tau_m} \Sigma^R(s, Z_{s^-}^{m,R}, \delta_s)\, dW_s + \sum_{i=1}^{D} \int_0^{t\wedge\tau_m} \int_0^{\infty} \Gamma_i^R(s, Z_{s^-}^{m,R}, \delta_s) \mathbf{1}_{\{z\leq\Lambda_s^i\}} \Pi^i(ds,dz).$$

(a) Global Lipschitz and bounded coefficients. By construction $b^R(t,z,\delta) = b(t,\pi_R(z),\delta)$ (and similarly for Σ^R, Γ_i^R), and π_R is 1-Lipschitz. Since Assumption 5.1 gives a Lipschitz constant L_R for (b,Σ,Γ_i) on the ball $\{\|z\| \le R\}$, we obtain for all $z, z' \in \mathbb{R}^m$,

$$\|b^R(t,z,\delta) - b^R(t,z',\delta)\| + \|\Sigma^R(t,z,\delta) - \Sigma^R(t,z',\delta)\|_F + \sum_{i=1}^{D} \|\Gamma_i^R(t,z,\delta) - \Gamma_i^R(t,z',\delta)\| \le L_R \|z - z'\|.$$

Moreover, since $\pi_R(z) \in \{\|u\| \le R\}$ for all z, the local boundedness part of Assumption 5.1 implies the uniform bound

$$\sup_{t\in[0,T],\, \delta\in\mathcal{D},\, z\in\mathbb{R}^m} \Big(\|b^R(t,z,\delta)\|^2 + \|\Sigma^R(t,z,\delta)\|_F^2 + \sum_{i=1}^{D} \|\Gamma_i^R(t,z,\delta)\|^2 \Big) \le C_R.$$

(b) The PRM integrands are predictable and square-integrable up to τ_m. The map $(s,\omega,z) \mapsto \Gamma_i^R(s, Z_{s^-}^{m,R}(\omega), \delta_s(\omega)) \mathbf{1}_{\{z \le \Lambda_s^i(\omega)\}}$ is predictable because $Z_{s^-}^{m,R}$ is left-continuous adapted, δ and Λ^i are predictable, and the indicator is measurable. Furthermore, using $\int_0^\infty \mathbf{1}_{\{z\le\Lambda_s^i\}}\, dz = \Lambda_s^i$ and the boundedness of Γ_i^R,

$$\mathbb{E}\int_0^{\tau_m}\!\!\int_0^\infty \|\Gamma_i^R(s, Z_{s^-}^{m,R}, \delta_s)\|^2 \mathbf{1}_{\{z\le\Lambda_s^i\}}\, dz\, ds = \mathbb{E}\int_0^{\tau_m} \|\Gamma_i^R(\cdot)\|^2\, \Lambda_s^i\, ds \le (\sup \|\Gamma_i^R\|^2)\, \mathbb{E}\int_0^{\tau_m} \Lambda_s^i\, ds \le (\sup \|\Gamma_i^R\|^2)\, m < \infty,$$

since by definition of τ_m one has $\int_0^{\tau_m} \Lambda_s^i\, ds \le m$.

(c) Conclusion (standard jump-SDE theorem). Items (a)-(b) put the equation in the standard regime of Brownian plus PRM-driven SDEs with globally Lipschitz coefficients and square-integrable jump integrands. Therefore there exists a unique càdlàg strong solution $Z^{m,R}$ on $[0,T]$ (with the integrals stopped at $t \wedge \tau_m$).

Step 2: on $[0,\theta]$ the truncated and original coefficients coincide. By definition of σ_R, for all $t \le \theta$ we have $\|Z_{t^-}^{m,R}\| \le R$, hence $\pi_R(Z_{t^-}^{m,R}) = Z_{t^-}^{m,R}$ and

$$b^R(t, Z_{t^-}^{m,R}, \delta_t) = b(t, Z_{t^-}^{m,R}, \delta_t), \quad \Sigma^R(t, Z_{t^-}^{m,R}, \delta_t) = \Sigma(t, Z_{t^-}^{m,R}, \delta_t), \quad \Gamma_i^R(t, Z_{t^-}^{m,R}, \delta_t) = \Gamma_i(t, Z_{t^-}^{m,R}, \delta_t).$$

Hence $Z_t := Z_{t\wedge\theta}^{m,R}$ satisfies (29) on $[0,\theta]$, proving (ii).

Step 3: pathwise uniqueness on $[0,\theta]$. Let Z and $\bar{Z}$ be two strong solutions of (29) on $[0,\theta]$ driven by the same noises $(W, \Pi^1, \dots, \Pi^D)$ and with the same initial condition Z_0. Recall $\theta = \theta_{m,R} = \tau_m \wedge \sigma_R$, and that by construction of Z (Step 2) one has $\|Z_t\| \le R$ for all $t \le \theta$.
To apply the local Lipschitz bounds simultaneously to Z and $\bar{Z}$, define the exit time for $\bar{Z}$,

$$\bar{\sigma}_R := \inf\{t \in [0,T] : \|\bar{Z}_t\| \ge R\} \wedge T, \qquad \hat{\theta} := \theta \wedge \bar{\sigma}_R.$$

Then on $[0,\hat{\theta}]$ both processes remain in the ball $\{\|z\| \le R\}$, so Assumption 5.1 applies to *both* trajectories. Define the difference process stopped at $\hat{\theta}$:

$$\Delta_t := Z_{t\wedge\hat{\theta}} - \bar{Z}_{t\wedge\hat{\theta}}.$$

Compensated form on $[0,\hat{\theta}]$. Write $\Pi^i = \tilde{\Pi}^i + ds\, dz$ and define the compensated drift

$$\tilde{b}(t,z,\delta) := b(t,z,\delta) + \sum_{i=1}^{D} \Gamma_i(t,z,\delta)\Lambda_t^i.$$

Then for $t \le \hat{\theta}$ we may write (in differential form)

$$dZ_t = \tilde{b}(t, Z_{t^-}, \delta_t)\, dt + \Sigma(t, Z_{t^-}, \delta_t)\, dW_t + \sum_{i=1}^{D} \int_0^\infty \Gamma_i(t, Z_{t^-}, \delta_t) \mathbf{1}_{\{z\le\Lambda_t^i\}} \tilde{\Pi}^i(dt, dz),$$

and analogously for $\bar{Z}$. Subtracting yields the compensated SDE for Δ:

$$\begin{aligned} d\Delta_t = {} & \mathbf{1}_{\{t\le\hat{\theta}\}} \Big(\tilde{b}(t, Z_{t^-}, \delta_t) - \tilde{b}(t, \bar{Z}_{t^-}, \delta_t) \Big)\, dt + \mathbf{1}_{\{t\le\hat{\theta}\}} \Big(\Sigma(t, Z_{t^-}, \delta_t) - \Sigma(t, \bar{Z}_{t^-}, \delta_t) \Big)\, dW_t \\ & + \mathbf{1}_{\{t\le\hat{\theta}\}} \sum_{i=1}^{D} \int_0^\infty \Big(\Gamma_i(t, Z_{t^-}, \delta_t) - \Gamma_i(t, \bar{Z}_{t^-}, \delta_t) \Big) \mathbf{1}_{\{z\le\Lambda_t^i\}} \tilde{\Pi}^i(dt, dz). \end{aligned} \tag{32}$$

Local Lipschitz bounds on $[0,\hat{\theta}]$. By Assumption 5.1, for $t \le \hat{\theta}$,

$$\|\Sigma(t, Z_{t^-}, \delta_t) - \Sigma(t, \bar{Z}_{t^-}, \delta_t)\|_F \le L_R \|\Delta_{t^-}\|, \qquad \|\Gamma_i(t, Z_{t^-}, \delta_t) - \Gamma_i(t, \bar{Z}_{t^-}, \delta_t)\| \le L_R \|\Delta_{t^-}\|.$$

Moreover,

$$\begin{aligned}\|\tilde{b}(t, Z_{t^-}, \delta_t) - \tilde{b}(t, \bar{Z}_{t^-}, \delta_t)\| &\le \|b(t, Z_{t^-}, \delta_t) - b(t, \bar{Z}_{t^-}, \delta_t)\| + \sum_{i=1}^{D} \|\Gamma_i(t, Z_{t^-}, \delta_t) - \Gamma_i(t, \bar{Z}_{t^-}, \delta_t)\| \Lambda_t^i \\ &\le L_R \|\Delta_{t^-}\| \Big(1 + \sum_{i=1}^{D} \Lambda_t^i\Big). \end{aligned} \tag{33}$$

Itô formula for $\|\Delta_t\|^2$. Let $g(x) = \|x\|^2$. Then $\nabla g(x) = 2x$ and $D^2 g(x) = 2I_m$. Applying the jump Itô formula to $g(\Delta_t)$ using (32) yields

$$\begin{aligned}\|\Delta_t\|^2 = \|\Delta_0\|^2 &+ \int_0^t \mathbf{1}_{\{s \le \hat{\theta}\}} 2\Delta_{s^-} \cdot \Big(\tilde{b}(s, Z_{s^-}, \delta_s) - \tilde{b}(s, \bar{Z}_{s^-}, \delta_s)\Big)\, ds \\ &+ \int_0^t \mathbf{1}_{\{s \le \hat{\theta}\}} \big\|\Sigma(s, Z_{s^-}, \delta_s) - \Sigma(s, \bar{Z}_{s^-}, \delta_s)\big\|_F^2\, ds \\ &+ \sum_{i=1}^{D} \int_0^t \mathbf{1}_{\{s \le \hat{\theta}\}} \big\|\Gamma_i(s, Z_{s^-}, \delta_s) - \Gamma_i(s, \bar{Z}_{s^-}, \delta_s)\big\|^2 \Lambda_s^i\, ds \\ &+ M_t, \end{aligned} \tag{34}$$

where the local martingale term M_t is

$$\begin{aligned} M_t := &\int_0^t \mathbf{1}_{\{s \le \hat{\theta}\}} 2\Delta_{s^-} \cdot \Big(\Sigma(s, Z_{s^-}, \delta_s) - \Sigma(s, \bar{Z}_{s^-}, \delta_s)\Big)\, dW_s \\ &+ \sum_{i=1}^{D} \int_0^t \mathbf{1}_{\{s \le \hat{\theta}\}} \int_0^\infty \Big(\|\Delta_{s^-} + \Delta\Gamma_i(s)\|^2 - \|\Delta_{s^-}\|^2\Big) \mathbf{1}_{\{z \le \Lambda_s^i\}}\, \tilde{\Pi}^i(ds, dz), \end{aligned} \tag{35}$$

with $\Delta\Gamma_i(s) := \Gamma_i(s, Z_{s^-}, \delta_s) - \Gamma_i(s, \bar{Z}_{s^-}, \delta_s)$.

Martingale justification. On $[0,\hat{\theta}]$ we have $\|Z\| \vee \|\bar{Z}\| \le R$, hence

$$\|\Sigma(Z) - \Sigma(\bar{Z})\|_F^2 \le L_R^2 \|\Delta\|^2 \le 4L_R^2 R^2, \qquad \|\Delta\Gamma_i\|^2 \le L_R^2 \|\Delta\|^2 \le 4L_R^2 R^2.$$

Also $\hat{\theta} \le \theta \le \tau_m$, so $\int_0^{\hat{\theta}} \Lambda_s^i\, ds \le m$ for each i. Therefore the stochastic integrands in (35) are square-integrable, and M_t is a true L^2-martingale on $[0,T]$ with $\mathbb{E}[M_t] = 0$.

Bounding the finite-variation term. Fix $s \le \hat{\theta}$ and set $\Delta := \Delta_{s^-}$. Using (33) and the Lipschitz bounds:

$$2\Delta \cdot (\tilde{b}(Z) - \tilde{b}(\bar{Z})) \le 2\|\Delta\|\, \|\tilde{b}(Z) - \tilde{b}(\bar{Z})\| \le 2L_R \|\Delta\|^2 \Big(1 + \sum_{i=1}^{D} \Lambda_s^i\Big),$$

$$\|\Sigma(Z) - \Sigma(\bar{Z})\|_F^2 \le L_R^2 \|\Delta\|^2 \le L_R^2 \|\Delta\|^2 \Big(1 + \sum_{i=1}^{D} \Lambda_s^i\Big),$$

and

$$\sum_{i=1}^{D} \|\Gamma_i(Z) - \Gamma_i(\bar{Z})\|^2 \Lambda_s^i \le L_R^2 \|\Delta\|^2 \sum_{i=1}^{D} \Lambda_s^i \le L_R^2 \|\Delta\|^2 \Big(1 + \sum_{i=1}^{D} \Lambda_s^i\Big).$$

Thus the finite-variation integrand in (34) satisfies, for $s \le \hat{\theta}$,

$$2\Delta_{s^-} \cdot (\tilde{b}(Z) - \tilde{b}(\bar{Z})) + \|\Sigma(Z) - \Sigma(\bar{Z})\|_F^2 + \sum_{i=1}^{D} \|\Gamma_i(Z) - \Gamma_i(\bar{Z})\|^2 \Lambda_s^i \le c_R \|\Delta_{s^-}\|^2 \Big(1 + \sum_{i=1}^{D} \Lambda_s^i\Big),$$

with the explicit choice $c_R := 2L_R + 2L_R^2$.

Define the increasing predictable process

$$A_t := \int_0^t \mathbf{1}_{\{s \le \hat{\theta}\}} \sum_{i=1}^{D} \Lambda_s^i\, ds = \int_0^{t \wedge \hat{\theta}} \sum_{i=1}^{D} \Lambda_s^i\, ds.$$

Remark 5.3 (Why we do not use an L^1-mass bound). Even though $\hat{\theta} \le \tau_m$ implies $\int_0^{\hat{\theta}} \sum_i \Lambda_s^i ds \le m$, one cannot in general bound $\int_0^t X_{s^-} \sum_i \Lambda_s^i \, ds$ by $m \int_0^t X_{s^-} \, ds$. Instead we treat $\sum_i \Lambda_s^i \, ds$ as the Stieltjes measure dA_s and absorb it via an integrating factor, which yields a valid Stieltjes–Gronwall argument.

Combining with (34) gives the pathwise inequality, for $t \in [0, T]$,

$$\|\Delta_t\|^2 \le \int_0^t c_R \|\Delta_{s^-}\|^2 \, ds + \int_0^t c_R \|\Delta_{s^-}\|^2 \, dA_s + M_t. \tag{36}$$

Integrating factor (Stieltjes–Gronwall). Let $X_t := \|\Delta_t\|^2$ and define the increasing finite-variation process

$$B_t := c_R t + c_R A_t \qquad \text{so that} \qquad dB_t = c_R \, dt + c_R \, dA_t.$$

Then (36) can be written compactly as

$$X_t \le \int_0^t X_{s^-} \, dB_s + M_t, \qquad t \in [0, T]. \tag{37}$$

Introduce the integrating factor $Y_t := e^{-B_t} X_t$. Since B has finite variation, integration by parts yields

$$dY_t = d(e^{-B_t} X_t) = e^{-B_{t^-}} \, dX_t - e^{-B_{t^-}} X_{t^-} \, dB_t.$$

Using (37) in differential form, $dX_t \le X_{t^-} \, dB_t + dM_t$, the finite-variation terms cancel and

$$dY_t \le e^{-B_{t^-}} \, dM_t.$$

Integrating from 0 to t gives

$$Y_t \le Y_0 + \int_0^t e^{-B_{s^-}} \, dM_s.$$

The stochastic integral is an L^2-martingale (predictable integrand bounded by 1 and M is L^2), hence has mean 0. Taking expectations yields

$$\mathbb{E}[Y_t] \le \mathbb{E}[Y_0].$$

But $Y_0 = X_0 = \|\Delta_0\|^2 = 0$ (same initial condition), and $Y_t \ge 0$, so $Y_t = 0$ a.s. for all t. Since $e^{-B_t} > 0$, this implies $X_t = \|\Delta_t\|^2 = 0$, i.e. $\Delta_t = 0$ a.s. for all t.

Removing the auxiliary stopping $\hat{\theta}$. If $\bar{\sigma}_R < \theta$, then at $t = \bar{\sigma}_R$ we have $\|\bar{Z}_t\| = R$ by definition, and since $\Delta_t = 0$ we also have $\|Z_t\| = R$. But $\bar{\sigma}_R < \theta \le \sigma_R$ implies $\bar{\sigma}_R < \sigma_R$, contradicting the definition of σ_R as the first exit time of Z from the ball of radius R. Hence $\mathbb{P}(\bar{\sigma}_R < \theta) = 0$, so $\bar{\sigma}_R \ge \theta$ a.s. Therefore $\hat{\theta} = \theta$ a.s., and $Z_t = \bar{Z}_t$ for all $t \le \theta$ a.s.

This proves pathwise uniqueness up to $\theta = \theta_{m,R}$, establishing (iii).

Step 4: $\mathbb{S}^2$ estimate up to θ. We write the stopped dynamics in compensated form

$$Z_{t\wedge\theta} = Z_0 + \int_0^{t\wedge\theta} \tilde{b}(s, Z_{s^-}, \delta_s) \, ds + \int_0^{t\wedge\theta} \Sigma(s, Z_{s^-}, \delta_s) \, dW_s + \sum_{i=1}^{D} \int_0^{t\wedge\theta} \int_0^{\infty} \Gamma_i(s, Z_{s^-}, \delta_s) \mathbf{1}_{\{z \le \Lambda_s^i\}} \, \tilde{\Pi}^i(ds, dz),$$

with $\tilde{b} = b + \sum_i \Gamma_i \Lambda^i$ as above.

On $[0, \theta]$ we have $\|Z_{s^-}\| \le R$, hence by Assumption 5.1, $\|b\|^2 + \|\Sigma\|_F^2 + \sum_i \|\Gamma_i\|^2 \le C_R$. The only nontrivial point is $\tilde{b}$ because it contains Λ. By Cauchy-Schwarz,

$$\Big\| \sum_{i=1}^{D} \Gamma_i(s, Z_{s^-}, \delta_s) \Lambda_s^i \Big\|^2 \le D \sum_{i=1}^{D} \|\Gamma_i(s, Z_{s^-}, \delta_s)\|^2 (\Lambda_s^i)^2 \le D C_R \sum_{i=1}^{D} (\Lambda_s^i)^2.$$

Integrating and using $\theta \le \tau_m$ gives

$$\int_0^{t\wedge\theta} \Big\| \sum_{i=1}^{D} \Gamma_i(s, Z_{s^-}, \delta_s) \Lambda_s^i \Big\|^2 ds \le D C_R \int_0^{t\wedge\tau_m} \sum_{i=1}^{D} (\Lambda_s^i)^2 ds \le D C_R \, m.$$

Similarly, the jump quadratic-variation terms satisfy

$$\int_0^{t\wedge\theta} \|\Gamma_i(s, Z_{s^-}, \delta_s)\|^2 \Lambda_s^i ds \le C_R \int_0^{t\wedge\tau_m} \Lambda_s^i ds \le C_R m.$$

Now apply the jump Itô formula to $\|Z_{t\wedge\theta}\|^2$, take suprema over $t \le T$, and take expectations. The martingale terms are controlled by BDG. All finite-variation terms are bounded by constants depending on (m, R, T) and $\mathbb{E}\|Z_0\|^2$, plus an integral of $\mathbb{E}[\sup_{u\le s} \|Z_{u\wedge\theta}\|^2]$. This yields a Gronwall inequality

$$\mathbb{E}\Big[\sup_{t\le T} \|Z_{t\wedge\theta}\|^2\Big] \le C_{m,R}\big(1 + \mathbb{E}\|Z_0\|^2\big) + C_{m,R} \int_0^T \mathbb{E}\Big[\sup_{u\le s} \|Z_{u\wedge\theta}\|^2\Big] ds,$$

for a finite constant $C_{m,R}$ depending only on (m, R, T) and the local constants in Assumption 5.1. By Gronwall, $\mathbb{E}[\sup_{t\le T} \|Z_{t\wedge\theta}\|^2] < \infty$, proving (iv). □

Remark 5.4 (From local to global: the exact checklist). Proposition 5.1 is intentionally local. A global strong solution on $[0, T]$ follows once we verify:

(i) $\tau_m \uparrow T$ a.s., equivalently $A_T < \infty$ a.s. (Lemma 5.2);
(ii) $\sigma_R \uparrow T$ a.s. (non-explosion of the state);
(iii) a standard gluing argument: by pathwise uniqueness, the local solutions agree on overlaps, hence define a unique global process on $\bigcup_{m,R}[0, \theta_{m,R}]$, which equals $[0, T]$ a.s. if (i)-(ii) hold.

Specialization to the Hawkes market-making model

Where Proposition 5.1 is applied. In the Hawkes market-making model, the derived intensities have the form $\Lambda_t^i = g^i(\delta_t)\lambda_t^\pm$ with $g^i \in [0, 1]$ and bounded δ. These are not exogenous in the full five-dimensional state, because they depend on λ. This is exactly where triangularity matters: we first construct $(N^\pm, \lambda^\pm)$ (Layer 1), then, conditional on that layer, we treat Λ as a given predictable input and apply Proposition 5.1 to the lower layers (Layer 2 and the resulting (S, Q, X) in Layer 3). The Hawkes layer itself is handled separately by the stability and well-posedness results established earlier.

Remark 5.5 (Why $\tau_m \uparrow T$ here). In our application $\Lambda_t^i = g^i(\delta_t)\lambda_t^\pm$ with $g^i \in [0, 1]$ and δ bounded. Assuming the Hawkes layer is non-explosive on $[0, T]$, the intensity paths $\lambda^\pm(\omega)$ are finite and càdlàg. A real-valued càdlàg function on a compact interval is bounded, hence $\sup_{t\le T} \lambda_t^\pm(\omega) < \infty$. Therefore each $\Lambda^i(\omega)$ is bounded on $[0, T]$, which implies $A_T = \int_0^T \sum_i (\Lambda_s^i + (\Lambda_s^i)^2)\, ds < \infty$ a.s. By Lemma 5.2, $\tau_m \uparrow T$ a.s.

Lemma 5.6 ($\mathbb{S}^2$ bound for the Hawkes market-making state). *Assume the Hawkes layer (N^+, N^-) is a linear Hawkes process with exponential kernels and is subcritical (as in Section 3.4 / Assumption 3.7), so that finite fourth moments on finite horizons hold; in particular, by Lemma 3.6,*

$$\mathbb{E}\big[(N_T^+)^4 + (N_T^-)^4\big] < \infty.$$

Assume $\mathcal{D}$ is bounded and $f_a, f_b \in [0, 1]$, $p^\pm \in [0, \bar{p}]$. Then the market-making state $Z_t = (S_t, Q_t, X_t, \lambda_t^+, \lambda_t^-)$ defined in Section 4 belongs to $\mathbb{S}^2([0, T])$. In particular, $\sup_{t\le T} \|Z_t\| < \infty$ a.s., hence $\sigma_R \uparrow T$ a.s. as $R \to \infty$.

Proof. The proof is by explicit domination of derived channels by the market-wide Hawkes counts, and by direct pathwise bounds.

(1) Midprice. By definition, $S_t = S_0 + \sigma W_t + \kappa(M_t^+ - M_t^-)$ and $M_T^\pm \le N_T^\pm$ a.s. (price moves are marked sub-events of market orders). Hence

$$\sup_{t\le T} |S_t| \le |S_0| + \sigma \sup_{t\le T} |W_t| + \kappa(N_T^+ + N_T^-).$$

Using BDG for $\sup |W|$ and the finite fourth moments of $N_T^\pm$ gives $\mathbb{E}[\sup_{t\le T} |S_t|^4] < \infty$, hence also $\mathbb{E}[\sup_{t\le T} |S_t|^2] < \infty$.

(2) Inventory. Since $Q_t = Q_0 + q_0(H_t^b - H_t^a)$ with $H_T^a \le N_T^+$ and $H_T^b \le N_T^-$ a.s.,

$$\sup_{t\le T} |Q_t| \le |Q_0| + q_0(N_T^+ + N_T^-),$$

so $\mathbb{E}[\sup_{t\le T} |Q_t|^4] < \infty$ and in particular $\mathbb{E}[\sup_{t\le T} |Q_t|^2] < \infty$.

(3) Cash. The cash process changes only at fill times. Each jump of H^a (resp. H^b) changes X by $(S_{t^-} + \delta_t^a)q_0$ (resp. $-(S_{t^-} - \delta_t^b)q_0$). Because $\mathcal{D}$ is bounded, $|\delta_t^{a,b}| \le \delta_{\max}$, hence pathwise

$$\sup_{t\le T}|X_t - X_0| \le q_0 \sum_{0<u\le T} (\,|S_{u^-}| + \delta_{\max}\,)\,\Delta(H_u^a + H_u^b) \le q_0(\delta_{\max} + \sup_{u\le T}|S_u|)\,(H_T^a + H_T^b).$$

Using $H_T^a + H_T^b \le N_T^+ + N_T^-$ and Cauchy-Schwarz,

$$\mathbb{E}\Big[\sup_{t\le T}|X_t - X_0|^2\Big] \le C\,\mathbb{E}\Big[(\delta_{\max} + \sup_{u\le T}|S_u|)^2\,(N_T^+ + N_T^-)^2\Big] < \infty,$$

because $\mathbb{E}[\sup|S|^4] < \infty$ from (1) and $\mathbb{E}[(N_T^+ + N_T^-)^4] < \infty$ by Lemma 3.6.

(4) Intensities. Under exponential kernels, (λ^+, λ^-) admits a linear Markovian embedding of the form $d\lambda_t = B(\mu - \lambda_t)dt + A\,dN_t$ with A, B constant matrices. Solving explicitly yields a pathwise bound

$$\sup_{t\le T}\|\lambda_t\| \le C\big(1 + N_T^+ + N_T^-\big),$$

for a deterministic constant $C = C(A, B, \mu, \lambda_0, T)$. Hence $\mathbb{E}[\sup_{t\le T}\|\lambda_t\|^2] < \infty$ under the finite fourth moments of $N_T^{\pm}$.

(5) Combine components. Summing the previous estimates gives

$$\mathbb{E}\Big[\sup_{t\le T}\big(|S_t|^2 + |Q_t|^2 + |X_t|^2 + \|\lambda_t\|^2\big)\Big] < \infty,$$

that is $Z \in \mathbb{S}^2([0,T])$.
Finally, if $\mathbb{E}[\sup_{t\le T}\|Z_t\|^2] < \infty$ then necessarily $\sup_{t\le T}\|Z_t\| < \infty$ a.s. (otherwise the expectation would be infinite), hence $\sigma_R \uparrow T$ a.s. as $R \to \infty$. □

Remark 5.7 (How this lemma is used). Lemma 5.6 serves two purposes.

(i) It provides a convenient global $\mathbb{S}^2$ bound for the concrete Hawkes market-making state on $[0,T]$. This is a sufficient integrability input for the L^2-martingale conditions appearing in Dynkin's formula and in the verification argument.

(ii) It implies non-explosion of the state on $[0,T]$ in the sense that $\sup_{t\le T}\|Z_t\| < \infty$ a.s., hence $\sigma_R \uparrow T$ a.s. as $R \to \infty$.

Together with Hawkes non-explosion (Section 3.4), which ensures $\tau_m \uparrow T$ a.s., this completes the local-to-global checklist in Remark 5.4. In the sequel we therefore work on $[0,T]$ without carrying the localising sequence explicitly.

Remark 5.8 (Why this is the right input for Dynkin / verification). Dynkin's identity requires (a) a non-explosive state on $[0,T]$ and (b) integrability strong enough that the Brownian and compensated-jump terms in Itô's formula are true martingales. Proposition 5.1 delivers (a)-(b) locally under τ_m and σ_R, and Lemma 5.6 verifies $\sigma_R \uparrow T$ in the Hawkes market-making model. The remaining ingredient is Hawkes stability (Section 3.4), which ensures both non-explosion and finiteness of the intensity mass (and hence $\tau_m \uparrow T$). Together these yield a global $\mathbb{S}^2$ state and justify Dynkin's formula and the verification step without bounded-jump assumptions.

5.4 Itô's formula and the drift/generator decomposition

Test functions. Split the state as $Z = (C, U)$, where C collects the coordinates with a *continuous* evolution (i.e. those appearing in finite-variation drift terms and/or in the diffusion term), while U collects the coordinates that evolve only through jumps. Let $y \subseteq C$ denote the sub-block that is actually driven by the Brownian motion (the nonzero rows of Σ).
Accordingly, for Itô's formula we assume $f(t,z)$ is C^1 in (t, C) and C^2 in the diffusive block y. No smoothness is required in the pure-jump coordinates U: they enter only through jump increments $f(t, z + \Gamma_i(t,z,\delta)) - f(t,z)$.

Proposition 5.2 (Itô formula for controlled jump–diffusions driven by counting processes). *Fix a predictable control δ and let Z be a càdlàg $\mathbb{F}$-adapted process solving* (23)*, i.e.*

$$dZ_t = b(t, Z_{t^-}, \delta_t)\,dt + \Sigma(t, Z_{t^-}, \delta_t)\,dW_t + \sum_{i=1}^{D}\Gamma_i(t, Z_{t^-}, \delta_t)\,dN_t^i,$$

where each N^i is a counting process admitting a predictable intensity $\lambda_t^i = \lambda^i(t, Z_{t^-}, \delta_t)$ and Doob–Meyer decomposition

$$dN_t^i = \lambda_t^i\, dt + dM_t^i, \qquad M_t^i := N_t^i - \int_0^t \lambda_s^i\, ds.$$

Let y denote the diffusive coordinates (rows of Σ that are nonzero), and let $f(t,z)$ be C^1 in t and C^2 in y (no smoothness is required in the pure-jump coordinates). Assume the jump increments

$$\Delta_i f(t,z) := f\big(t, z + \Gamma_i(t,z,\delta)\big) - f(t,z)$$

are integrable so that the stochastic integrals below are well-defined (after localisation if needed). Then

$$\begin{aligned} df(t,Z_t) = &\Big(\partial_t f + \nabla_C f \cdot b + \tfrac{1}{2}\mathrm{Tr}\big(\Sigma\Sigma^\top D_y^2 f\big)\Big)(t,Z_{t^-})\, dt \\ &+ \sum_{i=1}^D \lambda_t^i \Big[f\big(t, Z_{t^-} + \Gamma_i\big) - f(t,Z_{t^-})\Big]\, dt \\ &+ \nabla_C f(t,Z_{t^-}) \cdot \Sigma(t,Z_{t^-},\delta_t)\, dW_t + \sum_{i=1}^D \Big[f\big(t, Z_{t^-} + \Gamma_i\big) - f(t,Z_{t^-})\Big]\, dM_t^i. \end{aligned} \tag{38}$$

Proof. Fix δ and write the controlled state in the counting-process form (23). Let f satisfy the stated regularity. Since Z is a càdlàg semimartingale, the general Itô formula for semimartingales with jumps yields

$$\begin{aligned} df(t,Z_t) = \partial_t f(t,Z_{t^-})\, dt + \nabla_C f(t,Z_{t^-}) \cdot dZ_t + \frac{1}{2}\mathrm{Tr}\big(D_y^2 f(t,Z_{t^-})\, d[Z^c]_t\big) \\ + \sum_{0<s\le t} \Big(f(s,Z_s) - f(s,Z_{s^-}) - \nabla_C f(s,Z_{s^-}) \cdot \Delta Z_s\Big), \end{aligned}$$

where Z^c denotes the continuous local-martingale part of the diffusive coordinates of Z. Here $Z^c = \int_0^\cdot \Sigma(\cdot)\, dW$, hence $d[Z^c]_t = \Sigma\Sigma^\top(t, Z_{t^-}, \delta_t)\, dt$ on the diffusive block and the second-order term becomes $\frac{1}{2}\mathrm{Tr}(\Sigma\Sigma^\top D_y^2 f)(t, Z_{t^-})\, dt$. At any jump time of N^i, the state jumps by $\Delta Z_t = \Gamma_i(t, Z_{t^-}, \delta_t)$. Under the PRM/thinning construction used throughout the thesis (independent Poisson random measures and disjoint refinements of the base flow), different channels do not share jump times almost surely. In particular, simultaneous jumps occur with probability 0, so the jump sum decomposes channel-by-channel and can be written as

$$\sum_{0<s\le t} \Big(f(s,Z_s) - f(s,Z_{s^-}) - \nabla_C f(s,Z_{s^-}) \cdot \Delta Z_s\Big) = \sum_{i=1}^D \int_0^t \Big(\Delta_i f(s,Z_{s^-}) - \nabla_C f(s,Z_{s^-}) \cdot \Gamma_i(s,Z_{s^-},\delta_s)\Big)\, dN_s^i.$$

On the other hand, the term $\nabla_C f \cdot dZ_t$ contains the jump contribution $\sum_i \nabla_C f(t,Z_{t^-}) \cdot \Gamma_i(t,Z_{t^-},\delta_t)\, dN_t^i$. Adding the two cancels the linear term in Γ_i and yields the non-compensated decomposition

$$\begin{aligned} df(t,Z_t) = &\Big(\partial_t f + \nabla_C f \cdot b + \tfrac{1}{2}\mathrm{Tr}\big(\Sigma\Sigma^\top D_y^2 f\big)\Big)(t,Z_{t^-})\, dt \\ &+ \nabla_C f(t,Z_{t^-}) \cdot \Sigma(t,Z_{t^-},\delta_t)\, dW_t + \sum_{i=1}^D \Delta_i f(t,Z_{t^-})\, dN_t^i. \end{aligned}$$

Finally, apply Doob–Meyer to each counting process: $dN_t^i = \lambda_t^i\, dt + dM_t^i$. Substituting gives

$$\sum_{i=1}^D \Delta_i f(t,Z_{t^-})\, dN_t^i = \sum_{i=1}^D \lambda_t^i\, \Delta_i f(t,Z_{t^-})\, dt + \sum_{i=1}^D \Delta_i f(t,Z_{t^-})\, dM_t^i,$$

which yields (38). □

5.5 Controlled generator: definition

For a fixed control value $\delta \in \mathcal{D}$, the (time-inhomogeneous) spatial generator $\mathcal{L}^\delta$ is defined by

$$(\mathcal{L}^\delta \psi)(t,z) = \lim_{h \downarrow 0} \frac{1}{h}\, \mathbb{E}\big[\psi(t, Z_{t+h}) - \psi(t,Z_t) \mid Z_t = z,\ \delta_t = \delta\big], \tag{39}$$

whenever the limit exists. Comparing (38) with (39) shows that the generator is the drift operator:

$$(\mathcal{L}^{\delta} f)(t,z) = b(t,z,\delta)\cdot \nabla_C f(t,z) + \tfrac{1}{2}\mathrm{Tr}\big(\Sigma\Sigma^{\top}(t,z,\delta)\, D_y^2 f(t,z)\big) + \sum_{i=1}^{D} \lambda^i(t,z,\delta)\Big[f\big(t, z+\Gamma_i(t,z,\delta)\big) - f(t,z)\Big]. \tag{40}$$

5.6 Dynkin's formula

Define the (time-inhomogeneous) extended generator acting on test functions $\varphi(t,\cdot)$ by

$$(\mathcal{A}^{\delta}\varphi)(t,z) := \partial_t \varphi(t,z) + (\mathcal{L}^{\delta}\varphi)(t,z), \tag{41}$$

where $\mathcal{L}^{\delta}$ is the controlled spatial generator in (40) (or (44) for our model).
Dynkin's identity follows from Itô's formula (38) after taking conditional expectations. Since (38) contains *local* martingale terms, we first record integrability conditions ensuring these terms are true L^2-martingales (hence have zero conditional expectation).

Lemma 5.9 (L^2-martingale conditions). *Assume that the (global) state Z has been constructed on $[0,T]$ and satisfies $Z \in \mathbb{S}^2([0,T])$ (e.g. by Proposition 5.1 together with Remark 5.4 and Lemma 5.6).*
Let φ be a test function for which Itô's formula (38) applies and such that

$$\mathbb{E}\int_0^T \|\nabla_C \varphi(u,Z_u)\Sigma(u,Z_{u^-},\delta_u)\|^2\, du + \mathbb{E}\int_0^T \sum_{i=1}^{D} \lambda_u^i \left|\varphi(u,Z_{u^-}+\Gamma_i(u,Z_{u^-},\delta_u)) - \varphi(u,Z_{u^-})\right|^2 du < \infty. \tag{42}$$

Then the processes

$$\int_0^t \nabla_C \varphi(u,Z_u)\Sigma(u,Z_{u^-},\delta_u)\, dW_u \quad \text{and} \quad \int_0^t \big(\varphi(u,Z_{u^-}+\Gamma_i) - \varphi(u,Z_{u^-})\big)\, dM_u^i$$

are square-integrable martingales on $[0,T]$ for each $i=1,\dots,D$, hence have zero conditional expectation given $\mathcal{F}_t$.

Proof. The Brownian term is an L^2-martingale by Itô's isometry under the first integrability condition in (42). For the jump terms, $M_t^i := N_t^i - \int_0^t \lambda_s^i ds$ is the compensated martingale associated with N^i. By standard point-process integration theory (e.g. Brémaud (1981, Ch. II)), the stochastic integral against dM^i is an L^2-martingale if its predictable quadratic variation has finite expectation, which is ensured by the second condition in (42). □

Proposition 5.3 (Dynkin identity). *Assume the setting of Lemma 5.9 and let φ satisfy (42). Then for any $t \in [0,T]$ and any bounded stopping time $\tau \in [t,T]$,*

$$\mathbb{E}[\varphi(\tau,Z_\tau) \mid \mathcal{F}_t] = \varphi(t,Z_t) + \mathbb{E}\left[\int_t^{\tau} (\mathcal{A}^{\delta_u}\varphi)(u,Z_u)\, du \;\middle|\; \mathcal{F}_t\right]. \tag{43}$$

Proof. Integrate (38) from t to τ and take conditional expectations given $\mathcal{F}_t$. By Lemma 5.9, the Brownian and compensated jump terms are true L^2-martingales, so their conditional expectations vanish. Rearranging yields (43). □

Remark 5.10 (Conditioning on Z_t under feedback controls). Equation (43) is stated with conditioning on $\mathcal{F}_t$, which is the correct form for general predictable controls. If $\delta_u = \hat{\delta}(u,Z_{u^-})$ is Markov (feedback), then the right-hand side depends on the past only through Z_t, and one may equivalently condition on $Z_t = z$.

Remark 5.11 (Later use in verification). In the verification argument, one applies (43) with φ chosen as a smooth candidate value function.

5.7 Specialisation: generator for the Hawkes market-making state

We now specialise the abstract generator (40) to the market-making model of Section 4. The state is

$$Z_t = (S_t, Q_t, X_t, \lambda_t^+, \lambda_t^-),$$

where only S has a Brownian component and (λ^+, λ^-) follow the exponential-kernel Markov embedding.

Diffusive coordinates in this model. The diffusion acts only on the midprice coordinate S, with constant volatility σ. Thus, for test functions $f(t, z)$, the second-order term reduces to $\frac{1}{2}\sigma^2 \partial_{ss} f$. No diffusion acts on $(Q, X, \lambda^+, \lambda^-)$.

Jump channels. We use the four disjoint event types (ask hit, bid hit, and two miss flows). Their predictable intensities are the thinnings

$$\lambda_a = f_a(\delta^a)\lambda^+, \quad \lambda_b = f_b(\delta^b)\lambda^-, \quad \lambda_{+0} = (1 - f_a(\delta^a))\lambda^+, \quad \lambda_{-0} = (1 - f_b(\delta^b))\lambda^-,$$

and each buy/sell event carries an additional Bernoulli price-move mark with probabilities $p^{\pm}(z)$. Writing out the post-jump states and mixing over the price mark yields exactly (44). (We keep the explicit formula in the sequel for later use in the HJB derivation.)

Post-jump states. Write $z = (s, q, x, \lambda^+, \lambda^-)$. A buy event (either H^a or $N^{+,0}$) adds (a_{++}, a_{-+}) to (λ^+, λ^-) and may move s by $+\kappa$. A sell event adds (a_{+-}, a_{--}) and may move s by $-\kappa$. If the event executes against the market maker, q and x are updated as in (18).

Generator contributions (mixture over the price mark). For each jump type we distinguish the two post-jump states $z^{(\cdot,0)}$ (no price move) and $z^{(\cdot,1)}$ (price move), and weight them by $1 - p^{\pm}(z)$ and $p^{\pm}(z)$.
(i) Ask hit (market maker sells, buy market order executes).

$$z^{(a,0)} = \big(s,\ q{-}q_0,\ x{+}(s{+}\delta^a)q_0,\ \lambda^+{+}a_{++},\ \lambda^-{+}a_{-+}\big), \qquad z^{(a,1)} = \big(s{+}\kappa,\ q{-}q_0,\ x{+}(s{+}\delta^a)q_0,\ \lambda^+{+}a_{++},\ \lambda^-{+}a_{-+}\big).$$

Contribution:

$$\lambda_a(z,\delta)\Big(p^+(z)\big[f(z^{(a,1)}) - f(z)\big] + \big(1 - p^+(z)\big)\big[f(z^{(a,0)}) - f(z)\big]\Big).$$

(ii) Bid hit (market maker buys, sell market order executes).

$$z^{(b,0)} = \big(s,\ q{+}q_0,\ x{-}(s{-}\delta^b)q_0,\ \lambda^+{+}a_{+-},\ \lambda^-{+}a_{--}\big), \qquad z^{(b,1)} = \big(s{-}\kappa,\ q{+}q_0,\ x{-}(s{-}\delta^b)q_0,\ \lambda^+{+}a_{+-},\ \lambda^-{+}a_{--}\big).$$

Contribution:

$$\lambda_b(z,\delta)\Big(p^-(z)\big[f(z^{(b,1)}) - f(z)\big] + \big(1 - p^-(z)\big)\big[f(z^{(b,0)}) - f(z)\big]\Big).$$

(iii) Buy market order that does not hit the market maker.

$$z^{(+0,0)} = \big(s,\ q,\ x,\ \lambda^+ + a_{++},\ \lambda^- + a_{-+}\big), \qquad z^{(+0,1)} = \big(s + \kappa,\ q,\ x,\ \lambda^+ + a_{++},\ \lambda^- + a_{-+}\big).$$

Contribution:

$$\lambda_{+0}(z,\delta)\Big(p^+(z)\big[f(z^{(+0,1)}) - f(z)\big] + \big(1 - p^+(z)\big)\big[f(z^{(+0,0)}) - f(z)\big]\Big).$$

(iv) Sell market order that does not hit the market maker.

$$z^{(-0,0)} = \big(s,\ q,\ x,\ \lambda^+ + a_{+-},\ \lambda^- + a_{--}\big), \qquad z^{(-0,1)} = \big(s - \kappa,\ q,\ x,\ \lambda^+ + a_{+-},\ \lambda^- + a_{--}\big).$$

Contribution:

$$\lambda_{-0}(z,\delta)\Big(p^-(z)\big[f(z^{(-0,1)}) - f(z)\big] + \big(1 - p^-(z)\big)\big[f(z^{(-0,0)}) - f(z)\big]\Big).$$

Full generator. Summing the continuous and jump contributions yields, for fixed δ,

$$\begin{aligned}(\mathcal{L}^\delta f)(z) = \frac{1}{2}\sigma^2\,\partial_{ss}f(z) - \beta_+(\lambda^+ - \mu^+)\,\partial_{\lambda^+}f(z) - \beta_-(\lambda^- - \mu^-)\,\partial_{\lambda^-}f(z)\\ + \lambda_a(z,\delta)\Big(p^+(z)\big[f(z^{(a,1)}) - f(z)\big] + \big(1-p^+(z)\big)\big[f(z^{(a,0)}) - f(z)\big]\Big)\\ + \lambda_b(z,\delta)\Big(p^-(z)\big[f(z^{(b,1)}) - f(z)\big] + \big(1-p^-(z)\big)\big[f(z^{(b,0)}) - f(z)\big]\Big)\\ + \lambda_{+0}(z,\delta)\Big(p^+(z)\big[f(z^{(+0,1)}) - f(z)\big] + \big(1-p^+(z)\big)\big[f(z^{(+0,0)}) - f(z)\big]\Big)\\ + \lambda_{-0}(z,\delta)\Big(p^-(z)\big[f(z^{(-0,1)}) - f(z)\big] + \big(1-p^-(z)\big)\big[f(z^{(-0,0)}) - f(z)\big]\Big).\end{aligned} \tag{44}$$

Equation (44) is the standard controlled jump–diffusion generator (here written in a Hawkes/Doob–Meyer setting) used in dynamic programming (Pham, 2009), with point-process thinning as in Brémaud (1981); Daley and Vere-Jones (2003) and Hawkes Markov embedding as in Hawkes and Oakes (1974); Bacry et al. (2015).

6 Dynamic Programming Principle

This section states and proves the Dynamic Programming Principle (DPP) for the finite-horizon control problem introduced in Section 4. Together with Dynkin's formula from Section 5.6, it yields the Hamilton–Jacobi–Bellman (HJB) characterization of the value function.
Throughout Sections 6 to 8 we restrict to feedback controls $\delta_u = \hat{\delta}(u, Z_{u^-})$, so conditioning on $Z_t = z$ is legitimate (Remark 5.10).

6.1 Viscosity solutions

Even with bounded inventory, global smoothness of the value function is not guaranteed: the HJB involves a supremum over controls and nonlocal jump terms (state shifts induced by executions and Hawkes intensity updates), which can generate nonsmooth value functions and prevent classical differentiability. For full mathematical rigor, the appropriate PDE notion is therefore that of viscosity solutions. In this thesis we keep the classical derivation and verification argument (under sufficient regularity of a candidate solution), and we only record the viscosity perspective as a remark.

Without the viscosity framework one typically lacks (i) a proof that V solves the HJB in the appropriate weak sense and (ii) a comparison principle ensuring uniqueness. The present thesis therefore uses the classical verification route: if a sufficiently smooth solution to the HJB exists, it yields an optimal feedback control.

6.2 Stopping-time DPP for a terminal criterion

Let $(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ satisfy the usual conditions. For $0 \le t \le T$ let

$$\mathcal{T}_{t,T} := \{\theta : \Omega \to [t,T] \text{ stopping time w.r.t. } \mathbb{F}\}.$$

Let $Z^{t,z,\delta}$ be the controlled state started from $Z_t = z$ under $\delta \in \mathcal{A}_t$. We consider a purely terminal payoff

$$J(t,z;\delta) := \mathbb{E}\Big[G\Big(Z_T^{t,z,\delta}\Big)\,\Big|\,Z_t = z\Big], \qquad V(t,z) := \sup_{\delta\in\mathcal{A}_t} J(t,z;\delta). \tag{45}$$

Although $\mathcal{A}_t$ is the class of all admissible predictable controls, the Markov structure of Z implies that optimality can be achieved within (measurable) feedback controls; the verification theorem constructs such a feedback optimizer. (We consider in this thesis $G(z) = U(\widetilde{W}_T(z))$ and $f \equiv 0$.)

Assumption 6.1 (Standing assumptions for the stopping-time DPP)**.** Fix (t,z) and $\theta \in \mathcal{T}_{t,T}$.

(i) *Well-posedness and integrability.* For every $\delta \in \mathcal{A}_t$, the controlled state $Z^{t,z,\delta}$ is well-defined on $[t,T]$ and $J(t,z;\delta)$ in (45) is finite.

(ii) *Non-anticipativity.* Admissible controls are $\mathbb{F}$-predictable.

(iii) *Stability under concatenation.* If $\delta^1 \in \mathcal{A}_t$ and $\delta^2 \in \mathcal{A}_\theta$, then the concatenation $\delta := \delta^1 \otimes_\theta \delta^2$ defined by

$$(\delta^1 \otimes_\theta \delta^2)_u(\omega) := \begin{cases} \delta^1_u(\omega), & u \in [t, \theta(\omega)], \\ \delta^2_u(\omega), & u \in (\theta(\omega), T], \end{cases}$$

belongs to $\mathcal{A}_t$. Moreover, the payoff satisfies the flow property: for any $\delta \in \mathcal{A}_t$,

$$J(t,z;\delta) = \mathbb{E}\Big[J\Big(\theta, Z_\theta^{t,z,\delta}; \delta^\theta\Big) \,\Big|\, Z_t = z\Big],$$

where δ^θ denotes the post-θ restriction of δ, i.e.

$$\delta^\theta_u(\omega) := \delta_u(\omega)\mathbf{1}_{\{u>\theta(\omega)\}} \qquad (u \in [t,T]),$$

viewed as an element of $\mathcal{A}_\theta$ (values on $[t,\theta]$ are irrelevant).

(iv) *Measurable ε-optimal continuation.* For every $\varepsilon > 0$ there exists an $\mathcal{F}_\theta$-measurable map $\omega \mapsto \hat{\delta}^\varepsilon(\omega) \in \mathcal{A}_\theta$ such that

$$V\Big(\theta, Z_\theta^{t,z,\delta}\Big) \le J\Big(\theta, Z_\theta^{t,z,\delta}; \hat{\delta}^\varepsilon\Big) + \varepsilon \qquad \text{a.s.}$$

Remark 6.2 (Stability under concatenation & existence of an ε-optimal selector.). Assumption 6.1 is stated in the general form of Pham's stopping-time DPP. In our market-making model it is mild for the following reasons.
(iii) Stability under concatenation. By Definition 4.5, admissible controls are $\mathbb{F}$-predictable and take values in the closed set $\mathcal{D}$ (typically $\mathcal{D} = [0, \delta_{\max}]^2$ in the numerical scheme). If $\delta^1 \in \mathcal{A}_t$ and $\delta^2 \in \mathcal{A}_\theta$, then the concatenation $\delta_u := \delta^1_u \mathbf{1}_{u\le\theta} + \delta^2_u \mathbf{1}_{u>\theta}$ is $\mathbb{F}$-predictable and still $\mathcal{D}$-valued. Moreover, under feedback controls $\delta_u = \hat{\delta}(u, Z_{u^-})$ the controlled state Z is Markov, so the post-θ evolution depends on the past only through Z_θ (Remark 5.10), yielding the flow property $J(t,z;\delta) = \mathbb{E}\big[J(\theta, Z_\theta; \delta^\theta) \mid Z_t = z\big]$.
(iv) Existence of an ε-optimal selector. In the implementation we work with a finite control grid $D_h \subset D$ (tick-consistent discretization), so the maximizer exists by finiteness and can be chosen measurably by a deterministic tie-breaking rule (e.g. lexicographic minimizer among maximizers). For the continuous-control idealization, one may invoke standard measurable selection results under continuity of the payoff in δ; in this thesis we keep (iv) as a standing assumption in the same spirit as the classical DPP references.
In particular, all numerical results in Chapters 10–11 refer to the discretized action set D_h, so maximizers are attained (hence measurable by tie-breaking) and the verification argument applies to the implemented problem.

Theorem 6.3 (Stopping-time DPP). *We follow the structure contained in Pham (2009, Thm. 3.3.1) adapting it to our framework.*
Under Assumption 6.1, for any (t,z) and any $\theta \in \mathcal{T}_{t,T}$,

$$V(t,z) = \sup_{\delta\in\mathcal{A}_t} \mathbb{E}\Big[V\Big(\theta, Z_\theta^{t,z,\delta}\Big) \,\Big|\, Z_t = z\Big]. \tag{46}$$

Remark 6.4 (Specialization to our Hawkes-driven state). In our model the augmented state $Z_t = (S_t, Q_t, X_t, \lambda_t^+, \lambda_t^-)$ is finite-dimensional and Markov under exponential Hawkes kernels (Proposition 3.1). The objective is purely terminal, so (46) becomes exactly the time-consistency identity used to derive the HJB.

6.3 Proof of the stopping-time DPP

Proof of Theorem 6.3. Fix (t,z) and $\theta \in \mathcal{T}_{t,T}$.

Step 1: "$\le$". Fix any $\delta \in \mathcal{A}_t$. By the flow property,

$$J(t,z;\delta) = \mathbb{E}\Big[J\Big(\theta, Z_\theta^{t,z,\delta}; \delta^\theta\Big) \,\Big|\, Z_t = z\Big].$$

Since $V(\theta,\cdot)$ is the supremum over controls on $[\theta,T]$, $J(\theta,y;\delta^\theta) \le V(\theta,y)$ for all y, hence

$$J(t,z;\delta) \le \mathbb{E}\Big[V\Big(\theta, Z_\theta^{t,z,\delta}\Big) \,\Big|\, Z_t = z\Big].$$

Taking the supremum over $\delta \in \mathcal{A}_t$ yields

$$V(t,z) = \sup_{\delta\in\mathcal{A}_t} J(t,z;\delta) \le \sup_{\delta\in\mathcal{A}_t} \mathbb{E}\Big[V\Big(\theta, Z_\theta^{t,z,\delta}\Big) \,\Big|\, Z_t = z\Big].$$

Step 2: "$\geq$". Let $\varepsilon > 0$ and fix $\delta^1 \in \mathcal{A}_t$ (the pre-θ control). By Assumption 6.1(iv), there exists an $\mathcal{F}_\theta$-measurable selector $\hat{\delta}^\varepsilon(\theta, \cdot) \in \mathcal{A}_\theta$ such that

$$V\Big(\theta, Z_\theta^{t,z,\delta^1}\Big) \leq J\Big(\theta, Z_\theta^{t,z,\delta^1}; \hat{\delta}^\varepsilon(\theta, Z_\theta^{t,z,\delta^1})\Big) + \varepsilon \qquad \text{a.s.}$$

Define the concatenated control $\delta := \delta^1 \otimes_\theta \hat{\delta}^\varepsilon \in \mathcal{A}_t$. By stability under concatenation and the flow property, the post-θ segment of δ coincides with $\hat{\delta}^\varepsilon$ and the state restarts from Z_θ^{t,z,δ^1}.

$$J(t,z;\delta) = \mathbb{E}\Big[J\Big(\theta, Z_\theta^{t,z,\delta^1}; \hat{\delta}^\varepsilon(\theta, Z_\theta^{t,z,\delta^1})\Big) \,\Big|\, Z_t = z\Big].$$

Combining the last two displays gives

$$\mathbb{E}\Big[V\Big(\theta, Z_\theta^{t,z,\delta^1}\Big) \,\Big|\, Z_t = z\Big] \leq J(t,z;\delta) + \varepsilon \leq V(t,z) + \varepsilon.$$

Since $\delta^1 \in \mathcal{A}_t$ was arbitrary,

$$\sup_{\delta \in \mathcal{A}_t} \mathbb{E}\Big[V\Big(\theta, Z_\theta^{t,z,\delta}\Big) \,\Big|\, Z_t = z\Big] \leq V(t,z) + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ yields the reverse inequality.
Combining Steps 1–2 proves (46). □

Remark 6.5 (Running costs). If the payoff includes $\int_t^T f(s, Z_s, \delta_s)\, ds$, the DPP becomes $V(t,z) = \sup_\delta \mathbb{E}\big[\int_t^\theta f(\cdot)\, ds + V(\theta, Z_\theta) \mid Z_t = z\big]$ and the proof is identical after splitting the integral at θ.

7 Derivation of the Hamilton–Jacobi–Bellman equation

This section turns the stopping-time DPP into a local (infinitesimal) optimality condition. In finance terms, the value function $V(t,z)$ is the market maker's *continuation utility*: given the current state z (midprice, inventory, cash, and Hawkes intensities), it is the best achievable expected terminal utility from time t onward.
The Hamilton–Jacobi–Bellman equation is the mathematical expression of a simple idea: *under an optimal quoting policy, you cannot improve expected continuation utility by changing quotes over an infinitesimal time step.* In a diffusion-only model this becomes a drift condition involving the usual second-order operator. In our setting, the state is a jump–diffusion where jumps arrive with *predictable stochastic intensities* (Hawkes flow and predictable thinnings), so the relevant infinitesimal object is the controlled generator $\mathcal{L}^\delta$ constructed in Section 5.5 and specialized in (44). The jump terms in $\mathcal{L}^\delta$ encode, in one operator, all the microstructure bookkeeping: fills update (Q, X), Bernoulli marks move S, and Hawkes excitation updates (λ^+, λ^-).

7.1 Classical derivation via a two-inequalities argument

Fix $(t,z) \in [0,T) \times \mathcal{Z}$ and take a small $h > 0$ with $t + h \leq T$. Applying the DPP (46) with the deterministic stopping time $\theta = t + h$ gives

$$V(t,z) = \sup_{\delta \in \mathcal{A}_t} \mathbb{E}\Big[V\big(t+h, Z_{t+h}^{t,z,\delta}\big) \,\Big|\, Z_t = z\Big]. \tag{47}$$

Regularity standing assumption. Assume for the moment that V is smooth in the continuous coordinates of z (time, the diffusive price coordinate, and the Hawkes intensity coordinates) and satisfies the integrability conditions needed to apply Dynkin's formula to $\varphi = V$ on $[t, t+h]$.
Then, for any admissible control δ, Dynkin's formula (43) yields

$$\mathbb{E}\Big[V\big(t+h, Z_{t+h}^{t,z,\delta}\big) \,\Big|\, Z_t = z\Big] = V(t,z) + \mathbb{E}\left[\int_t^{t+h} \big(\partial_u V + \mathcal{L}^{\delta_u} V\big)\big(u, Z_u^{t,z,\delta}\big)\, du \,\middle|\, Z_t = z\right]. \tag{48}$$

Subsolution inequality ("≤ 0"). Fix an *action value* $\bar{\delta} \in \mathcal{D}$. Consider the subset of admissible controls that are frozen to $\bar{\delta}$ on $[t, t+h]$ (and are arbitrary after $t+h$):

$$\mathcal{A}_t^{\bar{\delta},h} := \{\delta \in \mathcal{A}_t : \ \delta_u = \bar{\delta} \text{ for all } u \in [t, t+h]\}.$$

Since (47) takes the supremum over *all* admissible controls, it also dominates the supremum over this restricted class, hence in particular

$$V(t,z) \ \geq \ \sup_{\delta \in \mathcal{A}_t^{\bar{\delta},h}} \mathbb{E}\Big[V\big(t+h, Z_{t+h}^{t,z,\delta}\big) \,\Big|\, Z_t = z\Big].$$

Pick any $\delta \in \mathcal{A}_t^{\bar{\delta},h}$ and plug (48) into the inequality $V(t,z) \geq \mathbb{E}[V(t+h, Z_{t+h}^{t,z,\delta}) \mid Z_t = z]$ to obtain

$$0 \ \geq \ \mathbb{E}\left[\int_t^{t+h} \big(\partial_u V + \mathcal{L}^{\bar{\delta}} V\big)\big(u, Z_u^{t,z,\delta}\big)\, du \,\middle|\, Z_t = z\right],$$

where we used that $\delta_u = \bar{\delta}$ on $[t, t+h]$. Divide by h and let $h \downarrow 0$ (formally) to get

$$\partial_t V(t,z) + (\mathcal{L}^{\bar{\delta}} V)(t,z) \leq 0.$$

Since $\bar{\delta} \in \mathcal{D}$ was arbitrary, taking the supremum yields

$$\partial_t V(t,z) + \sup_{\delta \in \mathcal{D}} (\mathcal{L}^{\delta} V)(t,z) \leq 0. \tag{49}$$

Supersolution inequality ("≥ 0"). Let $\varepsilon > 0$ and choose an ε-optimal control $\delta^{\varepsilon} \in \mathcal{A}_t$ such that

$$V(t,z) \leq \mathbb{E}\Big[V\big(t+h, Z_{t+h}^{t,z,\delta^{\varepsilon}}\big) \,\Big|\, Z_t = z\Big] + \varepsilon.$$

Using (48) with δ^{ε} gives

$$-\varepsilon \leq \mathbb{E}\left[\int_t^{t+h} \big(\partial_u V + \mathcal{L}^{\delta_u^{\varepsilon}} V\big)\big(u, Z_u^{t,z,\delta^{\varepsilon}}\big)\, du \,\middle|\, Z_t = z\right].$$

Divide by h and let $h \downarrow 0$ (formally), then let $\varepsilon \downarrow 0$, to obtain

$$\partial_t V(t,z) + \sup_{\delta \in \mathcal{D}} (\mathcal{L}^{\delta} V)(t,z) \geq 0. \tag{50}$$

HJB equation. Combining (49) and (50) yields the HJB equation (Pham, 2009; Fleming and Soner, 2006):

$$\boxed{\partial_t V(t,z) + \sup_{\delta \in \mathcal{D}} (\mathcal{L}^{\delta} V)(t,z) = 0, \qquad (t,z) \in [0,T) \times \mathcal{Z},} \tag{51}$$

with terminal condition

$$\boxed{V(T,z) = U\big(\widetilde{W}_T(z)\big).} \tag{52}$$

Economic reading of the operator. Equation (51) states that the optimal policy chooses quotes δ so that the *instantaneous expected change of continuation utility* is maximized and, at the optimum, exactly offsets the time decay $\partial_t V$. Concretely, $\mathcal{L}^{\delta} V$ decomposes into: (i) a diffusion term (inventory exposure to midprice risk), (ii) execution jump terms (spread capture and inventory/cash updates when you get filled), (iii) adverse-selection terms coming from price-move marks, and (iv) intensity-state drift/jumps (Hawkes mean reversion and excitation), which is the channel that makes quote asymmetry rational during one-sided bursts.

Remark 7.1 (Viscosity interpretation). The smoothness assumed above is generally unavailable; rigorously, the DPP implies that V is a viscosity solution of (51)–(52). The classical derivation is included to make the role of the generator explicit.

8 Verification theorem (classical solution implies optimality)

The HJB equation yields a pointwise drift condition: for any control δ, one has $\partial_t v + \mathcal{L}^\delta v \le 0$ if v solves the HJB (as a supremum), and if a control attains the supremum then equality holds. Dynkin's formula then turns this into a global inequality on expected terminal utility.

Theorem 8.1 (Verification theorem: terminal utility, Hawkes-driven Markov state)**.** *Assume we restrict to Markov feedback controls $\delta_u = \hat{\delta}(u, Z_{u^-})$, so that the controlled state Z is Markov. Let $\mathcal{L}^\delta$ be the corresponding generator.*
Let $v : [0,T] \times \mathcal{Z} \to \mathbb{R}$ satisfy:

- ***Regularity.***
  (i) v is C^1 in t, C^2 in the diffusive coordinate s, and C^1 in the Hawkes intensity coordinates (λ^+, λ^-). No differentiability is required in the pure-jump coordinates (q, x).
  (ii) For every admissible control δ, the Dynkin L^2 condition of Lemma 5.9 holds with $\phi = v$ along the controlled state.
  (iii) v satisfies the HJB equation and terminal condition in the classical sense.
- ***HJB.*** *For all $(t,z) \in [0,T) \times \mathcal{Z}$,*

$$\partial_t v(t,z) + \sup_{\delta \in \mathcal{D}} \big(\mathcal{L}^\delta v\big)(t,z) = 0, \qquad v(T,z) = U\big(\widetilde{W}_T(z)\big).$$

- ***Measurable maximizer.*** *There exists a measurable feedback selector $\delta^\star(t,z) \in \arg\max_{\delta \in \mathcal{D}} (\mathcal{L}^\delta v)(t,z)$, and the closed-loop control $\delta^\star_t := \delta^\star(t, Z_{t^-})$ is admissible.*

Then:

1. ***Upper bound.*** *For every admissible control $\delta \in \mathcal{A}_t$ and every initial (t,z),*

$$v(t,z) \ge J(t,z;\delta).$$

   Consequently, $v(t,z) \ge V(t,z)$.
2. ***Optimality.*** *Under the feedback control $\delta^\star$, equality holds:*

$$v(t,z) = J(t,z;\delta^\star) = V(t,z),$$

   hence $\delta^\star$ is optimal and v coincides with the value function.

References. Classical verification: Pham (2009, Thm. 3.5.2), Fleming and Soner (2006, Ch. V).

Proof of Theorem 8.1. Fix (t,z).

Step 1: upper bound for arbitrary controls. Let $\delta \in \mathcal{A}_t$ be any admissible control, and consider the corresponding state $Z^{t,z,\delta}$. Apply Dynkin's formula (43) to $\varphi = v$ over $[t,T]$:

$$\mathbb{E}\Big[v(T, Z_T^{t,z,\delta}) \,\Big|\, Z_t = z\Big] = v(t,z) + \mathbb{E}\left[\int_t^T (\partial_u v + \mathcal{L}^{\delta_u} v)\big(u, Z_u^{t,z,\delta}\big)\, du \,\middle|\, Z_t = z\right].$$

Since v solves the HJB as a supremum, for every u and every $\delta_u \in \mathcal{D}$ we have $\partial_u v + \mathcal{L}^{\delta_u} v \le \partial_u v + \sup_{\delta \in \mathcal{D}} \mathcal{L}^\delta v = 0$. Therefore the integral term is ≤ 0, and using the terminal condition $v(T,\cdot) = U(\widetilde{W}_T(\cdot))$ yields

$$v(t,z) \ge \mathbb{E}\Big[U\big(\widetilde{W}_T(Z_T^{t,z,\delta})\big) \,\Big|\, Z_t = z\Big] = J(t,z;\delta).$$

Taking the supremum over $\delta \in \mathcal{A}_t$ gives $v(t,z) \ge V(t,z)$.

Step 2: optimality under a maximizer. Assume now $\delta^\star(t,z)$ attains the supremum, and the closed-loop control $\delta^\star_t = \delta^\star(t, Z_{t^-})$ is admissible. Then, along the controlled trajectory, the HJB inequality becomes an equality: $\partial_u v + \mathcal{L}^{\delta^\star_u} v = 0$ a.s. for $u \in [t,T]$. Applying Dynkin's formula again with $\delta^\star$ yields

$$v(t,z) = \mathbb{E}\Big[v(T, Z_T^{t,z,\delta^\star}) \,\Big|\, Z_t = z\Big] = \mathbb{E}\Big[U\big(\widetilde{W}_T(Z_T^{t,z,\delta^\star})\big) \,\Big|\, Z_t = z\Big] = J(t,z;\delta^\star).$$

Combined with $v \ge V$ this implies $v(t,z) = V(t,z)$ and $\delta^\star$ is optimal. □

Remark 8.2 (When the supremum is not attained). If an argmax does not exist, one works with ε-maximizers and obtains ε-optimal controls, or uses relaxed controls; the verification idea is unchanged.

Theorem 8.1 is conditional: it does not provide a solution of the HJB, but it states that *any* sufficiently regular solution yields an upper bound on performance, and any feedback control attaining the supremum is optimal. In Section 9 we exploit CARA structure to reduce the HJB and construct such a candidate (numerically), after which Theorem 8.1 applies.

9 CARA factorization (dimension reduction)

This section performs an algebraic change of variables that exploits CARA utility. It is *not* a verification argument: it is a structural simplification of the HJB. Once the reduced equation is obtained and (numerically) solved, the verification theorem (Section 8) justifies optimality of the resulting feedback control.

Let $z = (s, q, x, \lambda^+, \lambda^-)$ and recall the terminal payoff $\widetilde{W}_T = x + qs - \frac{\eta}{2}q^2$. For CARA utility $U(w) = -e^{-\gamma w}$, the value function is translation-invariant in cash, and the dependence on the midprice level enters only through the marked-to-market term qs. This motivates factoring out $e^{-\gamma x}$ and $e^{-\gamma qs}$ and reducing the PDE dimension.

9.1 CARA factorization and terminal condition

Motivated by the translation invariance of CARA utility in cash and the marked-to-market term qs, we perform the exponential change of variables

$$V(t, s, q, x, \lambda^+, \lambda^-) = -e^{-\gamma x}\, e^{-\gamma qs}\, \Theta(t, q, \lambda^+, \lambda^-), \tag{53}$$

which *defines* the reduced unknown Θ. The terminal condition $V(T, z) = -e^{-\gamma(x+qs-\eta q^2/2)}$ then implies

$$\Theta(T, q, \lambda^+, \lambda^-) = \exp\Big(\frac{\gamma\eta}{2}q^2\Big). \tag{54}$$

9.2 Recap: controlled generator in our Hawkes market-making state

To keep this section self-contained, we restate the controlled generator specialized to our state $z = (s, q, x, \lambda^+, \lambda^-)$ and to the disjoint event decomposition (ask fill / bid fill / missed buy / missed sell). This is the operator appearing in the HJB, obtained from Dynkin's formula and Doob–Meyer compensation.

Let $\varphi = \varphi(t, s, q, x, \lambda^+, \lambda^-)$ be smooth in the continuous variables. The continuous part (diffusive midprice and mean-reverting Hawkes intensities) is

$$\mathcal{L}^{\delta,\mathrm{cont}}\varphi = \frac{1}{2}\sigma^2\, \partial_{ss}\varphi - \beta_+(\lambda^+ - \mu^+)\, \partial_{\lambda^+}\varphi - \beta_-(\lambda^- - \mu^-)\, \partial_{\lambda^-}\varphi.$$

For jumps, we use Bernoulli marks $J^+ \sim \mathrm{Bernoulli}(p^+)$ and $J^- \sim \mathrm{Bernoulli}(p^-)$ (modeling the probability that an incoming buy/sell market order moves the midprice by one tick). The post-jump intensity updates are

$$(\lambda^+, \lambda^-)^{(+)} := (\lambda^+ + a_{++},\ \lambda^- + a_{-+}), \qquad (\lambda^+, \lambda^-)^{(-)} := (\lambda^+ + a_{+-},\ \lambda^- + a_{--}),$$

and the controlled intensities are

$$\Lambda^a = f_a(\delta^a)\lambda^+, \quad \Lambda^{+,0} = (1 - f_a(\delta^a))\lambda^+, \quad \Lambda^b = f_b(\delta^b)\lambda^-, \quad \Lambda^{-,0} = (1 - f_b(\delta^b))\lambda^-.$$

Then the full controlled generator can be written as

$$\begin{aligned}
(\mathcal{L}^\delta\varphi)(t, z) = {} & \mathcal{L}^{\delta,\mathrm{cont}}\varphi(t, z) \\
& + \Lambda^a\, \mathbb{E}\Big[\varphi\big(t,\ s + \kappa J^+,\ q - q_0,\ x + (s + \delta^a)q_0,\ (\lambda^+, \lambda^-)^{(+)}\big) - \varphi(t, z)\Big] \\
& + \Lambda^b\, \mathbb{E}\Big[\varphi\big(t,\ s - \kappa J^-,\ q + q_0,\ x - (s - \delta^b)q_0,\ (\lambda^+, \lambda^-)^{(-)}\big) - \varphi(t, z)\Big] \\
& + \Lambda^{+,0}\, \mathbb{E}\Big[\varphi\big(t,\ s + \kappa J^+,\ q,\ x,\ (\lambda^+, \lambda^-)^{(+)}\big) - \varphi(t, z)\Big] \\
& + \Lambda^{-,0}\, \mathbb{E}\Big[\varphi\big(t,\ s - \kappa J^-,\ q,\ x,\ (\lambda^+, \lambda^-)^{(-)}\big) - \varphi(t, z)\Big].
\end{aligned} \tag{55}$$

Equation (55) is the only input needed for the CARA reduction: we now plug the ansatz (53) into $\partial_t V + \sup_{\delta\in\mathcal{D}} \mathcal{L}^\delta V = 0$ and simplify each term by computing the ratios $V(t, z')/V(t, z)$.

9.3 Derivation of the reduced HJB

Write $E(t,s,q,x) := e^{-\gamma x}e^{-\gamma q s}$, so $V = -E\,\Theta$. Since Θ does not depend on s or x, we have

$$\partial_t V = -E\,\partial_t\Theta, \tag{56}$$
$$\partial_s V = -E\,(-\gamma q)\Theta = -\gamma q\,V, \tag{57}$$
$$\partial_{ss} V = \gamma^2 q^2\,V, \tag{58}$$
$$\partial_{\lambda^\pm} V = -E\,\partial_{\lambda^\pm}\Theta. \tag{59}$$

Therefore the *continuous* part of the generator contributes

$$\frac{1}{2}\sigma^2\partial_{ss}V - \beta_+(\lambda^+ - \mu^+)\partial_{\lambda^+}V - \beta_-(\lambda^- - \mu^-)\partial_{\lambda^-}V = -E\Big(\frac{1}{2}\sigma^2\gamma^2 q^2\,\Theta - \beta_+(\lambda^+ - \mu^+)\partial_{\lambda^+}\Theta - \beta_-(\lambda^- - \mu^-)\partial_{\lambda^-}\Theta\Big).$$

Jump terms via ratios. For each event type, the generator includes a term of the form $\Lambda^i(t,z)\,\mathbb{E}[\,V(t,z+\Delta z^i) - V(t,z)\mid \mathcal{F}_{t^-}\,]$, where the expectation is taken over the Bernoulli price-move mark $J^\pm$. Under the ansatz, each jump contribution can be written as $-E$ times a corresponding operator acting on Θ. The simplification comes from computing ratios $V(t,z')/V(t,z)$, in which the midprice level s cancels.

Example: ask execution H^a. On an ask fill, inventory decreases $q \mapsto q - q_0$ and cash increases $x \mapsto x + (s+\delta^a)q_0$. If $J^+ = 1$, the midprice jumps $s \mapsto s+\kappa$; otherwise s is unchanged. A direct computation gives

$$\frac{V\big(t, s+\kappa J^+,\, q-q_0,\, x+(s+\delta^a)q_0,\, (\lambda^+,\lambda^-)^{(+)}\big)}{V(t,s,q,x,\lambda^+,\lambda^-)} = e^{-\gamma q_0\delta^a}\,e^{-\gamma\kappa J^+(q-q_0)}\,\frac{\Theta\big(t,q-q_0,(\lambda^+,\lambda^-)^{(+)}\big)}{\Theta(t,q,\lambda^+,\lambda^-)}.$$

Taking conditional expectation over J^+ yields $(1-p^+) + p^+e^{-\gamma\kappa(q-q_0)}$.

Reduced HJB for Θ. Plugging (53) into the full HJB, using (56)–(59), and dividing by $-E$ yields a reduced equation on $(t,q,\lambda^+,\lambda^-)$:

$$\begin{aligned} 0 = {}& \partial_t\Theta + \frac{1}{2}\sigma^2\gamma^2 q^2\,\Theta - \beta_+(\lambda^+ - \mu^+)\,\partial_{\lambda^+}\Theta - \beta_-(\lambda^- - \mu^-)\,\partial_{\lambda^-}\Theta \\ & + \sup_{\delta=(\delta^a,\delta^b)\in\mathcal{D}} \Big\{\mathcal{J}^\delta_a\Theta + \mathcal{J}^\delta_b\Theta + \mathcal{J}^\delta_{+0}\Theta + \mathcal{J}^\delta_{-0}\Theta\Big\}, \end{aligned} \tag{60}$$

with terminal condition (54). The four jump operators correspond to the four disjoint event types:

$$\mathcal{J}^\delta_a\Theta = \Lambda^a\Big(e^{-\gamma q_0\delta^a}\big[(1-p^+) + p^+e^{-\gamma\kappa(q-q_0)}\big]\,\Theta\big(t,q-q_0,(\lambda^+,\lambda^-)^{(+)}\big) - \Theta(t,q,\lambda^+,\lambda^-)\Big),$$
$$\mathcal{J}^\delta_b\Theta = \Lambda^b\Big(e^{-\gamma q_0\delta^b}\big[(1-p^-) + p^-e^{+\gamma\kappa(q+q_0)}\big]\,\Theta\big(t,q+q_0,(\lambda^+,\lambda^-)^{(-)}\big) - \Theta(t,q,\lambda^+,\lambda^-)\Big),$$
$$\mathcal{J}^\delta_{+0}\Theta = \Lambda^{+,0}\Big(\big[(1-p^+) + p^+e^{-\gamma\kappa q}\big]\,\Theta\big(t,q,(\lambda^+,\lambda^-)^{(+)}\big) - \Theta(t,q,\lambda^+,\lambda^-)\Big),$$
$$\mathcal{J}^\delta_{-0}\Theta = \Lambda^{-,0}\Big(\big[(1-p^-) + p^-e^{+\gamma\kappa q}\big]\,\Theta\big(t,q,(\lambda^+,\lambda^-)^{(-)}\big) - \Theta(t,q,\lambda^+,\lambda^-)\Big).$$

Why the reduced HJB does not require S_t as a grid variable. Equation (60) determines optimal offsets $(\delta^{a\star},\delta^{b\star})$ as functions of $(t,q,\lambda^+,\lambda^-)$. The midprice level is then used only to recenter offsets into absolute quotes: $p^a_t = S_t + \delta^{a\star}_t$, $p^b_t = S_t - \delta^{b\star}_t$.

Remark 9.1 (Diffusive inventory risk as an endogenous running penalty). Although the objective includes only a terminal quadratic penalty, CARA utility combined with a diffusive midprice component endogenises a running inventory-risk term in the HJB through the quadratic variation of qS_t. Under the factorisation $V = -e^{-\gamma x}e^{-\gamma q s}\Theta$, this appears as the term $\frac{1}{2}\sigma^2\gamma^2q^2\Theta$ in the reduced equation. Equivalently, defining the certainty-equivalent transform $\psi := -\frac{1}{\gamma}\log\Theta$ converts this contribution into an additive penalty $\frac{1}{2}\,\gamma\,\sigma^2q^2$ in the ψ-equation.

Remark 9.2. We assume the market-order flow

$$(N^+, N^-)$$

is in its stationary regime (consistent with calibration on long samples), so that moment bounds apply uniformly on finite horizons.

Lemma 9.3 (Exponential moment of Hawkes activity on $[0,T]$)**.** *Assume the bivariate linear Hawkes process (N^+,N^-) is* subcritical *in the sense of Section 3.4, i.e. the spectral radius of the matrix of kernel masses $K_{ij} := \|\phi_{ij}\|_{L^1(\mathbb{R}_+)}$ satisfies $\rho(K)<1$, and assume (N^+,N^-) is in its stationary regime. Then, for every finite horizon $T<\infty$ there exists $\eta_T>0$ such that*

$$\mathbb{E}\Big[\exp\big(\eta\,(N_T^+ + N_T^-)\big)\Big] < \infty \qquad \text{for all } \eta \in [0,\eta_T).$$

Proof. Under $\rho(K)<1$, Brémaud–Massoulié ensure existence of a *stationary* linear Hawkes process. Leblanc (2024) proves (Theorem 1) that there exists $\xi>0$ (depending only on the kernel-mass matrix) such that for every bounded Borel set $B\subset\mathbb{R}$, $\mathbb{E}[\exp(\xi\,N(B))]<\infty$, where $N(B)=N^+(B)+N^-(B)$. Taking $B=[0,T]$ yields the claim with $\eta_T := \xi$. □

Remark 9.4 (Why Lemma 9.3 is the exponential strengthening of Lemma 3.6). Lemma 3.6 provides finite *polynomial* moments on $[0,T]$ (in particular, finite fourth moments), which is enough for the S^2 estimates and Dynkin-type L^2 arguments. For CARA utility, however, the verification step requires *exponential* integrability of the activity, i.e. finiteness of $\mathbb{E}[\exp(\eta(N_T^+ + N_T^-))]$ for some $\eta>0$. Lemma 9.3 supplies exactly this stronger property under the same subcriticality condition, using the sharp exponential-moment theorem of Leblanc (2024).

Assumption 9.5 (Finite control grid in the implemented problem)**.** In the numerical implementation we restrict controls to a finite set $\mathcal{D}_h\subset\mathcal{D}$. Accordingly, $\sup_{\delta\in\mathcal{D}}$ is replaced by $\max_{\delta\in\mathcal{D}_h}$. A deterministic tie-breaking rule yields a measurable maximiser on $\mathcal{D}_h$.

9.4 Algebraic lifting of classical solutions

Corollary 9.6 (Algebraic lifting: reduced HJB to full HJB)**.** *Let Θ be a positive classical solution of* (60)–(54)*. Define v by the CARA ansatz* (53)*. Then v is a classical solution of the full HJB. This is an algebraic equivalence of PDEs for classical solutions; identifying v with the value function requires the verification argument of Section 8.*

Proof. Fix an arbitrary control value $\delta=(\delta^a,\delta^b)\in\mathcal{D}$ and write $z=(s,q,x,\lambda^+,\lambda^-)$. We show that for this fixed δ,

$$\big(\partial_t+\mathcal{L}^\delta\big)v(t,z) = -\,E(t,s,q,x)\,\Big(\partial_t\Theta+\mathcal{G}^\delta\Theta\Big)(t,q,\lambda^+,\lambda^-), \tag{61}$$

where $\mathcal{G}^\delta$ is precisely the reduced operator appearing inside (60). Taking a supremum over δ then yields the full HJB.

Step 1: continuous terms.
Since Θ does not depend on s or x,

$$\partial_t v = -E\,\partial_t\Theta, \qquad \partial_s v = -\gamma q\,v, \qquad \partial_{ss} v = \gamma^2 q^2\,v, \qquad \partial_{\lambda^\pm} v = -E\,\partial_{\lambda^\pm}\Theta.$$

Hence the continuous part of the generator gives

$$\frac{1}{2}\sigma^2\partial_{ss}v-\beta_+(\lambda^+-\mu^+)\partial_{\lambda^+}v-\beta_-(\lambda^--\mu^-)\partial_{\lambda^-}v = -\,E\Big(\frac{1}{2}\sigma^2\gamma^2q^2\,\Theta-\beta_+(\lambda^+-\mu^+)\partial_{\lambda^+}\Theta-\beta_-(\lambda^--\mu^-)\partial_{\lambda^-}\Theta\Big).$$

Step 2: jump terms via ratios.
Using (55), each jump contribution has the form $\Lambda^i\,\mathbb{E}[v(t,z+\Delta z^i)-v(t,z)]$, where the expectation is over $J^\pm$. Under the CARA ansatz, each bracket equals $-E$ times the corresponding bracket for Θ. The ask-fill computation above yields exactly $-E\,\mathcal{J}_a^\delta\Theta$, and the other channels are analogous, producing $-E\,\mathcal{J}_b^\delta\Theta$, $-E\,\mathcal{J}_{+0}^\delta\Theta$, and $-E\,\mathcal{J}_{-0}^\delta\Theta$.

Step 3: conclude.
Collecting Step 1 and Step 2 yields (61) with

$$\mathcal{G}^\delta\Theta := \frac{1}{2}\sigma^2\gamma^2q^2\,\Theta-\beta_+(\lambda^+-\mu^+)\partial_{\lambda^+}\Theta-\beta_-(\lambda^--\mu^-)\partial_{\lambda^-}\Theta+\mathcal{J}_a^\delta\Theta+\mathcal{J}_b^\delta\Theta+\mathcal{J}_{+0}^\delta\Theta+\mathcal{J}_{-0}^\delta\Theta.$$

Taking $\sup_{\delta\in\mathcal{D}}$ in (61) gives the full HJB, and the terminal condition follows by substituting (54) into (53). □

9.5 Integrability for Dynkin and verification under CARA

Lemma 9.7 (Dynkin L^2 condition for the CARA candidate)**.** *Assume: (i) $|Q_t| \le Q_{\max}$ a.s. on $[0,T]$, (ii) $\delta_t \in [0,\delta_{\max}]^2$, (iii) Lemma 9.3 holds; let $\eta_T > 0$ denote the corresponding constant,and (iv) $\int_0^T \mathbb{E}[(\lambda_t^i)^2]\,dt < \infty$ for each jump channel i.*
Let κ be the tick size and q_0 the order size, and set

$$C_{\text{jump}} := q_0(\delta_{\max} + \kappa) + Q_{\max}\kappa.$$

Define the marked-to-market wealth process $\mathcal{W}_t := X_t + Q_t S_t$, and let J_t be the number of jump times of $\mathcal{W}$ up to time t. Then $J_T \le N_T^+ + N_T^-$ a.s., so we may take $c = 1$ in the bound $J_T \le c(N_T^+ + N_T^-)$.
If $4\gamma\, C_{\text{jump}} < \eta_T$, then

$$\sup_{t\le T} \mathbb{E}\Big[e^{4\gamma|\mathcal{W}_t|}\Big] < \infty.$$

Moreover, if Θ is bounded, then $v(t,s,q,x,\lambda) := -e^{-\gamma(x+qs)}\Theta(t,q,\lambda)$ satisfies the Dynkin L^2 condition of Lemma 5.9.

Proof. **Step 1: exponential moment of $\mathcal{W}$.**
Write $\mathcal{W}_t = X_t + Q_t S_t$. Each maker fill changes $\mathcal{W}$ by $\Delta\mathcal{W} = q_0\delta_t^{a/b}$, hence $|\Delta\mathcal{W}| \le q_0\delta_{\max}$. Each midprice move changes $\mathcal{W}$ by $\Delta\mathcal{W} = Q_{t^-}\Delta S$ with $|\Delta S| = \kappa$, hence $|\Delta\mathcal{W}| \le Q_{\max}\kappa$. If a price move coincides with a fill, the cross term $\Delta Q_t\,\Delta S_t$ contributes at most $q_0\kappa$. Therefore every jump of $\mathcal{W}$ has magnitude at most C_{jump}, and

$$|\mathcal{W}_t| \le |\mathcal{W}_0| + \left|\sigma\int_0^t Q_s\,dW_s\right| + C_{\text{jump}}\,J_t,$$

where W is the Brownian motion driving S.
Since $|Q_s| \le Q_{\max}$, conditioning on the predictable process Q yields

$$\mathbb{E}\left[\exp\left(\alpha\sigma\int_0^t Q_s\,dW_s\right)\Big|\,(Q_s)_{s\le t}\right] = \exp\left(\frac{1}{2}\alpha^2\sigma^2\int_0^t Q_s^2\,ds\right) \le \exp\left(\frac{1}{2}\alpha^2\sigma^2 Q_{\max}^2 T\right).$$

Using $e^{a|u|} \le e^{au} + e^{-au}$, we obtain the unconditional bound

$$\mathbb{E}\left[\exp\left(\alpha\left|\sigma\int_0^t Q_s\,dW_s\right|\right)\right] \le 2\exp\left(\frac{1}{2}\alpha^2\sigma^2 Q_{\max}^2 T\right).$$

Next, $\mathcal{W}$ can only jump at order-arrival times, hence $J_T \le N_T^+ + N_T^-$ a.s. By Lemma 9.3, for $\eta = 4\gamma C_{\text{jump}} < \eta_T$ we have

$$\mathbb{E}\Big[e^{\eta(N_T^+ + N_T^-)}\Big] < \infty.$$

Since $J_T \le N_T^+ + N_T^-$ a.s., it follows that

$$\mathbb{E}\big[e^{4\gamma C_{\text{jump}} J_T}\big] \le \mathbb{E}\Big[e^{4\gamma C_{\text{jump}}(N_T^+ + N_T^-)}\Big] < \infty.$$

Combining the above bounds yields $\sup_{t\le T}\mathbb{E}[e^{4\gamma|\mathcal{W}_t|}] < \infty$.

Step 2: Dynkin L^2 condition for the CARA candidate.
Let $v(t,s,q,x,\lambda) = -e^{-\gamma(x+qs)}\Theta(t,q,\lambda)$ with bounded Θ. Along the controlled state, $|v(t,Z_t)| \le \|\Theta\|_\infty e^{\gamma|\mathcal{W}_t|}$, hence $v(t,Z_t)^2 \le \|\Theta\|_\infty^2 e^{2\gamma|\mathcal{W}_t|}$, which is integrable on $[0,T]$ by Step 1.
For the Brownian term, only s diffuses and $\partial_s v = -\gamma q\, v$, so

$$\mathbb{E}\int_0^T |\sigma\partial_s v(t,Z_t)|^2\,dt \le \sigma^2\gamma^2 Q_{\max}^2\int_0^T \mathbb{E}[v(t,Z_t)^2]\,dt < \infty.$$

For the jump part, any jump channel i changes the exponent argument $x + qs$ by a bounded amount. In the worst case, a fill and a one-tick move occur together, producing the cross term $\Delta Q_t\,\Delta S_t$; by bounded inventory and bounded controls, the total change is bounded by C_{jump}. Therefore, for every jump channel i,

$$|v(t,Z_{t^-} + \Gamma_i) - v(t,Z_{t^-})| \le |v(t,Z_{t^-})|\,(e^{\gamma C_{\text{jump}}} + 1),$$

and hence

$$\lambda_t^i \, |v(t, Z_{t^-} + \Gamma_i) - v(t, Z_{t^-})|^2 \le C \, \lambda_t^i \, v(t, Z_{t^-})^2.$$

Using Cauchy–Schwarz and assumption (iv),

$$\mathbb{E} \int_0^T \lambda_t^i \, v(t, Z_{t^-})^2 \, dt \le \left(\int_0^T \mathbb{E}[(\lambda_t^i)^2] \, dt \right)^{1/2} \left(\int_0^T \mathbb{E}[v(t, Z_t)^4] \, dt \right)^{1/2} < \infty,$$

where $\mathbb{E}[v^4] < \infty$ follows from bounded Θ and Step 1. Summing over i gives

$$\mathbb{E} \int_0^T \sum_i \lambda_t^i \, |v(t, Z_{t^-} + \Gamma_i) - v(t, Z_{t^-})|^2 \, dt < \infty,$$

which is exactly the Dynkin L^2 condition of Lemma 5.9 with $\phi = v$. □

Proposition 9.1 (Verification in reduced variables (implemented finite grid))**.** *Assume the hypotheses of Theorem 8.1, Lemma 9.7, and the finite-grid restriction of Assumption 9.5. Let Θ be a bounded classical solution of the reduced HJB where $\sup_{\delta \in \mathcal{D}}$ is replaced by $\max_{\delta \in \mathcal{D}_h}$, with terminal condition (54). Let $\delta^\star(t, q, \lambda) \in \mathcal{D}_h$ be the maximiser selected by the tie-breaking rule, and define $v = -e^{-\gamma(x+qs)}\Theta$ and the feedback control $\delta_t^\star = \delta^\star(t, Q_{t^-}, \lambda_{t^-})$. Then v is the value function of the finite-grid control problem and $\delta^\star$ is optimal for that problem.*

Proof. By Corollary 9.6, v is a classical solution of the corresponding full HJB (with $\max_{\mathcal{D}_h}$). Lemma 9.7 verifies the Dynkin integrability condition of Lemma 5.9 for $\phi = v$. Therefore Theorem 8.1 applies and yields optimality of $\delta^\star$ and $v = V$ for the finite-grid problem. □

Consequently, once a classical solution Θ of (60) is obtained (numerically in Section 10), the function v defined by (53) is a classical HJB candidate for the full problem, and the implemented feedback maximiser on $\mathcal{D}_h$ is optimal for the discretised control problem by Proposition 9.1.

10 Empirical pipeline and implementation conventions

This section documents the empirical-to-model pipeline *as implemented* in the accompanying codebase. Its purpose is operational rather than theoretical: it makes explicit the **data transformations**, **calibration procedures**, and **model-validation diagnostics** that map raw LOBSTER messages into (i) calibrated point-process dynamics and (ii) an executable market-making policy evaluated in a backtesting loop.
The implementation is a substantial component of this thesis: it is the mechanism that ensures that (a) the mathematical model is *identifiable from data*, (b) the numerical HJB layer is solved on grids consistent with the observed tick structure, and (c) the resulting policy can be stress-tested in both replay and simulation regimes with sign-consistent bookkeeping.
Throughout, time is measured in seconds and prices live on the native tick grid of the dataset. All transformations are deterministic once an episode window and preprocessing options are fixed.

Implementation map (modules). For readability and reproducibility, the codebase is modularized as follows:

- `OrderBook.py`: LOBSTER ingestion, episode slicing, and event extraction utilities;
- `event_cleaning.py`: tie handling and deterministic timestamp preprocessing;
- `HawkesCalibrator.py`: exponential-kernel Hawkes MLE calibration (windowed fits, stability checks);
- `microstructure_calibration.py`: reduced-form fill proxy and event-level marking (impact) calibration;
- `hjb_theta_solver.py`: IMEX HJB solver on tick-consistent grids and policy export;
- `control_frame.py`: executable policy wrapper and backtesting interfaces;
- `diagnostics.py`, `PointProcessPlotSuite.py`, `thesis_figures_v2.py`: figure generation and validation plots saved into `Images/`.

10.1 Episode definition and time origin

An *episode* is a fixed intraday window

$$[t_{\text{start}},\, t_{\text{start}} + T],$$

selected from a single trading day. All event timestamps are shifted to the episode clock

$$\tau := t - t_{\text{start}} \in [0, T].$$

This shift is purely representational (it preserves event ordering), but it is essential for: (i) aligning calibration and control on the same finite horizon, and (ii) making diagnostics comparable across episodes.

10.2 Event extraction: a LOBSTER-consistent event dictionary

We represent the LOBSTER message stream as a multivariate point process. The raw message data are assumed to contain at minimum:

$$\texttt{time} \in \mathbb{R}_+, \qquad \texttt{lob_action} \in \{1,2,3,4,5\}, \qquad \texttt{lob_dir} \in \{+1,-1\}.$$

We follow the standard LOBSTER conventions:

- `lob_action`=1: limit order submission;
- `lob_action`$\in \{2,3\}$: cancellation / deletion (both treated as liquidity removal);
- `lob_action`$\in \{4,5\}$: execution events (marketable order removes standing liquidity);
- `lob_dir`=+1 refers to the *bid* side (a buy limit order is placed/removed/executed);
- `lob_dir`=-1 refers to the *ask* side (a sell limit order is placed/removed/executed).

Direction convention. LOBSTER provides a side indicator `lob_dir` $\in \{+1,-1\}$. In this thesis we adopt the *standing-side* interpretation used by the implementation: for executions, `lob_dir` is treated as the side of the resting limit order that is removed by the trade. Consequently, `lob_dir`=+1 corresponds to a standing bid being executed (sell aggressor), and `lob_dir`=-1 corresponds to a standing ask being executed (buy aggressor). This mapping is enforced end-to-end (calibration → control → P&L) by `OrderBook.to_hawkes_events_10d_robust`. *Remark:* if an alternative feed encodes aggressor-side direction, the mapping between (E_a, E_b) and (N^+, N^-) must be swapped.

Event types. Event times are stored as a dictionary of sorted arrays. The core components are:

$$\begin{aligned}
L_b &: \{\texttt{lob_action} = 1,\ \texttt{lob_dir} = +1\}, & C_b &: \{\texttt{lob_action} \in \{2,3\},\ \texttt{lob_dir} = +1\},\\
E_b &: \{\texttt{lob_action} \in \{4,5\},\ \texttt{lob_dir} = +1\}, & HE_b &: \{\texttt{lob_action} = 5,\ \texttt{lob_dir} = +1\},\\
L_a &: \{\texttt{lob_action} = 1,\ \texttt{lob_dir} = -1\}, & C_a &: \{\texttt{lob_action} \in \{2,3\},\ \texttt{lob_dir} = -1\},\\
E_a &: \{\texttt{lob_action} \in \{4,5\},\ \texttt{lob_dir} = -1\}, & HE_a &: \{\texttt{lob_action} = 5,\ \texttt{lob_dir} = -1\}.
\end{aligned}$$

For compatibility with alternative notations, the implementation also exposes $D_b := C_b$ and $D_a := C_a$ as aliases (not distinct data streams).

Tie handling and deterministic ordering. LOBSTER timestamps can contain duplicates. Exponential-kernel Hawkes recursions require a non-decreasing merged timeline; some numerical routines benefit from strict monotonicity. The pipeline can optionally:

1. collapse exact duplicates within each component, and
2. apply a deterministic global tie-break by adding $k\varepsilon$ within each same-time group ordered by component index (`event_cleaning.prepare_hawkes_events`).

This is a numerical device: it does not encode economic information and only perturbs timestamps at an artificial scale ε.

This deterministic perturbation only affects events that are exactly simultaneous in the input timestamp resolution and leaves all order statistics and counts unchanged; it is used solely to avoid undefined terms in recursions that assume $t_{k+1} > t_k$.

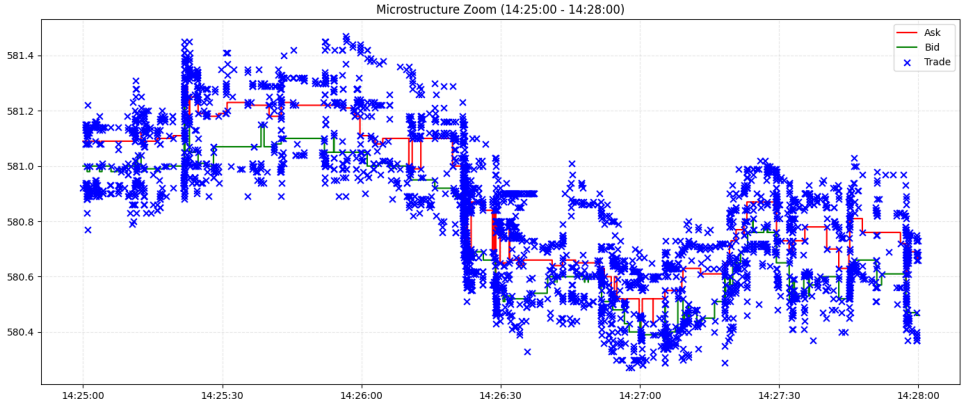


Figure 3: Micro-zoom of the episode, illustrating the alignment between midprice moves and execution timestamps.

10.3 Reduction to a 2D aggressive execution flow

The control layer does not attempt to model the full 8D/10D message taxonomy. Instead, it uses the **aggressive execution flow** (market orders) on each side as the primitive source of order-flow risk.
Using the standing-side convention, define

$$N^+ := N^{E_a}, \qquad N^- := N^{E_b},$$

so that:

- N^+ counts *buy* market orders (they execute standing asks, hence come from E_a);
- N^- counts *sell* market orders (they execute standing bids, hence come from E_b).

Interpretation. This reduction is a modelling choice that targets the dominant control-relevant channel: aggressive flow drives (i) the market maker's fill opportunities and (ii) short-horizon directional pressure. Other message types remain available for diagnostics and feature engineering, but are not promoted to HJB state variables.

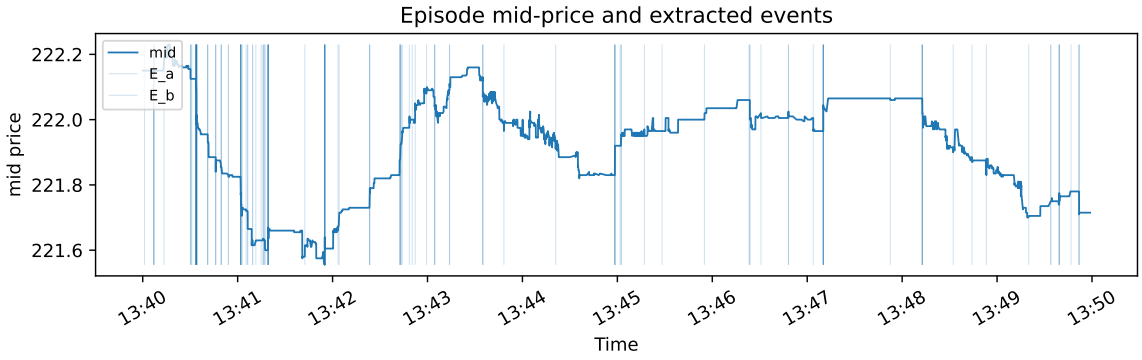


Figure 4: Episode-level view of the midprice and detected aggressive execution events (buy/sell aggressors).

10.4 Empirical evidence supporting the modelling reductions

The two most consequential empirical modelling choices are: (i) representing aggressive flow as a self-exciting point process (Hawkes rather than Poisson), and (ii) focusing the control state on the 2D execution streams (N^+, N^-).

Clustering across time scales (Hawkes motivation). A Poisson process with constant intensity has *Fano factor* equal to 1 for counts in fixed bins. Let C_Δ denote the event count in bins of width Δ and define

$$\mathrm{Fano}(\Delta) := \frac{\mathrm{Var}(C_\Delta)}{\mathbb{E}[C_\Delta]}.$$

Empirically, Fano(Δ) $\gg 1$ and increases with Δ, indicating overdispersion and burstiness, consistent with clustering dynamics.

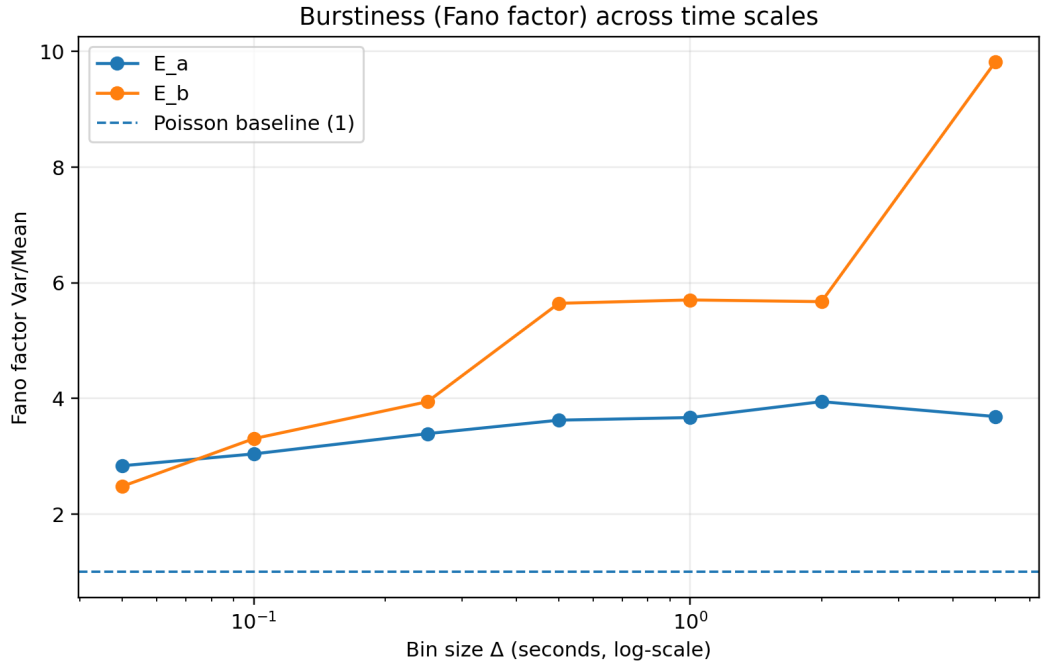


Figure 5: Burstiness across time scales: Fano factor for execution streams. Values above the Poisson baseline (= 1) indicate overdispersion (clustering).

Why executions for control (2D reduction check). To support the choice of aggressive executions as control-relevant primitives, we measure an *event-type lift*. For a fixed horizon τ, define

$$\text{lift(type)} := \frac{\mathbb{P}(\text{one-tick mid move within } [t, t+\tau] \mid \text{event of that type at } t)}{\mathbb{P}(\text{one-tick mid move within } [t, t+\tau])}.$$

A lift above 1 means that conditioning on that event type increases the short-horizon move propensity relative to the baseline. This diagnostic is not a causal statement: it is a pragmatic check that the reduced event state is statistically aligned with short-horizon price pressure on the time scales relevant for quoting. For interpretability, the baseline probability is computed using the same sampling scheme as the conditional probabilities (event-time conditioning vs time-grid conditioning is discussed in the text).

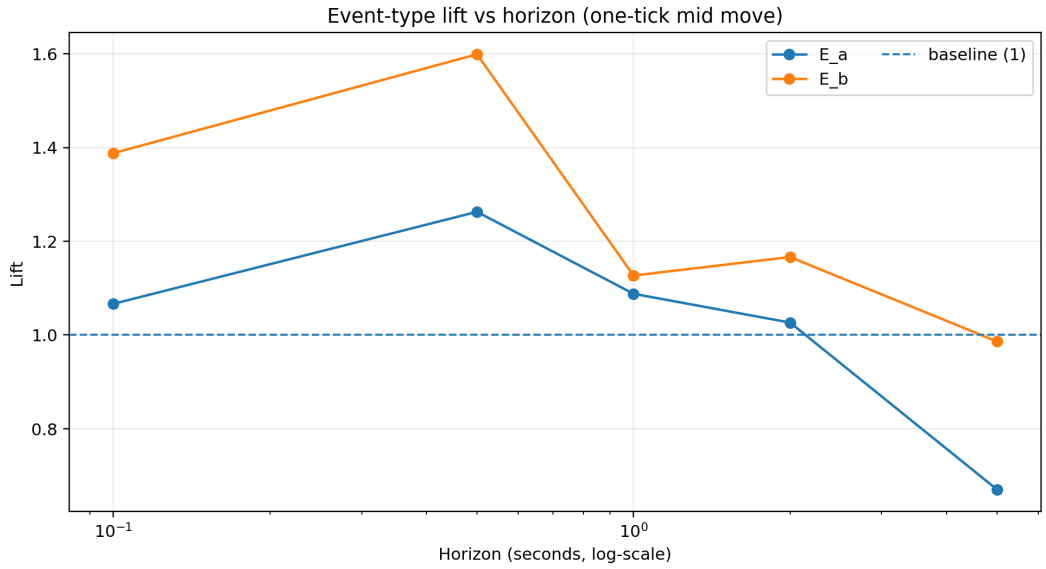
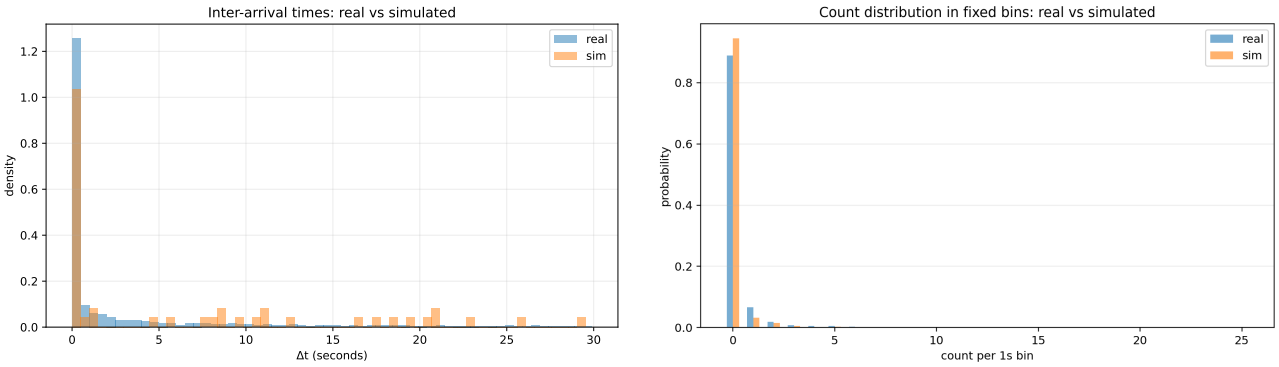


Figure 6: Event-type lift: probability of observing a one-tick mid move within a short horizon after an event, normalized by the unconditional baseline. (This figure must be computed using the *full* pre-reduction event dictionary to avoid zero-count components.)

Simulation sanity checks (real vs simulated stylized facts). As the simulator/backtester generates event streams from the fitted Hawkes model, we validate that simulated episodes reproduce basic distributional signatures of the data at the marginal level: inter-arrival times and counts in fixed bins. These diagnostics are intentionally low-level: they are designed to catch gross miscalibration, near-critical blow-ups, or sign/clock inconsistencies before policy evaluation.



(a) Inter-arrival times: data vs simulation.

(b) Counts per fixed bin: data vs simulation.

Figure 7: Distributional sanity checks for the event simulator (marginal comparisons).

Parameter stability across intraday windows (robustness). To assess whether the fitted Hawkes parameters are stable across nearby windows (and not a single-window artefact), we track fitted parameters and/or branching summaries over rolling calibration windows.

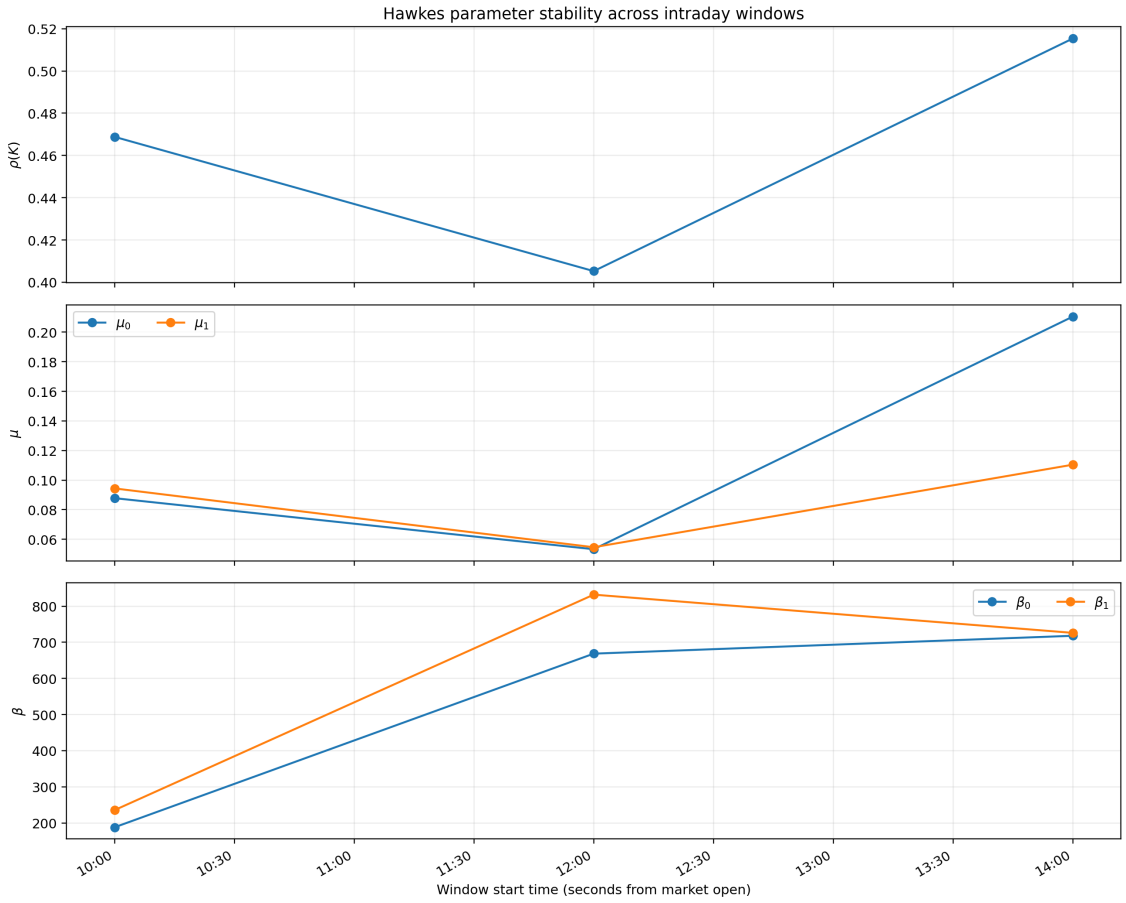


Figure 8: Stability over rolling calibration windows: fitted Hawkes summaries across intraday subwindows.

10.5 Hawkes calibration on intraday windows

We fit a bivariate Hawkes process with exponential kernels to (N^+, N^-) on fixed intraday windows. This choice reflects a control-driven trade-off: exponential kernels restrict the response shape, but they yield a finite-dimensional Markovian intensity state used directly in the HJB.
Let $c_k \in \{+, -\}$ be the component of the k-th event and t_k its episode time. The model is:

$$\lambda_i(t) = \mu_i + \sum_{j \in \{+,-\}} A_{ij} \sum_{t_k^j < t} e^{-\beta_i (t - t_k^j)}, \qquad i \in \{+, -\},$$

with $\mu_i \geq 0$, $A_{ij} \geq 0$, $\beta_i > 0$. We use **source-specific decay** β_i (so that thw Markow embedding uses two-dimensional state

$$(\lambda^+, \lambda^-)$$

.
The branching matrix is

$$K_{ij} := \int_0^\infty A_{ij} e^{-\beta_i u} \, du = \frac{A_{ij}}{\beta_i}., \qquad \rho(K) < 1.$$

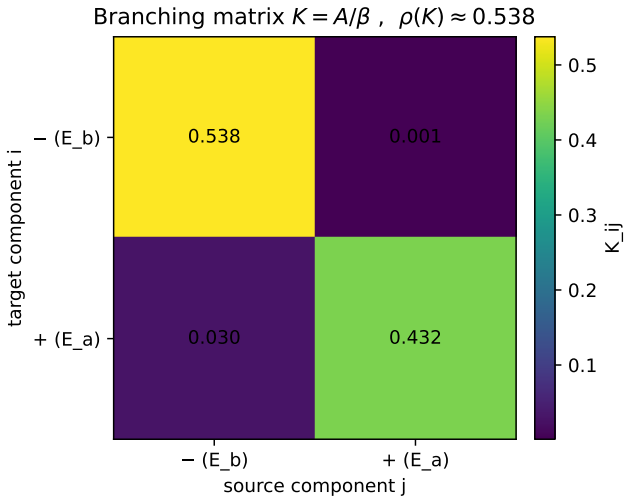


Figure 9: Estimated branching matrix K (endogeneity/cross-excitation summary).

Likelihood (used for estimation). On $[0, T]$ the log-likelihood is

$$\log \mathcal{L}(\theta) = \sum_{k=1}^{n} \log \lambda_{c_k}(t_k) \; - \sum_{i \in \{+,-\}} \int_0^T \lambda_i(t)\, dt,$$

and the exponential kernel yields the closed form

$$\int_0^T \lambda_i(t)\, dt = \mu_i T + \sum_j A_{ij} \sum_{t_k^j < T} \frac{1 - e^{-\beta_i (T - t_k^j)}}{\beta_i}.$$

Stability enforcement (estimation vs simulation). During fitting we do *not* project parameters onto the stable set by rescaling A. Instead, the optimizer is constrained by a hard barrier: if $\rho(K) \geq \rho_{\max}$, the objective is penalized by a large barrier term (equivalently, the candidate is rejected).
Consequently, estimates with $\rho(K)$ close to $\rho_{\max}$ are interpreted as "near-critical fits" rather than artefacts of projection.
When simulating from the fitted model, the code can optionally rescale A to a stricter target $\rho_{\text{sim}} < \rho_{\max}$ (default $\rho_{\text{sim}} = 0.9$) to avoid near-critical long-memory artefacts on finite horizons:

$$A_{\text{used}} = \begin{cases} A, & \rho(K) \leq \rho_{\text{sim}}, \\ A \cdot \dfrac{\rho_{\text{sim}}}{\rho(K)}, & \rho(K) > \rho_{\text{sim}}. \end{cases}$$

Optional intraday deseasonalization (choice). The wrapper `LOBHawkes.fit_windows` can estimate a piecewise-constant activity profile $s(t)$ with mean 1 and apply a time-change $S(t) = \int_0^t s(u)\, du$ before fitting. In the baseline experiments shown here we set `deseasonalize=False` to keep the intensity state interpretable in physical time.

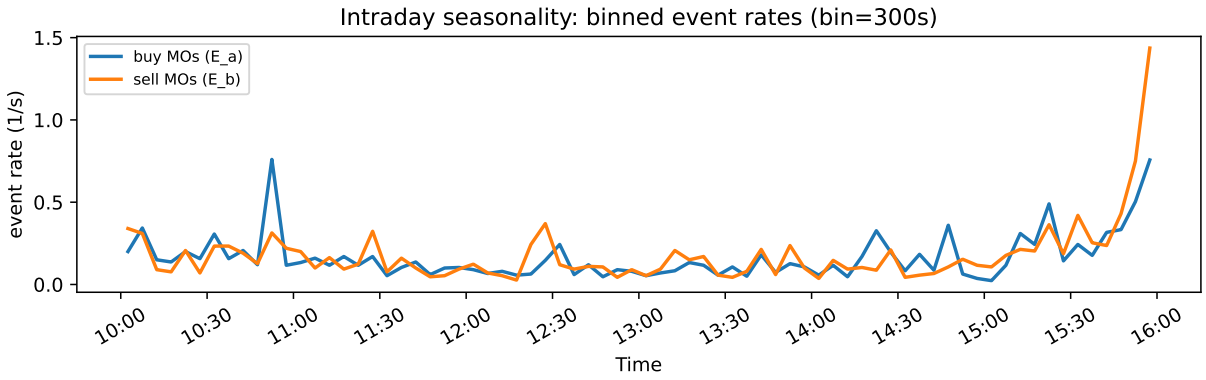


Figure 10: Hawkes diagnostics: endogeneity (K) and intraday activity profile.

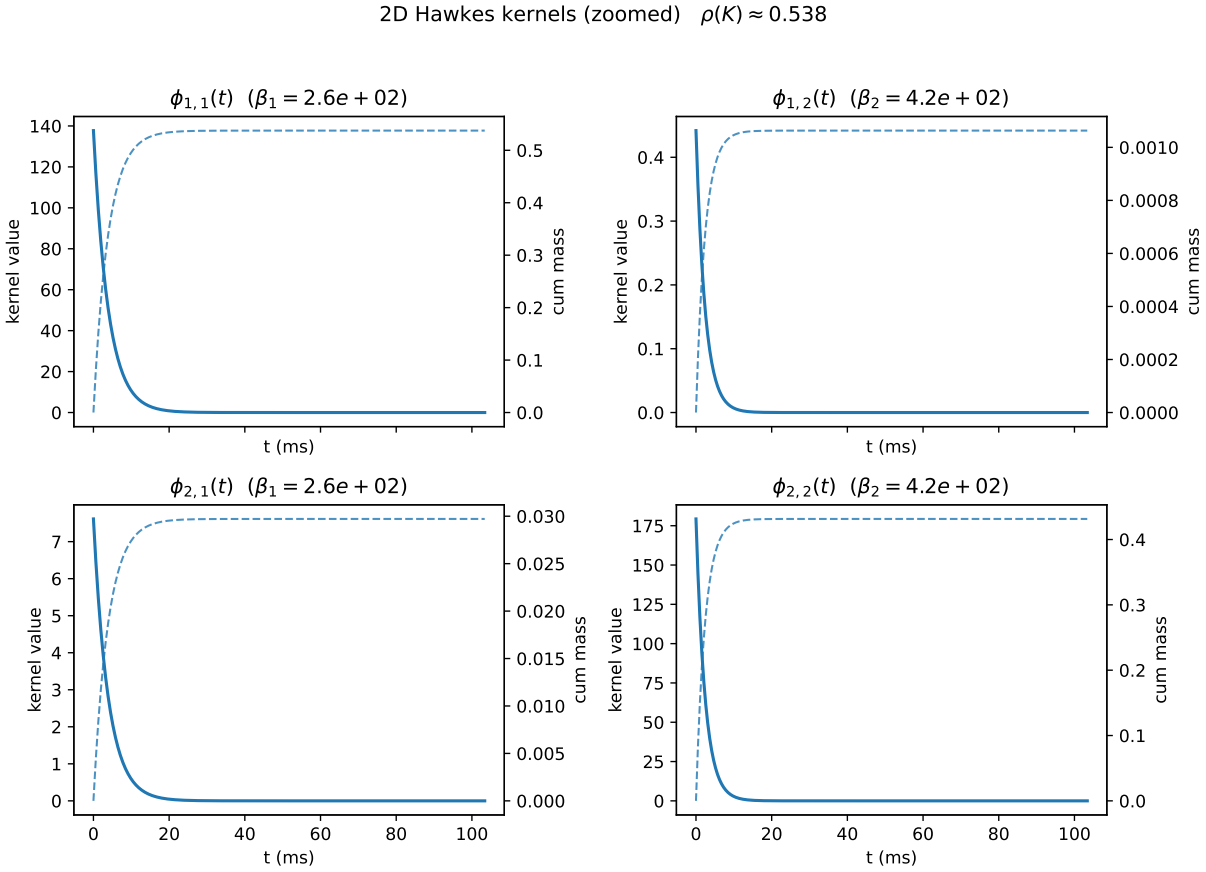


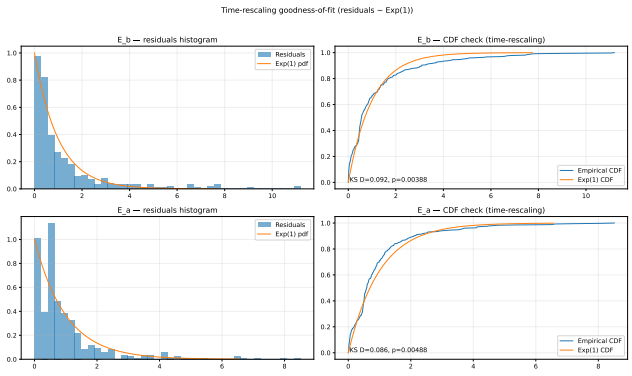
Figure 11: Estimated exponential responses and decay scales for the 2D execution Hawkes model.

Estimated kernel shapes and reaction time scales. Figure 11 visualizes the fitted exponential kernels $\phi_{ij}(t) = A_{ij}e^{-\beta_i t}$ and their cumulative mass. Because the exponential form has a one-parameter decay, the model implies a *characteristic memory scale* of order $1/\beta_i$ for each source component j. Empirically, the fitted responses concentrate their mass on short horizons: the cumulative curves saturate quickly, indicating that most self- and cross-excitation effects are realized within a few milliseconds[1]. This is consistent with a microstructure interpretation where aggressive flow arrives in bursts, and intensities revert rapidly after a cluster.

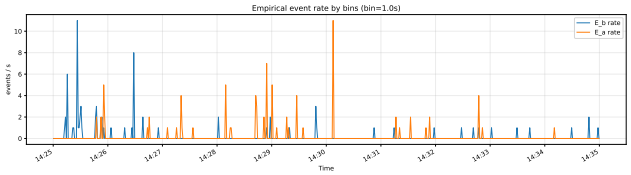
Beyond visual intuition, the kernel masses summarize endogeneity through the branching matrix $K_{ij} = A_{ij}/\beta_i$. The diagonal terms quantify self-excitation (same-side clustering), while off-diagonals quantify cross-excitation (spillover from buy to sell pressure and vice versa). This decomposition is operationally important for control:

[1]For visualization we display t in milliseconds; the underlying calibration uses seconds throughout.

the HJB state (λ^+, λ^-) responds differently to symmetric bursts (both sides excited) versus imbalanced bursts (one side dominates), and the optimal quotes react accordingly.



(a) Time-rescaling diagnostics (Ogata transform).



(b) Binned event rates: data vs fitted intensity.

Figure 12: Goodness-of-fit diagnostics for the fitted Hawkes model. Under correct specification, the time-rescaled inter-arrival times are i.i.d. Exp(1) and pass standard QQ/KS checks.

Goodness-of-fit diagnostics and what they do (and do not) guarantee. Figure 12 reports two complementary checks. First, the Ogata time-rescaling theorem implies that under correct specification the transformed inter-arrival times should be i.i.d. Exp(1). Accordingly, the QQ/KS-style diagnostics in Figure 12(a) are interpreted as a *calibration sanity check*: major deviations would signal misspecification (e.g. wrong kernel shape), numerical instability (e.g. near-critical fits), or implementation issues (e.g. timestamp tie handling). Second, the binned rate comparison in Figure 12(b) checks whether the fitted intensity tracks the observed activity at coarse time resolution.
These diagnostics are intentionally low-level: passing them does not imply the Hawkes model is a structural truth of the market, but it does indicate that the fitted process captures the *first-order* timing regularities that matter for the downstream simulation/control loop. In particular, it reduces the risk that policy behaviour is driven by a broken event clock rather than genuine state-dependence.

10.6 Reduced microstructure calibration

The simulator/backtester requires two episode-calibrated reduced-form ingredients:

1. a **fill-proxy curve** as a function of quote distance to mid, and
2. an **event-level marking probability** that an aggressive order triggers a one-tick mid move.

Both are designed to be (i) cheap to estimate, (ii) stable on short windows, and (iii) aligned with the control parametrization (quotes expressed as offsets from mid in ticks).

(i) Fill model via level-by-level consumption (proxy, not queue position). For each execution message (`lob_action`$\in \{4, 5\}$) we observe midprice m, execution price p_{exec}, and marketable size V (`lob_size`). The code infers the aggressor side via a touch-consistency rule: buy market order if $p_{\text{exec}} \geq a_1 - \varepsilon$, sell market order if $p_{\text{exec}} \leq b_1 + \varepsilon$.
For a buy market order we scan ask levels $\ell = 1, \ldots, L$ and construct samples

$$\delta_\ell := a_\ell - m \geq 0, \qquad \widehat{p}_\ell := \text{clip}\left(\frac{V - \sum_{r<\ell} q_r^{ask}}{q_\ell^{ask}}, 0, 1\right),$$

where q_ℓ^{ask} is displayed depth at level ℓ (sell market orders are handled symmetrically). The quantity $\widehat{p}_\ell$ is a *consumption fraction proxy*; it is not interpreted as a literal queue-position fill probability.
To align with quoting offsets, define a "touch distance" δ_0 as the median of level-1 deltas and set

$$d := \max(\delta - \delta_0, 0).$$

We then fit by least squares

$$f(\delta) = A_{\text{fill}} e^{-k_{\text{fill}}\, d}, \qquad 0 < A_{\text{fill}} \leq 1,\ k_{\text{fill}} > 0,$$

implemented in `microstructure_calibration.calibrate_fill_probability`.

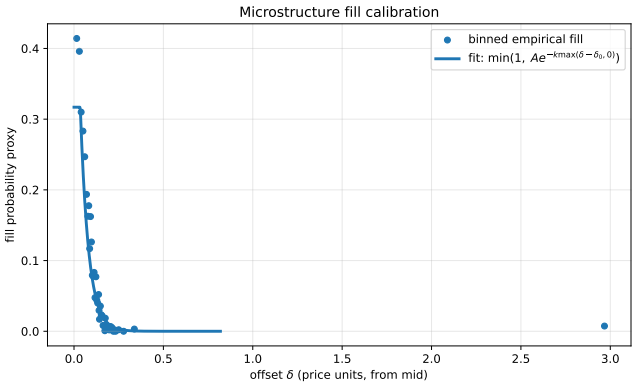


Figure 13: Fill-proxy calibration: consumption-fraction samples $(\delta, \widehat{p})$ and fitted curve $f(\delta)$.

(ii) One-tick marking (impact) model at event level. Using only executions, define aggressor sign

$$s := -\,\texttt{lob_dir} \in \{+1, -1\},$$

so $s = +1$ for buy market orders (hit ask) and $s = -1$ for sell market orders (hit bid). Let $\rho \in [-1, 1]$ be a contemporaneous feature (preferably multi-level imbalance `rho_5L`).
For a fixed horizon $h \geq 1$ measured in *event rows*, define $\Delta_h m := m_{t+h} - m_t$ and label

$$y := \mathbf{1}\Big\{(s = +1 \;\&\; \Delta_h m \geq \text{tick}) \;\text{ or }\; (s = -1 \;\&\; \Delta_h m \leq -\text{tick})\Big\}.$$

We fit by MLE the logistic form

$$\mathbb{P}(y = 1 \mid \rho, s) = \frac{\bar{p}}{1 + \exp(-s\, k_{\text{imp}}(\rho - \theta_{\text{imp}}))}, \qquad 0 < \bar{p} < 1,\; k_{\text{imp}} \geq 0,\; \theta_{\text{imp}} \in [-1, 1],$$

implemented in `microstructure_calibration.calibrate_price_impact` with robust fallbacks when identification is weak.

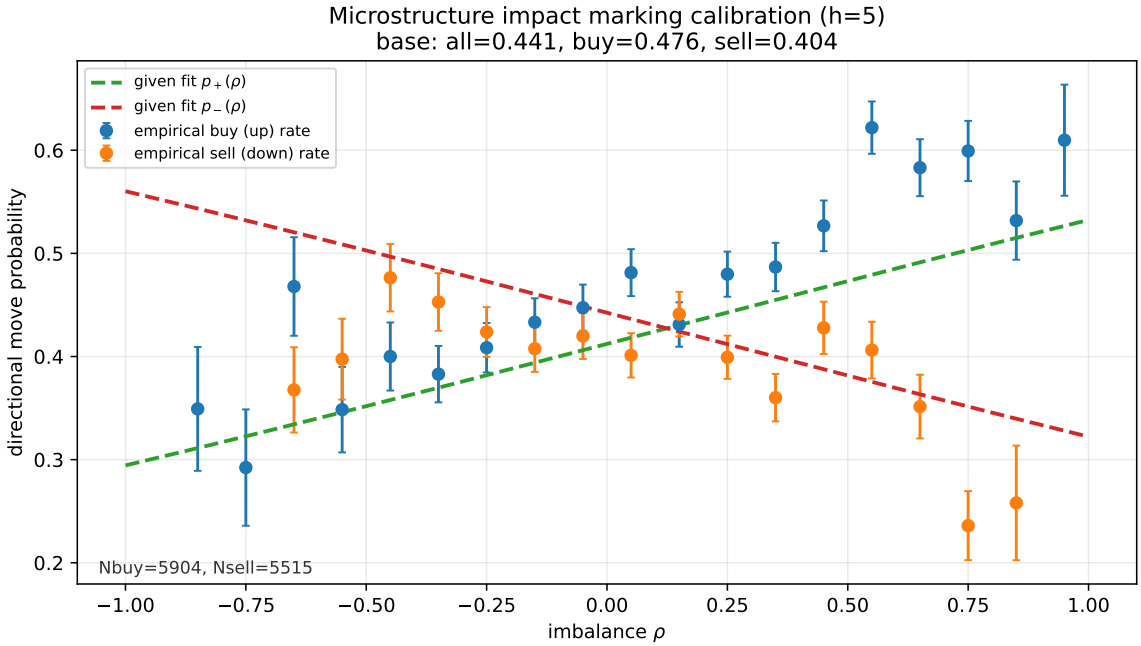


Figure 14: Event-level marking calibration: directional mid-move labels vs feature ρ, and fitted logistic form.

Scope. These submodels are deliberately reduced-form: they translate quoting distance into (i) a fill-rate proxy and (ii) a short-horizon adverse-move probability, without modelling latent queue position. They should

be read as *conditional proxies* estimated from observed executions under the historical market environment, not as structural queue-based execution probabilities; in particular, selection effects are possible because we only observe fills when trades occur. The purpose here is pragmatic: obtain stable mappings that preserve the intensity-based control structure and keep the state dimension aligned with the HJB formulation.

10.7 Numerical HJB solution: discretization choices that match the data

The control problem is solved in the reduced Θ representation. The numerical layer is designed to match the empirical constraints of the dataset:

- **State variables:** inventory q on a bounded grid and Hawkes intensities (λ^+, λ^-) on bounded grids centered around fitted stationary scales;
- **Tick-consistent controls:** quote offsets discretized on the empirical grid (step = `quote_tick`), so the HJB maximization is a finite search;
- **Time stepping:** an IMEX scheme stabilizes stiff intensity drifts while keeping jump/control terms explicit;
- **Interpolation:** post-jump intensity values are interpolated and clamped to grid bounds to avoid out-of-domain evaluation.

The solver exports an executable policy object mapping $(t, q, \lambda^+, \lambda^-) \mapsto (\delta_a^\star, \delta_b^\star)$, evaluated off-grid by deterministic interpolation.

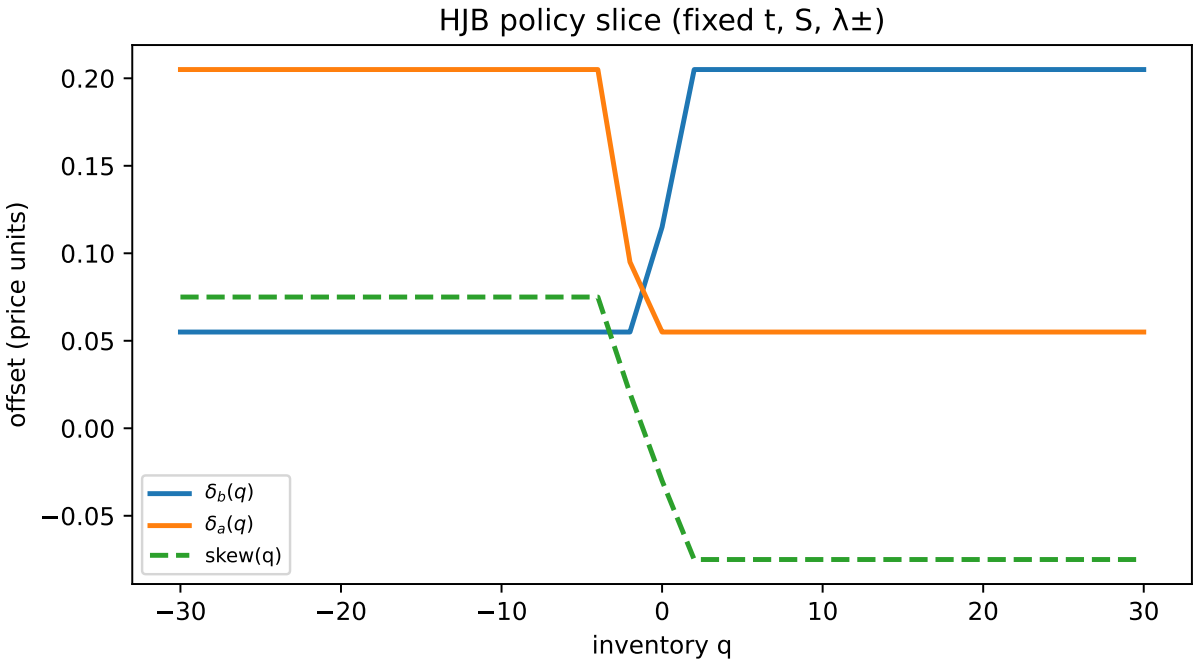


Figure 15: Optimal quote offsets produced by the HJB policy on representative state slices.

10.8 Backtesting modes and sign-consistent bookkeeping

The backtester supports two evaluation modes:

- **Replay mode:** the midprice path is taken from data and the policy is evaluated on the realized trajectory, isolating execution mechanics from price-model mismatch;
- **Simulation mode:** events are generated by the fitted Hawkes model, fills by thinning using $f(\delta)$, and price moves by Bernoulli marking using the fitted impact model.

Event ordering. When a single order-flow event carries both an execution outcome and a price-move mark, cashflows are computed using pre-jump values (e.g. S_{t^-}) and the midprice update is applied afterwards. This matches the predictable-control convention and avoids look-ahead bias.

Remark 10.1. In replay mode, the midprice path S_t is treated as exogenous and fixed by the data: the policy selects offsets using $(q, \lambda^+, \lambda^-)$, quotes are centered on the historical S_t, and P&L is marked-to-mid along the realized trajectory.

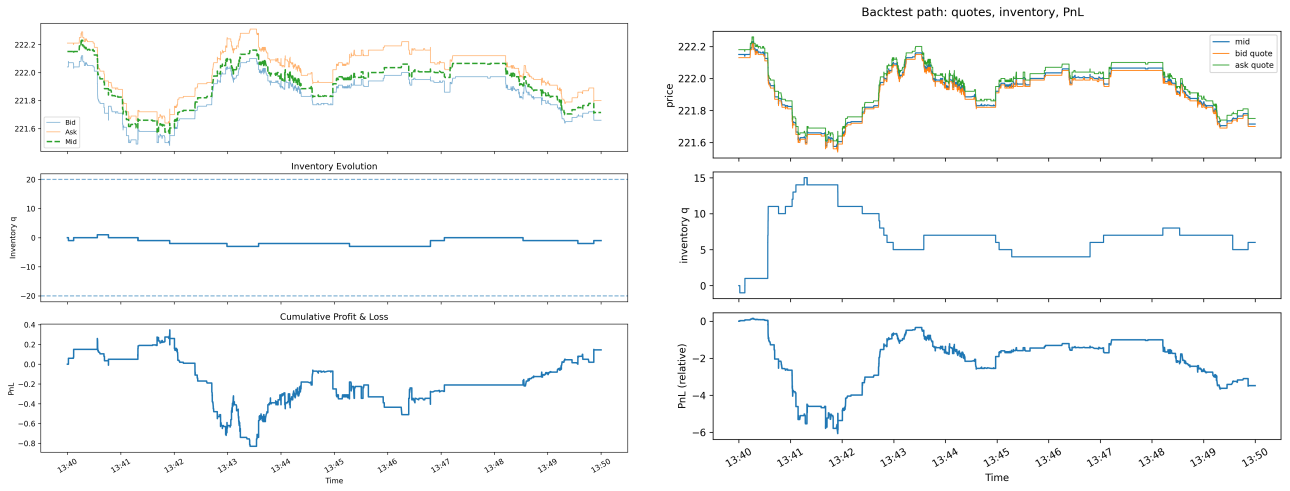
Why replay vs simulation matters. Replay mode is the most conservative evaluation: it keeps the realized midprice path fixed and asks whether the policy can extract spread while controlling inventory under realistic price moves, without benefiting from a self-generated simulator. Simulation mode is the most *model-consistent* evaluation: it tests whether the policy behaves as intended inside the fitted Hawkes + microstructure proxy environment, and it enables controlled stress tests (e.g. varying endogeneity via $\rho(K)$, or changing the fill/impact parameters). Using both is essential: a strategy that looks strong in simulation may simply be overfitting the simulator, while a strategy that fails in replay mode may be exposing model mismatch rather than a control failure.

Single sign convention across layers (calibration → control → P&L). The implementation uses one consistent mapping:

- N^+ counts *buy* market orders (hit ask); if the market maker is executed, this is a *sell* fill.
- N^- counts *sell* market orders (hit bid); if the market maker is executed, this is a *buy* fill.

This is enforced explicitly because sign inconsistencies are a common silent failure mode in market-making implementations.

Benchmark policy (fixed skew around mid). To contextualize the HJB policy, we compare it to a simple baseline that quotes at fixed offsets around the midprice ("FixedSkew"): the bid and ask distances are constants (in ticks) possibly with a static inventory tilt. This benchmark is deliberately crude: it does not react to the Hawkes state (λ^+, λ^-) and therefore cannot adapt to transient bursts or intensity imbalance. The purpose of the comparison is not to claim universal dominance, but to isolate the incremental value of *state-dependent* quoting driven by self-exciting order flow.
Concretely, FixedSkew quotes $\delta_b(t) \equiv \bar{\delta}_b$ and $\delta_a(t) \equiv \bar{\delta}_a$ (optionally with a static inventory tilt $\pm\eta q_t$), without dependence on (λ^+, λ^-).



(a) HJB policy path: mid, quotes, inventory, and P&L.

(b) FixedSkew benchmark path on the same episode.

Figure 16: Replay-mode path diagnostics: state evolution and realized P&L under HJB vs a static quoting benchmark.

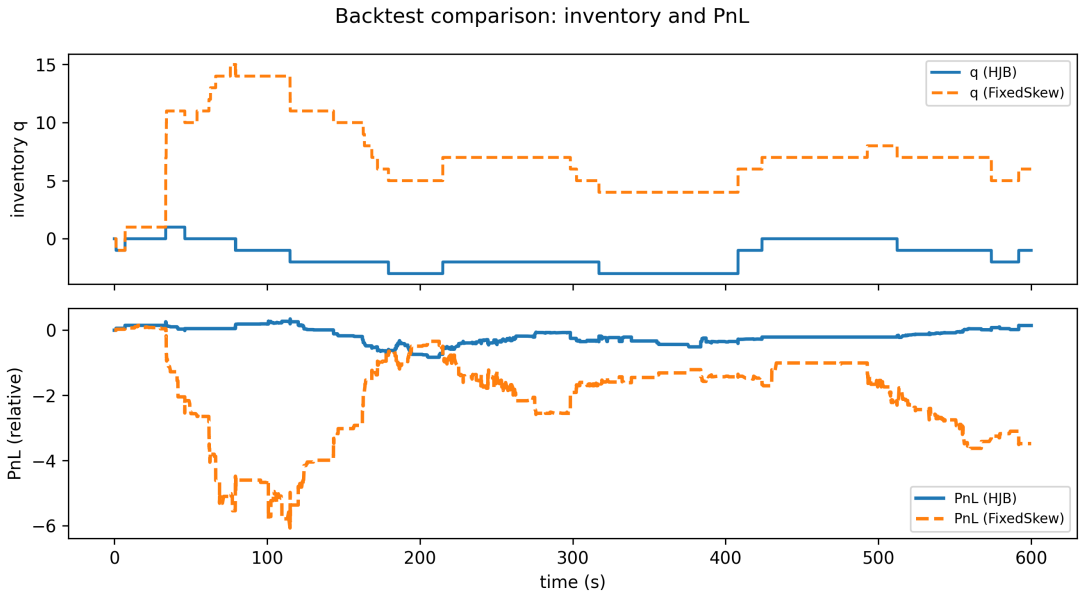


Figure 17: Direct comparison: inventory trajectories and cumulative P&L for HJB vs FixedSkew on the same episode.

State dependence shows up as inventory control and asymmetry in quote placement. Figures 16–17 make the qualitative difference explicit. The FixedSkew benchmark is mechanically consistent but insensitive: it tends to accumulate inventory when the realized flow is persistently one-sided, because it does not adjust to intensity imbalance. In contrast, the HJB policy reacts to the Hawkes state by widening/protecting on the side exposed to short-horizon adverse selection, while leaning quotes to encourage rebalancing when inventory is large. This is precisely the mechanism the control problem is designed to encode.

Bookkeeping identity (self-financing decomposition). Let X_t denote cash and q_t inventory (positive = long). Let m_t be the midprice on the episode clock, and let trades occur at prices p_k with signed size Δq_k (positive for buys, negative for sells). The self-financing update is

$$X_{t_k} = X_{t_k^-} - p_k\,\Delta q_k, \qquad q_{t_k} = q_{t_k^-} + \Delta q_k.$$

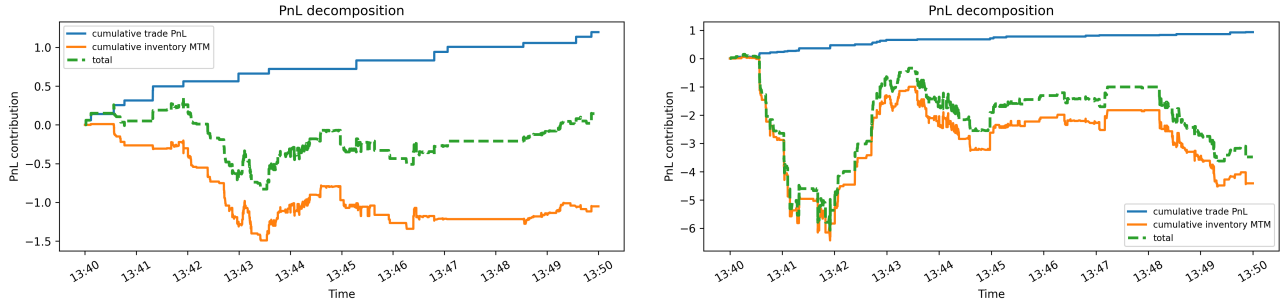
The marked-to-mid equity ("total P&L") is

$$\Pi_t := X_t + q_t m_t.$$

This yields the standard decomposition

$$\Pi_t - \Pi_0 = \underbrace{\sum_{k:\,t_k \le t} (m_{t_k} - p_k)\,\Delta q_k}_{\text{trade P\&L (spread capture / execution)}} + \underbrace{\int_0^t q_{s^-}\,dm_s}_{\text{inventory MTM (price exposure)}}.$$

In the discrete replay implementation, the second term is computed as the cumulative sum of q times midprice increments, and it is the channel through which adverse selection materializes as drawdowns.



(a) HJB: trade P&L vs inventory MTM contribution.

(b) FixedSkew: trade P&L vs inventory MTM contribution.

Figure 18: P&L decomposition: spread capture (trade P&L) versus mark-to-mid inventory component (adverse selection / drift exposure).

P&L decomposition clarifies where performance is coming from. Figure 18 separates the two dominant contributions: (i) trade P&L (spread capture plus any execution-price effects encoded in the fill proxy), and (ii) inventory mark-to-mid (exposure to midprice moves while holding inventory). In episodes with directional moves, a strategy can appear profitable on trade P&L while still losing overall due to inventory MTM drawdowns. The decomposition therefore functions as a diagnostic for *adverse selection versus spread capture*: when MTM dominates, the strategy is effectively being "run over" by short-horizon price pressure, and the natural remedy is more protection (larger offsets / stronger skew / tighter inventory penalties).

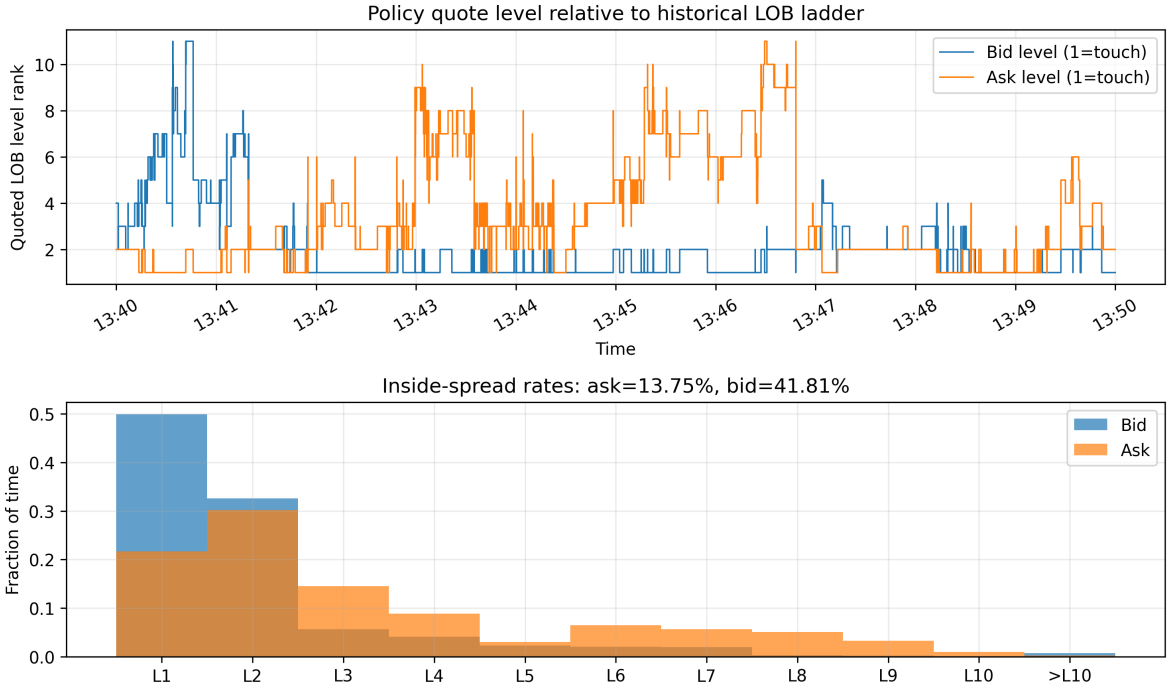


Figure 19: Quote placement relative to the historical LOB ladder: quoted level rank (1 = touch) and inside-spread occupancy over time.

Quote levels, aggressiveness, and execution risk. Figure 19 translates the control into an interpretable microstructure object: where the policy places quotes relative to the realized book ladder. Time spent at level 1 (touch) corresponds to high execution probability but also higher adverse selection risk, while deeper levels trade off fill rate for protection. The reported inside-spread occupancy rates quantify how often the strategy steps inside the spread; this is a useful empirical proxy for "aggressiveness" and should be read jointly with the P&L decomposition: inside-spread quoting can increase trade P&L in calm periods but amplify MTM losses during directional bursts.

Quote level rank. Let $b_\ell(t)$ (descending in ℓ) and $a_\ell(t)$ (ascending in ℓ) denote the historical bid/ask ladders. For a bid quote $\hat{b}(t)$, define the bid rank as the smallest ℓ such that $\hat{b}(t) \geq b_\ell(t)$. For an ask quote $\hat{a}(t)$,

define the ask rank as the smallest ℓ such that $\hat{a}(t) \leq a_\ell(t)$. Thus rank 1 corresponds to quoting at the touch; larger ranks correspond to deeper placement.

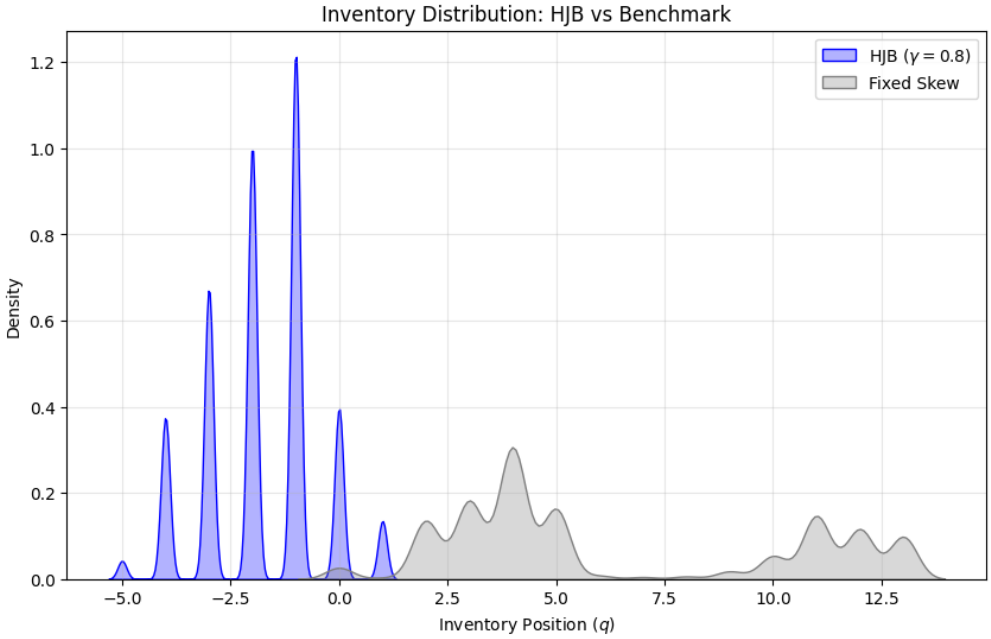


Figure 20: Inventory distribution over the episode: HJB policy versus FixedSkew benchmark.

Inventory placement as a risk signature. Figure 20 summarizes a central control objective: keeping inventory concentrated near zero. Compared to FixedSkew, the HJB policy produces a tighter inventory distribution (more mass near $q = 0$), reflecting state-dependent skewing that actively offsets persistent one-sided flow. This matters because inventory is the channel through which adverse selection is monetized into MTM drawdowns; therefore, the inventory distribution is not just a risk metric but a structural diagnostic of whether the Hawkes state is being exploited for protection rather than only for opportunistic spread capture.

10.9 End-to-end pipeline: what is fixed, what is estimated, what is assumed

Given a fixed episode window and preprocessing options, the pipeline is:

1. **Data → events:** map LOBSTER message codes to the multivariate event dictionary and shift to $\tau \in [0, T]$ (section 10.2);
2. **Dimension reduction:** select aggressive executions (E_a, E_b) as the bivariate control-relevant flow (N^+, N^-) (section 10.3);
3. **Hawkes fit:** estimate (μ, A, β) with stability enforced by $\rho(K) < \rho_{\max}$ (section 10.5);
4. **Microstructure fit:** estimate fill proxy $f(\delta)$ and marking parameters from executions (section 10.6);
5. **HJB solve → policy:** solve the reduced HJB on tick-consistent grids and export a quoting policy (section 10.7);
6. **Evaluation:** replay-mode and simulation-mode backtests with diagnostics and sign-consistent bookkeeping (section 10.8).

Core assumptions (and why they are made).

- **Exponential Hawkes kernels:** chosen for a Markov intensity state that makes HJB control feasible;
- **2D aggressive-flow reduction:** assumes executions carry the dominant short-horizon signal for fills and adverse selection;
- **Reduced microstructure proxies:** fill/impact are calibrated from observable consumption and mid-move patterns, trading structural realism for stability across episodes;
- **Tick-consistent control discretization:** ensures the numerical control problem respects the empirical price grid.

10.10 Code availability and reproducibility

All experiments, calibration routines, numerical solvers, and backtesting utilities referenced in this thesis are implemented in a public repository:

`github.com/andreatrdt/HawkesOptimalControl`

The repository contains the end-to-end pipeline described in section 10, including: (i) LOBSTER[2] message preprocessing and event extraction, (ii) Hawkes calibration and diagnostics, (iii) reduced-form microstructure calibration, (iv) HJB numerical solution and policy export, and (v) replay/simulation backtests with sign-consistent bookkeeping.

11 Results and empirical findings

This thesis developed an end-to-end pipeline for Hawkes-driven optimal market making on limit-order-book data, linking (i) a calibrated bivariate Hawkes model for aggressive execution flow, (ii) reduced-form microstructure proxies for fills and short-horizon adverse moves, (iii) a numerically solved HJB in a Markovian intensity state (exponential kernels), and (iv) an executable quoting policy evaluated in replay and simulation regimes with sign-consistent bookkeeping.

Order-flow clustering and the Hawkes mechanism. Across intraday windows, aggressive executions are strongly overdispersed relative to a Poisson benchmark, with Fano factors well above 1 and increasing with the bin size (Figure 5). This is consistent with bursty clustering and motivates modelling the two-sided execution flow using self-exciting intensities.

Calibration stability and well-posedness in practice. Rolling-window estimation produces parameter values that are broadly stable on short horizons while still exhibiting episodes with near-critical behaviour. The stability diagnostics (Figure 8) and observed values of $\rho(K)$ provide a practical check that the fitted dynamics remain in the subcritical region required for simulation and control.

Model-consistency checks before control. The simulator driven by the fitted Hawkes model reproduces key marginal timing signatures of the data (inter-arrival and binned-count distributions; Figure 7). These tests do not certify structural correctness, but they materially reduce the risk that subsequent policy evaluation is dominated by calibration or implementation errors.

State-dependent quoting and backtest behaviour. The HJB policy produces systematic variation of quote offsets with inventory and intensity imbalance (Figure 15). Qualitatively, the policy becomes more protective when short-horizon order-flow pressure is high and skews when inventory is away from target, aligning with the economic trade-off between spread capture and adverse selection. Replay-mode backtests isolate execution mechanics under the realized midprice path, while simulation-mode tests the policy under the fitted stylised environment, enabling controlled stress-testing of the endogeneity and microstructure parameters.

Reproducibility as an empirical result. A concrete deliverable of the thesis is a reproducible *data* $\rightarrow$ *calibration* $\rightarrow$ *policy* $\rightarrow$ *backtest* workflow. The modular implementation (Section 10) makes each transformation explicit and enforces consistent sign conventions, so that diagnostic plots and performance summaries can be regenerated from a fixed episode window without hidden manual steps.

12 Conclusion

Market making is often framed as a simple exchange: provide liquidity, earn the spread, and keep inventory from turning that spread into risk. The empirical reality in electronic limit order books is harsher: executions arrive in bursts, activity is endogenous, and the moments that increase fill opportunities are frequently the same moments in which adverse selection becomes most severe. This thesis took that observation as a design constraint and asked whether one can build a market-making control problem that is (i) genuinely data-anchored, (ii) explicitly

[2]The LOBSTER dataset is distributed under license and is therefore not included in the repository.

state-dependent in the sense that the state includes a self-exciting order-flow signal, and (iii) still solvable and executable on a tick-consistent grid.

The proposed answer is an end-to-end pipeline that links limit-order-book data to an implementable quoting policy through a model deliberately structured for control. On the modelling side, aggressive buy/sell execution flow is represented by a bivariate Hawkes process, and exponential kernels are used not as a claim of structural truth but as a tractability device: they yield a finite-dimensional Markov embedding for the intensity state. The market maker's interaction with the flow is encoded through predictable mechanisms—thinning for quote-dependent executions and Bernoulli marking for short-horizon price moves—so that both fills and adverse selection are expressed within the same compensator-based point-process framework. This produces a controlled Markov state in which dynamic programming is legitimate, an HJB equation can be derived, and (via CARA factorization) the dimensionality reduces to a numerically manageable form that can be solved with a stable, tick-aligned IMEX discretization and exported as a feedback policy.

A second contribution is theoretical: the thesis does not simply reuse the classical diffusion-based market-making arguments, but rebuilds the control layer in a Hawkes-driven jump setting and makes explicit the mathematical conditions under which the resulting dynamic programming principle is valid. In particular, the work develops self-contained proofs that are typically assumed or only sketched in the literature when moving from Poisson/Lévy arrivals to self-exciting point processes: (i) a Markovian embedding of the Hawkes intensities (under exponential kernels) that yields a controlled jump–diffusion state with an explicit generator; (ii) the semimartingale/compensator representation needed to justify Dynkin-type identities for the controlled process; and (iii) a rigorous verification route connecting the HJB solution to an admissible feedback control in this non-Lévy environment, with the required integrability and non-explosion conditions stated in a form that matches the calibrated/simulated model. These results are not presented as abstract generality for its own sake: they are the steps that make the numerical solution meaningful, and they close the gap between "writing down an HJB" and having a well-posed control problem in a self-exciting order-flow model.

Empirically, the thesis makes the data layer part of the argument rather than an afterthought. The event representation is constructed in a LOBSTER-consistent way and carried through calibration, control interpretation, and P&L bookkeeping with a single sign convention, removing a common source of silent failure in market-making implementations. Diagnostics motivate the Hawkes choice by showing pronounced overdispersion and clustering relative to a Poisson benchmark, and goodness-of-fit checks provide baseline evidence that the fitted model captures first-order timing regularities well enough to serve as a stylised simulator. Windowed calibration further highlights a practical message: parameters can be locally stable while still approaching near-critical regimes, so stability and subcriticality checks are not theoretical decoration but operational requirements for simulation and control.

On the control side, the main outcome is a policy mechanism rather than a single headline P&L number. Solving the HJB with the intensity state produces quote offsets that adapt systematically to both inventory and intensity imbalance: the policy becomes asymmetric in one-sided bursts, protecting the vulnerable side against adverse selection while leaning quotes to promote rebalancing when inventory deviates from target. The evaluation framework strengthens interpretation by separating replay mode (robustness under the realized midprice path, minimizing reliance on the simulator) from simulation mode (model-consistent stress tests inside the calibrated environment), and by decomposing performance into trade P&L versus mark-to-mid inventory effects so that losses can be attributed to adverse selection and exposure rather than to execution scarcity. Compared to a static fixed-skew benchmark, the consistent qualitative finding is that state dependence in (λ^+, λ^-) translates into tighter inventory behaviour and a more controlled exposure profile, which is exactly the economic channel the model is designed to capture.

Overall, the thesis does not claim a full-fidelity limit-order-book simulator; it claims a controlled and diagnosable laboratory in which empirically calibrated self-exciting order flow drives both opportunity and risk, and in which the optimal response is computed, executed, and audited end-to-end. In that sense, its value is methodological, empirical, and theoretical at once: it makes explicit what is learned from data, what is assumed for tractability, and what must be proved to justify the control step—providing a clean base on which more realistic price formation, richer event dictionaries, queue-aware execution, marked Hawkes specifications, and broader PDE generality (e.g., viscosity formulations when smoothness cannot be expected) can be added without losing the central "calibrate–control–verify" logic developed here.

13 Future work and development directions

The framework developed in this thesis was designed around a deliberate trade-off: keep the market model rich enough to capture the two dominant microstructural forces for a market maker—*bursty order flow* and *state-dependent adverse selection*—while remaining simple enough to yield a controlled Markov state and a numerically solvable HJB. Future work naturally follows the fault lines of that trade-off. Some directions increase **market realism** (how prices form and how orders actually execute), others increase **mathematical generality** (weakening smoothness and Markov assumptions), and others address **scalability and robustness** (higher-dimensional state representations and out-of-sample performance).

13.1 A more realistic midprice model (beyond one-tick Bernoulli marking)

In the current setup, price formation is compressed into a simple mechanism: each aggressive trade may trigger a one-tick move, and the probability of a move depends on the Hawkes state through imbalance. This is effective for isolating the *directional* component of adverse selection, but it also hard-codes two restrictive features: tick moves are binary and of fixed size, and midprice dynamics are driven by a reduced-form marking rather than by explicit quote/queue dynamics.

A first natural upgrade is to move from Bernoulli marks to *multi-tick* or even continuous marks, by calibrating a conditional distribution for the size of midprice changes given event sign, intensity imbalance, and possibly additional LOB covariates. This leads to a compound point-process price model, in which the control problem remains interpretable (fills generate inventory exposure to marked price moves) but the distribution of adverse price jumps becomes closer to what is seen in data. In parallel, one can enrich the diffusive component by letting drift and/or volatility depend on the Hawkes state, separating background microstructure noise from systematic flow pressure. Finally, a more structural direction is to model price changes as their own event types within a multivariate (possibly marked) Hawkes system, so that feedback between flow and price is learned rather than imposed ex post through marking.

At the highest realism level, one would replace midprice by a queue-reactive object such as the microprice, or a depletion-based mechanism where best quotes move when queues are exhausted. This is a qualitatively different step because it requires tracking at least best-level depth and introduces additional state variables, but it is also the direction that most directly connects adverse selection to observable order-book conditions.

13.2 From 2D aggressive flow to richer event dictionaries

The two-dimensional reduction to aggressive buy/sell market orders is computationally convenient and empirically meaningful, but it ignores cancellations and submissions that shape queue risk and short-horizon liquidity conditions. A richer event dictionary would allow the state to reflect whether the book is currently *fragile* (thin depth, high cancellation pressure) or *resilient* (rapid replenishment), which in turn affects both execution quality and adverse selection.

One extension is to promote additional streams—for instance, limit submissions and cancellations by side, and possibly best-quote changes— into a higher-dimensional Hawkes state. Another, complementary direction is to allow intensities to depend explicitly on observed LOB state (spread, depth, imbalance), leading to state-dependent Hawkes models that better capture regime changes and intraday seasonality. Finally, kernel flexibility can be improved: power-law kernels capture long memory, while sum-of-exponentials kernels retain a finite-dimensional Markov embedding and can approximate slow decay at moderate computational cost. In practice, sum-of-exponentials with a sparse excitation structure is often the sweet spot between realism and tractability.

13.3 Control and execution realism: queue position, latency, and partial fills

Execution modelling remains the largest gap between a tractable control problem and the behaviour of real market-making systems. In the present thesis, the control affects execution via a reduced-form fill probability depending on quote distance from mid. This is appropriate for building a clean end-to-end loop, but it abstracts away *queue position* and *competition*, which are central in electronic markets: being at the best quote is not enough; where you sit in the queue often dominates fill likelihood and adverse selection exposure.

A more realistic control layer would track best-level queue sizes and the agent's queue position, with executions occurring when incoming flow and cancellations deplete the queue ahead of the agent. This brings cancellations and limit submissions back into the execution mechanism in a structural way. Another crucial upgrade is to incorporate latency and order-update constraints: refresh limits, cancellation costs, and minimum quote lifetimes typically make optimal policies less reactive and often more robust out-of-sample. Finally, inventory

management can be made more realistic by replacing symmetric bounds with explicit risk limits and liquidation mechanics; similarly, a more structural P&L attribution would separate temporary and permanent impact and model execution-price dynamics directly, rather than representing adverse selection only via midprice marking.

13.4 Stronger mathematical generality: viscosity solutions and weak DPP

Several of the realism upgrades above introduce non-smooth features: discrete decision sets, hard inventory limits, queue-position dynamics, and possibly degenerate diffusions. In such settings, classical smooth-solution arguments for the HJB can fail even when the control problem is well-posed. A standard way forward is to move to a viscosity-solution characterization of the value function, which provides a robust notion of solution under weak regularity and is compatible with monotone numerical schemes.
Relatedly, one can aim for a weaker form of the dynamic programming principle (DPP) that requires minimal measurability and integrability assumptions for controlled point processes. This broadens the admissible control class and reduces dependence on strong Markovian representations. While these extensions increase mathematical overhead, they also make the theory align better with the discontinuities inherent to market microstructure.

13.5 Numerics: higher dimensions, approximation, and verification

As soon as the state dimension grows beyond a handful of variables, full grid-based PDE methods become expensive. The most promising route to keep Markov structure while scaling is to exploit *structure*: sum-of-exponentials kernels preserve finite-dimensionality, and if the excitation matrix is sparse (or approximately sparse), one can reduce computation substantially. In parallel, approximate dynamic programming methods can be used to learn policies from simulation on Hawkes states and benchmark them against HJB-based policies in the current framework.
Even within grid methods, there is room for sharper numerics. Adaptive discretization can concentrate resolution where intensities and inventories spend most of their mass, while error control can quantify the effect of interpolation and boundary clamping near inventory limits. Finally, numerical verification should be treated as part of the methodology: forward-simulation estimates of the value under the computed policy provide a practical consistency check for the PDE solution and can reveal discretization artefacts early.

13.6 Empirical robustness and cross-asset generalization

Beyond modelling choices, the key empirical question is stability: how much do calibrated parameters and resulting policies vary across days, regimes, and assets? A principled next step is systematic out-of-sample evaluation using train/test splits across trading days, volatility regimes, and intraday periods. Cross-asset transfer is equally informative: assets with different tick sizes and liquidity profiles should induce different parameter distributions and potentially different optimal quoting behaviour, and mapping these differences is a way to separate universal mechanisms from asset-specific quirks.
Finally, the Hawkes model itself should be benchmarked against alternative intensity models using predictive likelihood and standard goodness-of-fit diagnostics. The goal is not to declare a single "true" microstructure model, but to understand which modelling ingredients are necessary for a policy to remain stable once it leaves the calibration window.

Closing outlook. The thesis provides a controllable and diagnosable environment in which self-exciting order flow drives both opportunities and risk, and in which optimal quoting is computed rather than prescribed. The most valuable next steps are those that increase realism without breaking this *calibrate–diagnose–control–verify* loop. Concretely, three upgrades stand out: enrich the price-marking mechanism beyond one-tick moves; introduce best-level queue dynamics to connect execution directly to observable LOB conditions; and extend Hawkes kernels via sum-of-exponentials to capture longer memory while preserving a finite-dimensional state for control.

References

F. Abergel, M. Anane, A. Chakraborti, A. Jedidi, and I. Muni Toke. *Limit Order Books*. Cambridge University Press, 2016.

M. Avellaneda and S. Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008. doi: 10.1080/14697680701381228.

E. Bacry, I. Mastromatteo, and J.-F. Muzy. Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(1):1550005, 2015. doi: 10.1142/S2382626615500057.

P. Brémaud. *Point Processes and Queues: Martingale Dynamics*. Springer, 1981. doi: 10.1007/978-1-4684-9477-8.

P. Brémaud and L. Massoulié. Stability of nonlinear hawkes processes. *The Annals of Probability*, 24(3): 1563–1588, 1996. doi: 10.1214/aop/1065725193.

Á. Cartea, S. Jaimungal, and J. Penalva. *Algorithmic and High-Frequency Trading*. Cambridge University Press, 2015.

R. Cont, S. Stoikov, and R. Talreja. A stochastic model for order book dynamics. *Operations Research*, 58(3): 549–563, 2010. doi: 10.1287/opre.1090.0806.

L. Cui, A. Hawkes, and H. Yi. An elementary derivation of moments of hawkes processes. *Advances in Applied Probability*, 52(1):102–137, 2020. doi: 10.1017/apr.2019.53.

D. J. Daley and D. Vere-Jones. *An Introduction to the Theory of Point Processes. Volume I: Elementary Theory and Methods*. Springer, 2 edition, 2003. doi: 10.1007/978-0-387-49835-5.

W. H. Fleming and H. M. Soner. *Controlled Markov Processes and Viscosity Solutions*, volume 25 of *Stochastic Modelling and Applied Probability*. Springer Science & Business Media, 2 edition, 2006. doi: 10.1007/0-387-31071-1.

O. Guéant, C.-A. Lehalle, and J. Fernandez-Tapia. Dealing with the inventory risk: a solution to the market making problem. *Mathematics and Financial Economics*, 7:477–507, 2013. doi: 10.1007/s11579-012-0087-0.

A. G. Hawkes. Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1):83–90, 1971. doi: 10.1093/biomet/58.1.83.

A. G. Hawkes and D. Oakes. A cluster process representation of a self-exciting process. *Journal of Applied Probability*, 11(3):493–503, 1974. doi: 10.1017/S0021900200096273.

W. Huang, C.-A. Lehalle, and M. Rosenbaum. Simulating and analyzing order book data: The queue-reactive model. *Journal of the American Statistical Association*, 110(509):107–122, 2015. doi: 10.1080/01621459.2014.982278.

K. Jain, N. Firoozye, J. Kochems, and P. Treleaven. Limit order book simulations: A review, 2024. revised Mar 2024.

T. Leblanc. Exponential moments for hawkes processes under minimal assumptions. *Electronic Communications in Probability*, 2024. doi: 10.1214/24-ECP625. Also available as arXiv:2505.15253.

X. Lu and F. Abergel. High-dimensional hawkes processes for limit order books: modelling, empirical analysis and numerical calibration. *Quantitative Finance*, 18(2):249–264, 2018. doi: 10.1080/14697688.2017.1403142.

Y. Ogata. On lewis' simulation method for point processes. *IEEE Transactions on Information Theory*, 27(1): 23–31, 1981. doi: 10.1109/TIT.1981.1056305.

H. Pham. *Continuous-time Stochastic Control and Optimization with Financial Applications*. Springer, 2009. doi: 10.1007/978-3-540-89500-8.

E. Smith, J. D. Farmer, L. Gillemot, and S. Krishnamurthy. Statistical theory of the continuous double auction. *Quantitative Finance*, 3(6):481–514, 2003. doi: 10.1088/1469-7688/3/6/307.